Futures Mechanics and Carry

Where a futures price and its return come from, and the rolls, series and margin a trader has to get right

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Abstract

A futures position earns the change in the spot price plus a carry that the trader can read from the curve in advance. Most failures with futures come from the mechanics: expiry dates, stitched price series and margin.

A futures contract costs nothing up front. The trader posts margin, which remains the trader's money, and the exchange settles gains and losses in cash every day. The exposure is far larger than the margin. One standard S&P 500 contract controls about 325,000 dollars of stock against a deposit of about 25,000. For a storable asset, the futures price equals the spot price plus financing and storage costs, minus any income until delivery. The futures price is therefore not a forecast. Only for non-storable underlyings such as the VIX does it act as a forecast, and a biased one. The slope of the curve is carry. In contango a long position loses a little each day as the contract converges to spot. In backwardation it gains. With crude oil at 70 dollars and the next contract at 70.80, a long position pays about 14% a year if the price stays flat.

Rolling to the next contract neither makes nor loses money. The curve's cost is paid daily, not at the roll. A fully cash-collateralized futures position replicates the underlying asset and is not leveraged. In the chapter's worked example, futures plus interest on the cash earned 233 dollars against 232 in dividends from the shares. No single futures price history exists. Each method of stitching contracts preserves one property and breaks another. Back-adjustment can produce negative prices that never traded, and an unadjusted splice books each roll gap as a gain or loss.

The chapter's practical conclusions are these. Use a series with correct returns for research and the actual front contract for sizing. At 10 times leverage, an ordinary 3% move triggers a margin call, and exchanges raise margin after large moves. Keep margin use to about a quarter or a third of the account. Micro contracts provide sizing precision, not lower cost: a 100,000-dollar account needs 0.29 standard S&P 500 contracts, or about three micros. Carry predicts returns across asset classes, but carry trades tend to crash together. A product that collected VIX futures carry lost about 96% of its value in one day in February 2018.

What this is. For an individual with a large enough account, futures are among the cheapest and most liquid ways to hold equity indices, government bonds, currencies and commodities. For a US taxpayer they are often the most tax-efficient way as well. They are also the native instrument of trend-following and carry strategies. Yet many people trade futures without knowing where the price comes from. They may not know why a "continuous" chart can show prices below zero that no contract ever traded at, or what the margin they posted actually buys.

This chapter builds futures from the contract up. It covers what the specification sheet says, and why the futures price differs from the spot price. It covers where a futures position's return comes from, and what rolling does and does not cost. It shows how to turn a sequence of expiring contracts into a usable price history, how margin and leverage work, and what micro contracts change for a small account. Carry, the return a futures position earns if nothing happens, runs through all of it, because it is the part of the return that can be seen in advance.

How to read this chapter. §1 is the conceptual core: the contract, the three ideas the rest depends on, and one position followed for a quarter. §2 dissects a contract specification, with a reference table of the contracts an individual is most likely to trade. §3 is history, and §4 is the bibliography. §5 and §6 are the economics: what sets the futures price, and what a futures position earns. §7 to §10 are the plumbing: rolling, continuous price series, margin, and micro contracts and position sizing. §11 turns carry into a trading input. §12 shows that most of the chapter reduces to a handful of identities. §13 is the build: the data needed and about 60 lines of code. §14 catalogues how things go wrong, with case studies, and §15 is the synthesis.

Different readers can start in different places.

- Readers who already trade futures and want the parts that are usually skipped should read §1.1, §6.3, §8.4 and §9.3.
- Readers about to place a first futures trade should read §1, §2, §9 and §10 first, and §7 before the first expiry.
- Readers who build backtests should read §7.5, §8 and §13, which change results.

Appendix A defines the market and statistical vocabulary that the main text relies on, built up in dependency order.

Objectives. After this chapter, you should be able to:

- compute what one contract controls, and what a round trip in it costs;
- derive the futures price from the cost of carry, and say where that argument fails;
- split a futures return into spot return, carry and curve shock, and explain why the roll itself costs nothing;
- choose a roll rule and a continuous series that suit the job, and read an adjusted price level correctly;
- size a position by risk, and compute its distance to liquidation;
- measure carry in each asset class, and use it as a signal or as a breakeven.

Relationship to the other notes. Systematic Trading Strategies treats carry as a strategy family and computes the minimum account size for a diversified futures portfolio. This chapter supplies the mechanics underneath both. Trend-Following in Financial Markets uses futures throughout, and summarises the stitching problem that §8 treats in full. Bonds and Bond Markets covers Treasury futures, the cheapest-to-deliver bond and the cash–futures basis trade, which §5.2 only sketches. Implied Volatility explains the VIX, on which VIX futures settle. Value at Risk explains the risk measure that exchanges use to set margin. Each note stands alone.

Where the numbers come from. Contract specifications are the exchanges' own, checked against their published pages in 2026. Specifications change, and §2.3 says how to check them. Worked examples use round illustrative prices, not quotes. They are labelled **[Computed]** when the number follows from the chapter's own arithmetic. The three figures are drawn from stylised or simulated markets, not market data, because each one illustrates a mechanism rather than a history. Their generators are `figures/fm_*.py` in the source repository.

A warning about scope. [Practice] Nothing here is investment advice. Futures can lose more than the money deposited against them. The rules described are US exchange rules as of 2026 unless stated otherwise, and brokers add their own rules on top.

Epistemic tags. The chapters in this collection flag claims by status:

- **[Fact]** — replicated across independent datasets or implementations; broad agreement.
- **[Contested]** — documented, but with live disagreement about magnitude, robustness, or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner convention; the evidence may be private or absent. A [Practice] claim is not a debunked one.

This chapter adds a fifth tag:

- **[Computed]** — arithmetic from the chapter's own worked examples.

Tags appear only where the status changes what a reader should do. A tag governs the sentence or clause it opens. Untagged sentences are definitions, derivations, or contract terms, true by construction or by rulebook rather than by evidence. The dated events in the history of §3 are also untagged. Each one is sourced where it recurs later.

Notation. The table lists every symbol that recurs in the chapter, with the section that defines or first uses it. Symbols used in only one section are defined where they appear. All rates are continuously compounded annual rates unless stated otherwise.

Symbol	Meaning	Defined in
S_t	**Spot price** of the underlying at time t: the price for immediate delivery, or the index level	§1.5, §5.1
$F_t(T)$; F_t	**Futures price** at time t of the contract that expires at T; the futures price of whichever contract is held	§1.2, §1.5
$\tau = T - t$	**Time to expiry**, in years	§1.5
$F^{(1)}, F^{(2)}, \ldots$	Prices of the first contract (the nearest, or after a roll date the contract held), the second, and so on	§6.2
$T_1 < T_2 < \ldots$; $\tau_1 < \tau_2 < \ldots$	Their expiries, and their times to expiry	§6.2, §11.1
r	**Financing rate**: the risk-free rate, or the **repo** rate on a loan secured by the asset itself	§1.5, §5.1
q	**Income yield** the asset pays its holder: dividends, coupons, or a foreign interest rate	§1.5, §5.1

Symbol	Meaning	Defined in
u	**Storage cost**: warehousing and insurance	§5.1
y	**Convenience yield**: the implicit benefit of having the physical commodity on hand	§5.1
$c = r + u - q - y$	**Net cost of carry**: positive when holding the asset costs money, negative when it pays	§1.5, §5.1
$c_t(T)$	Observed slope of the curve: the annualised log gap between a contract and spot	§6.2
$\kappa = -c$	**Carry**: the return a long futures position earns per year if spot does not move and the curve keeps its slope	§1.5, §6.2
U; I	Storage cost and income in dollars, rather than as yields	§5.1, §5.6
$F_t - S_t$	The **basis** (§12.3 lists the other things "basis" means)	§5.4
$\pi_t(T)$	**Risk premium** in price units: the profit a long position expects for bearing the risk	§5.4
r^{impl}; r^{bill}	Implied financing rate of an index future; Treasury bill rate (simple)	§5.2; §6.1
m	Contract **multiplier**: dollars per point of the futures price (50 for the E-mini S&P 500, 1,000 for crude oil quoted per barrel)	§1.2
N	Number of contracts, signed: positive for long, negative for short	§1.2
NmF_t	**Notional**, or **exposure**, of a position	§1.5
h; mh	**Tick**, the minimum price change; **tick value** in dollars	§2.6
A	Account **equity**, in dollars	§9.1
M^{M}	**Maintenance margin** per contract	§9.1
$\mu = M^{\text{M}}/(mF_t)$	Maintenance margin as a fraction of one contract's notional	§9.1
$L = \lvert N \rvert mF_t/A$	**Notional leverage**	§9.1
$x^\star$	Adverse price move, as a fraction, that takes an account to liquidation	§9.1
σ; $\sigma^\star$	Annualised volatility of a contract's percentage returns; annual volatility the account targets	§10.2
w	Share of the risk budget given to one market	§10.2
$N^\star$	Ideal, unrounded number of contracts that the risk budget implies	§10.2
b	Half-width of the no-trade buffer around $N^\star$	§10.4
s	Trading signal, scaled to lie between −1 and 1	§12.1
Δ; Δx	Short holding period, in years; change in x over it	§1.5, §6.1
$E_t[\cdot]$	Expectation given the information available at t	§5.4
ln	Natural logarithm; returns written $\Delta \ln F$ are log returns	§1.5

Three notes on reading the symbols. First, c has two readings. In the cost-of-carry model it is $r + u - q - y$. From §6.2 on, and in the second identity of §1.5 and in §5.3, $c_t(T)$ is the observed slope of the curve. The two are equal when the cost-of-carry relation holds. Second, where the period Δ multiplies a rate, it is written after the rate, so $\kappa\,\Delta$ is the carry earned over the period, not a change in κ. Third, the chapter follows the collection's reserved letters: S_t is always the spot price, r the financing rate, and σ a volatility.

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1. What a futures contract is

1.1 The wrong intuitions

Many mistakes with futures come from a picture carried over from buying shares. Eight parts of that picture need replacing before the rest of the chapter makes sense. The table lists each one, the correction, and the section that explains it.

The intuition	What is actually true	Where
The futures price is the market's forecast of the future spot price	For anything that can be stored or financed, the futures price is mostly today's spot price plus the cost of carrying the asset to expiry. Only a few contracts, such as VIX futures, price a forecast, and that forecast is biased	§5
Contango loses money at the roll, when the trader "sells low and buys high"	The roll swaps one contract for another at market prices and has no value of its own. The cost of contango is paid every day, as the contract held drifts down toward spot	§6.3
Contango is bearish and backwardation bullish	The slope gives the carry, the return if spot does not move. A market in contango can rally, and a backwardated one can collapse. The slope is a cost or a credit, not a direction	§5.4, §6.2

The intuition	What is actually true	Where
Buying a contract costs the margin	Buying a contract costs nothing. Margin is a deposit that still belongs to the trader and can earn interest. The exposure is the full notional value	§1.2, §9
Futures are leveraged, so they are dangerous	Leverage is notional exposure divided by equity, and the trader chooses it. One E-mini contract held against 330,000 dollars of cash is unlevered	§9.3
A continuous futures chart is a price history	A continuous chart is a construction. Each method preserves one property and breaks another, and some show prices that never traded, including negative ones	§8
An index future tracks the index	An index future tracks the index's total return, dividends included, minus a financing rate	§5.2, §6.2
Micro contracts are a cheap way to trade	Per dollar of exposure, micros cost more to trade. What they buy is granularity	§10

1.2 The contract: an agreement, settled every day

A **futures contract** is a standardised, exchange-traded agreement to buy or sell a fixed quantity of something at a price agreed today. Delivery, or cash settlement, happens at a fixed later date. The buyer is **long** and gains if the futures price rises. The seller is **short** and gains if it falls. Two features separate a futures contract from buying the asset itself.

Entering costs nothing. No money changes hands for the contract when the trade is made. Instead, both sides post **initial margin** with their broker. Initial margin is a performance bond: collateral against future losses, not a down payment. It still belongs to the trader who posted it.

Gains and losses are paid every day. At the end of each trading day the exchange sets a **settlement price**. Every open position is marked to that price. The day's change in value then moves between accounts in cash, as **variation margin**. The cash flows from losers to winners, through the clearing house, before the next day's open. A position of N contracts receives

$$\text{variation margin}_t = N\, m\, (F_t - F_{t-1})$$

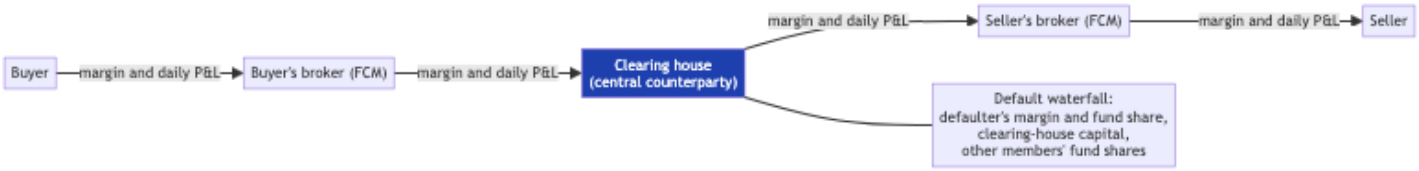
on trading day t, where F_t is that day's settlement price and F_{t-1} the previous day's. For a long position ($N > 0$), the amount is negative when the futures price fell, and a negative amount is a payment. Because gains and losses are settled daily, the contract's value is reset to zero every evening. The position is marked to the new settlement price, and the next day's gain or loss is measured from there. A trader who holds a contract overnight therefore holds an exposure to the next day's price change, not an asset with a value.

Most contracts never reach delivery. Traders close them, or **roll** them into a later contract (§7), before expiry. Only a small fraction, commonly put at under 2%, ends in physical delivery. A contract that reaches expiry settles in one of two ways:

- **Physical delivery.** The short delivers the commodity, bond or currency, and the long pays the final settlement price for it. Examples are barrels of crude oil at Cushing, Oklahoma; Treasury notes from a deliverable basket; and euros paid into a bank account.
- **Cash settlement.** No goods change hands. The final variation-margin payment is computed against a reference price, such as the opening prices of the 500 stocks in the S&P 500 on expiry morning.

1.3 Who stands behind it

The two parties to a futures trade never face each other. Once the trade is matched, the exchange's **clearing house** steps between them by **novation**. It becomes the buyer to every seller and the seller to every buyer. A trader's own counterparty is their **futures commission merchant** (FCM). The FCM is the broker, and it is either a member of the clearing house or works through one. The diagram shows the chain.



The clearing house turns a web of credit exposures between pairs of traders into a hub. It protects itself with three resources: margin, a guarantee fund contributed by its members, and its own capital. If a member defaults, these absorb the losses in a fixed order called the **default waterfall**. First come the defaulter's own margin and its share of the guarantee fund. Next comes a tranche of the clearing house's capital. Last come the other members' shares of the fund.

An FCM must keep customer money in **segregated accounts**, separate from the firm's own money. **[Fact]** This protection is strong but not absolute. When MF Global failed in October 2011, about 1.6 billion dollars of segregated customer money was missing, and customers were made whole only years later. The risk an individual actually bears is the broker's, not the exchange's (§14.2).

1.4 Forward versus futures

A **forward** is the same agreement made privately between two parties. It has no exchange and no daily settlement, and the whole gain or loss is paid at expiry. If interest rates were known in advance, forward and futures prices would be identical (Cox, Ingersoll & Ross, 1981). The two prices differ only when interest rates move with the futures price.

Suppose the futures price tends to rise when rates rise. The long then receives variation margin when it can be invested at a high rate, and pays variation margin when it can be borrowed cheaply. That timing has value to the long, so the futures price settles above the forward price. If the futures price tends to fall when rates rise, the futures price settles below the forward price.

[Fact] For equity, commodity and currency futures the difference is negligible. For long-dated interest-rate futures, such as SOFR futures several years out, the difference is a measurable **convexity adjustment**. A SOFR future is quoted as 100 minus the rate, so its price falls when rates rise. The futures price therefore sits below the forward price, and the futures rate sits above the forward rate.

Daily settlement has two practical consequences for a hedger. Consider a short futures hedge on a stock portfolio. The hedge pays and receives cash every day, while the portfolio's gain or loss stays unrealised. The hedge therefore needs a cash buffer. Second, each day's futures gain earns interest until the hedge is lifted, and each day's loss costs interest. The futures leg therefore grows slightly faster than the portfolio it offsets. A precise hedger compensates by scaling the contract count down by the discount factor to the hedge's end. This adjustment is called **tailing the hedge**. For an individual the buffer matters, and the tailing does not.

1.5 The spine: price, return, collateral

Three ideas generate nearly everything in this chapter. Each is an identity, or close to one. Each answers a question a trader should be able to answer about any contract: what sets its price, what it earns, and how much of it to hold.

1. The price: spot, carried forward. For anything that can be bought today and held, the futures price equals the spot price grown at the net cost of carry until expiry:

$$F_t(T) \;=\; S_t\, e^{c\,\tau}, \qquad c = r + u - q - y.$$

Financing and storage make holding the asset expensive, so they push the futures price above spot. Income and convenience make holding the asset rewarding, so they pull the futures price below spot. The sign of c therefore decides whether the curve slopes up (**contango**) or down (**backwardation**). For things that cannot be stored, such as electricity or the VIX, this link breaks. There the futures price is an expectation adjusted by a risk premium (§5.4).

2. The return: spot move plus carry. Over a short period Δ, a futures position earns three things: the log return of the spot price, the carry, and a term for any change in the slope of the curve:

$$\Delta \ln F \;=\; \Delta \ln S \;+\; \kappa_t\,\Delta \;+\; (c_{t+\Delta} - c_t)(\tau - \Delta).$$

Carry, $\kappa_t = -c_t$, is the part of the return known in advance: what the position earns if nothing else happens. In contango, carry is negative. It is paid daily, as the contract converges to spot, and not at the roll (§6). The last term is the change in

the slope c over the period, multiplied by the time the contract has left to expiry. A steeper or flatter curve therefore moves a distant contract more than a near one. The term is small for a contract a few weeks from expiry. When c is read as the observed slope of the curve, rather than as the model's $r + u - q - y$, the equation is an exact identity (§6.2).

3. The collateral: exposure is notional, margin is a bond. A futures position is exposure to NmF dollars of the underlying, acquired with no money. The trader's cash, including the margin deposit, stays the trader's and can earn interest separately, provided it sits in Treasury bills or with a broker that pays interest (§9.5). A fully collateralised futures position is one backed by cash equal to its notional. It earns

$$\text{total return} \;=\; \text{interest on the collateral} \;+\; \text{futures excess return},$$

where the **futures excess return** is the change in the futures price as a fraction of the notional, $\Delta F/F$ (§6.1). Leverage is then a choice made through the number of contracts, not a feature of the instrument. Margin caps the leverage a trader may take and decides when the trader is forced out. It does not measure risk (§9).

In short, the first identity says what a buyer pays for, and the second says what the position earns. The third says how much exposure the account holds and what can force it to stop holding it. Every later section elaborates one of the three.

1.6 A worked instance: one quarter of a Micro E-mini position

An account of 100,000 dollars wants exposure to about 100,000 dollars of the S&P 500. It buys three **Micro E-mini S&P 500** contracts (symbol MES, multiplier 5 dollars). Assume the index is at 6,500, the financing rate is 3.75%, the index's dividend yield is 1.2%, and the December contract has 80 days to expiry. **[Computed]**

The price. The first identity, with $c = r - q = 2.55\%$, gives

$$F = 6{,}500 \times e^{0.0255 \times 80/365} = 6{,}536.43.$$

The 36-point premium over the index equals the financing a buyer of the stocks would pay over 80 days, less the dividends that buyer would receive. Three contracts give an exposure of $3 \times 5 \times 6{,}536.43 \approx 98{,}000$ dollars.

The cash. The broker requires initial margin of roughly 2,500 dollars a contract, about 7,500 in total, held as cash in the account. The other 92,500 stays with the trader, who can keep it in Treasury bills or a money-market fund. None of the 100,000 has been spent.

The first week. Suppose the contract settles over five days as the table shows. Each point is worth $3 \times 5 = 15$ dollars to the position.

Day	Settlement	Change (points)	Variation margin (dollars)	Cumulative
Entry	6,536.50			
1	6,588.25	+51.75	+776.25	+776.25
2	6,489.00	−99.25	−1,488.75	−712.50
3	6,507.75	+18.75	+281.25	−431.25
4	6,377.50	−130.25	−1,953.75	−2,385.00
5	6,441.25	+63.75	+956.25	−1,428.75

Cash moved every evening. The week's total, −1,428.75, is exactly $15 \times (6{,}441.25 - 6{,}536.50)$. Daily settlement changes when the trader is paid, not how much, apart from interest on the payments (§1.4).

The quarter, if the index goes nowhere. Now suppose instead that the index stays at 6,500 for 72 days. With 8 days left, the contract is then worth $6{,}500 \times e^{0.0255 \times 8/365} = 6{,}503.63$. The futures position has lost $15 \times (6{,}503.63 - 6{,}536.43) = 492$ dollars. This loss is carry: the contract converged toward the index a little every day.

Meanwhile the account's cash earned interest. Count only the part that matches the notional, 98,046 dollars, because that is what an investor buying the stocks would have spent. The remaining 1,954 dollars earns the same interest either way. Assume the margin cash also earns the bill rate (§9.5). The collateral then earned $98{,}046 \times 3.75\% \times 72/365 = 725$ dollars of interest. This figure uses simple interest, the way a money-market fund pays it. Continuous compounding, the convention of the price formula, adds about 3 dollars over a quarter.

Futures plus bills therefore made $725 - 492 = 233$ dollars. An investor who had instead bought 98,046 dollars of the index's stocks would have collected $98{,}046 \times 1.2\% \times 72/365 = 232$ dollars of dividends, on a price that also went nowhere. **[Computed]** The two positions agree to within a dollar: futures plus cash is the index, dividends included.

The roll. With 8 days left, the trader rolls into March. The March contract has 99 days to expiry and is worth $6{,}500 \times e^{0.0255 \times 99/365} = 6{,}545.11$, which is 41.48 points above the December contract. Selling December at 6,503.63 and buying March at 6,545.11 neither makes nor loses money, because the trader swaps one exposure for another at market prices. If the index stays at 6,500, the 41-point gap is the next quarter's carry. It is paid in the same way, daily, over the 91 days until the March contract in turn has 8 days left.

1.7 What a futures contract is not

The table separates a futures contract from six instruments it is often confused with.

A futures contract is not	Because
An option	Both sides are obliged to trade. The payoff is linear in the price, with no premium and no limit on the loss
A loan or a share	A contract pays no dividend or coupon and owns nothing. It is a commitment to trade at a fixed price
A forecast	Its price is mostly spot plus carry (§5.4)
An exchange-traded fund	A fund holds assets or futures and charges a fee. A contract is a standardised obligation with no manager, no fee, and an expiry
Bought "on margin"	Securities margin is a loan that charges interest. Futures margin is a deposit that can earn interest (§9)
A continuous instrument	Each contract expires. A continuous series is a construction (§8)

§1 Key takeaways

1. A futures contract costs nothing to enter. Margin is a performance bond that still belongs to the trader, and the exposure is the full notional value, NmF.
2. Gains and losses settle in cash every day. Daily settlement changes when the trader is paid, not how much (apart from interest on the payments, §1.4). It does mean that losses must be funded immediately.
3. For any asset that can be held, the futures price is the spot price carried forward at the net cost of carry: financing plus storage, minus income and convenience.
4. A futures position earns the spot move plus carry. Carry is the part known in advance, and it accrues daily, not at the roll.
5. Futures plus cash in Treasury bills is the underlying asset, income included, to within a financing spread (§5.2). The worked Micro E-mini quarter matches to within a dollar.
6. Leverage is the notional exposure of the contracts held divided by the cash behind them. The trader chooses it; the instrument does not impose it.
7. The clearing house removes counterparty risk between traders. It does not remove the risk of the trader's own broker.

2. Reading a contract specification

The exchange that lists a futures contract defines it on a one-page specification. Most operational mistakes an individual makes with futures come from a field on that page that went unread. Two common examples are an unwanted delivery notice and a position 10 times the intended size. This section goes through the fields, then collects the contracts an individual is most likely to trade.

2.1 The fields, and why each one matters

The table lists the fields of a specification, with the values for two contracts: the E-mini S&P 500, which is cash settled, and WTI crude oil, which is physically delivered.

Field	What it says	E-mini S&P 500 (ES)	WTI crude oil (CL)
Underlying	What is bought or sold	The S&P 500 index	Light sweet crude oil delivered at Cushing, Oklahoma
Contract unit and multiplier	How much one contract controls; dollars per point of price	50 dollars × index	1,000 barrels; 1,000 dollars per dollar of price
Price quotation	The units of the price	Index points	Dollars and cents per barrel
Minimum tick	The smallest price change, and its dollar value	0.25 points = 12.50 dollars	0.01 dollars = 10 dollars
Listed months	Which expiries trade	March, June, September, December	Every month, for years ahead
Trading hours	When it trades	Nearly 23 hours a day, Sunday evening to Friday afternoon (US Central time)	The same
Last trading day	When trading in an expiry stops	9:30 a.m. Eastern on the third Friday of the contract month	Three business days before the 25th calendar day of the month *before* the contract month
Settlement	How an open position at expiry is closed	Cash, against a special opening quotation of the index	Physical delivery of crude at Cushing
Daily settlement price	The price that sets daily variation margin	Set by the exchange from trading in a short window near the close	The same
Price limits	How far it may move in a session	Tiered limits tied to stock-market circuit breakers (§2.5)	Dynamic circuit breakers rather than fixed limits

Five of those fields cause most of the trouble.

The multiplier sets the size. One E-mini S&P 500 contract at an index level of 6,500 is a 325,000-dollar position. One crude oil contract at 65 dollars is 65,000 dollars of oil. Before trading a contract, compute mF, the exposure that one contract adds. This number is often far larger than the margin on the same page suggests.

The tick value sets the cost unit. Exchanges quote bid–ask spreads in ticks, and in the liquid contracts the spread is usually one tick wide. Crossing the spread once costs half a tick, measured from the midpoint. A round trip that buys at the offer and sells at the bid therefore costs one tick. §2.6 converts this cost into basis points.

The contract month is the delivery month, not the trading month. The "January 2027" crude oil contract stops trading around 20 December 2026, because delivery happens during January. Energy contracts therefore expire earlier than the month in the symbol suggests, and grain and metal contracts reach first notice earlier too. Traders who look only at the month in the symbol are routinely caught out.

Physical delivery is real. A long position in a physically delivered contract can be assigned delivery (§2.4). This happens if the position is still open once the notice period starts or, for crude oil, after the last trading day. An individual cannot take 1,000 barrels of crude at a pipeline hub. Brokers protect themselves by closing customer positions before that point, usually at a time and price of the broker's choosing.

The settlement price is not the last trade. Daily profit and loss, margin, and most data vendors' "close" all use the exchange's settlement price. The exchange computes that price from trading in a defined window. A backtest that uses last-trade prices for some markets and settlement prices for others mixes two different clocks.

2.2 Month codes and symbols

A futures symbol has three parts: a **root**, a **month code**, and a **year**. Every exchange uses the same fixed alphabet of month codes:

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
F	G	H	J	K	M	N	Q	U	V	X	Z

So ESZ6 is the December 2026 E-mini S&P 500, and CLF7 is the January 2027 crude oil contract. A single-digit year is ambiguous across decades, and data vendors and brokers resolve it differently: ESZ26, ESZ2026, or a separate field holding `202612`. The same root can also be listed on several exchanges, or with several multipliers. Brokers therefore distinguish a contract's **symbol** from a unique **contract identifier** and, for some products, from a **trading class**. **[Practice]** Key every stored price on the exchange, the root, and the contract month in `YYYYMM` form, never on a display symbol.

2.3 The contracts an individual is most likely to trade

The two tables below give the specifications of the main US-listed contracts and their micro versions, plus two European benchmarks. **[Practice]** Specifications change. Exchanges revise tick sizes, add months, and redesign settlement procedures. Check the exchange's own page before trading a contract for the first time. CME's pages follow the pattern `cmegroup.com/markets/<asset>/<group>/<product>.contractSpecs.html`.

Equity indices and rates:

Contract	Exchange	Unit	Tick (dollar value)	Months	Settlement
E-mini S&P 500 (ES)	CME	50 × index	0.25 (12.50)	H M U Z	Cash
Micro E-mini S&P 500 (MES)	CME	5 × index	0.25 (1.25)	H M U Z	Cash
E-mini Nasdaq-100 (NQ)	CME	20 × index	0.25 (5.00)	H M U Z	Cash
Micro E-mini Nasdaq-100 (MNQ)	CME	2 × index	0.25 (0.50)	H M U Z	Cash
E-mini Russell 2000 (RTY)	CME	50 × index	0.10 (5.00)	H M U Z	Cash
Micro E-mini Russell 2000 (M2K)	CME	5 × index	0.10 (0.50)	H M U Z	Cash
E-mini Dow (YM)	CBOT	5 × index	1 (5.00)	H M U Z	Cash
Micro E-mini Dow (MYM)	CBOT	0.50 × index	1 (0.50)	H M U Z	Cash
Euro Stoxx 50 (FESX)	Eurex	10 euros × index	1 (10 euros)	H M U Z	Cash
2-Year Treasury note (ZT)	CBOT	200,000 face	1/8 of 1/32 (7.8125)	H M U Z	Physical, deliverable basket
5-Year Treasury note (ZF)	CBOT	100,000 face	1/4 of 1/32 (7.8125)	H M U Z	Physical, deliverable basket
10-Year Treasury note (ZN)	CBOT	100,000 face	1/2 of 1/32 (15.625)	H M U Z	Physical, deliverable basket
Treasury bond (ZB)	CBOT	100,000 face	1/32 (31.25)	H M U Z	Physical, deliverable basket
Three-month SOFR (SR3)	CME	25 per basis point	0.0025 (6.25) near; 0.005 (12.50) other	Quarterly, years out	Cash
Euro-Bund (FGBL)	Eurex	100,000 euros face	0.01 (10 euros)	H M U Z	Physical, deliverable basket

Commodities, currencies, crypto and volatility:

Contract	Exchange	Unit	Tick (dollar value)	Months	Settlement
WTI crude oil (CL)	NYMEX	1,000 barrels	0.01 (10.00)	Every month	Physical
Micro WTI crude oil (MCL)	NYMEX	100 barrels	0.01 (1.00)	Every month	Cash, to the CL settlement
Henry Hub natural gas (NG)	NYMEX	10,000 MMBtu	0.001 (10.00)	Every month	Physical

Contract	Exchange	Unit	Tick (dollar value)	Months	Settlement
Gold (GC)	COMEX	100 troy ounces	0.10 (10.00)	Active: G J M Q V Z	Physical
Micro gold (MGC)	COMEX	10 troy ounces	0.10 (1.00)	G J M Q V Z	Physical
Silver (SI)	COMEX	5,000 troy ounces	0.005 (25.00)	Active: H K N U Z	Physical
Copper (HG)	COMEX	25,000 pounds	0.0005 (12.50)	Active: H K N U Z	Physical
Corn (ZC)	CBOT	5,000 bushels	1/4 cent (12.50)	H K N U Z	Physical
Soybeans (ZS)	CBOT	5,000 bushels	1/4 cent (12.50)	F H K N Q U X	Physical
Euro FX (6E)	CME	125,000 euros	0.00005 (6.25)	H M U Z	Physical
Micro EUR/USD (M6E)	CME	12,500 euros	0.0001 (1.25)	H M U Z	Physical
Japanese yen (6J)	CME	12,500,000 yen	0.0000005 (6.25)	H M U Z	Physical
Bitcoin (BTC)	CME	5 bitcoin	5 dollars (25.00)	Every month	Cash, to a reference rate
Micro bitcoin (MBT)	CME	0.1 bitcoin	5 dollars (0.50)	Every month	Cash, to a reference rate
VIX futures (VX)	Cboe Futures Exchange	1,000 × VIX	0.05 (50.00)	Every month	Cash, to a special opening quotation

Two patterns in the tables matter for practice. First, the micro contracts are a tenth the size of their parents for equity indices, currencies, gold and crude oil, and a fiftieth for bitcoin. A micro is not always settled the same way as its parent, though. The micro crude contract is cash settled against the physically delivered parent, and it stops trading a business day earlier (CME, Micro WTI fact card). Micro gold, by contrast, is physically delivered like its parent. Second, “active months” matter in metals. Gold lists every month, but liquidity sits in the six months marked active, and a position in an inactive month is hard to exit.

2.4 Last trading day, notice, and delivery

Physically delivered contracts have two dates that bound the life of a long position.

The **first notice day** (FND) is the first day on which a short may notify the clearing house that it intends to deliver. The clearing house then assigns the delivery to a long, usually the one with the oldest open position. For grains and metals, the FND falls around the last business day of the month before the contract month. For Treasury futures, notices can begin just before the delivery month starts, and the short chooses when within the month to deliver. The **last trading day** (LTD) is the day trading stops. Crude oil has no notice period before the LTD. It delivers after the LTD to whoever is still long.

[Practice] Retail brokers do not let customers take delivery. They require customers to close long positions in physically delivered contracts some days before the FND or LTD, and they liquidate positions that remain open. The practical rule follows: **know the FND and LTD of every physically delivered contract held, and roll several business days before whichever comes first.** The cash-settled micro crude contract exists largely so that small traders need not worry about delivery.

Cash-settled contracts have their own expiry hazard. The E-mini S&P 500 settles to a **special opening quotation**, computed from the opening trade of each constituent stock on expiry morning. Stocks open at different times and prices. The settlement value can therefore differ materially from both the previous evening’s close and the index’s first published print. A position left to expire is a bet on that number.

2.5 Settlement prices, limits, and hours

Settlement prices. Exchanges compute the daily settlement price from trades and quotes in a short window near the close. Each contract has its own window. For thinly traded deferred months, the exchange derives the settlement price from spreads to the front month. Margin, profit and loss, and the “close” in most historical data all use this number.

Price limits. Some contracts cannot trade beyond a band around the previous settlement price. During the stock market’s hours, US equity index futures carry tiered limits of 7%, 13% and 20%, tied to the stock market’s own circuit breakers. Outside those hours they have a static 7% band with a 3.5% dynamic circuit breaker. This arrangement replaced a 5% overnight limit in October 2020 (CME, price limits FAQ). Grain contracts have daily limits that the exchange resets twice a year. **[Fact]** A market at its limit can be **locked**: no one will trade at the limit price, so a trader who needs to exit cannot. A stop order therefore does not guarantee an exit at any price.

Hours. CME's electronic markets trade nearly 23 hours a day from Sunday evening to Friday afternoon, with a one-hour break each day. The overnight session is real, and a large share of index moves happen in it. Outside US daytime hours, though, liquidity is thinner and spreads are wider. **[Practice]** Rolls and large orders belong in the main session.

2.6 Ticks as a cost unit

The useful measure of trading cost is basis points of notional. Consider a round trip that crosses a one-tick spread and pays a commission k per contract on each side. Its cost is the tick value plus two commissions, divided by the notional:

$$\text{cost (bp)} \;=\; 10^4 \times \frac{mh + 2k}{mF}.$$

Assume commissions of about 2.25 dollars a side for the E-mini and 0.60 for the micro, as illustrative 2026 retail figures that include exchange fees. At an index level of 6,500, a round trip in one E-mini costs $12.50 + 4.50 = 17.00$ dollars on 325,000 of exposure, or 0.52 basis points. A round trip in one micro costs $1.25 + 1.20 = 2.45$ dollars on 32,500, or 0.75 basis points. **[Computed]** The micro is about 45% more expensive per dollar of exposure, because commissions do not scale down by a factor of 10.

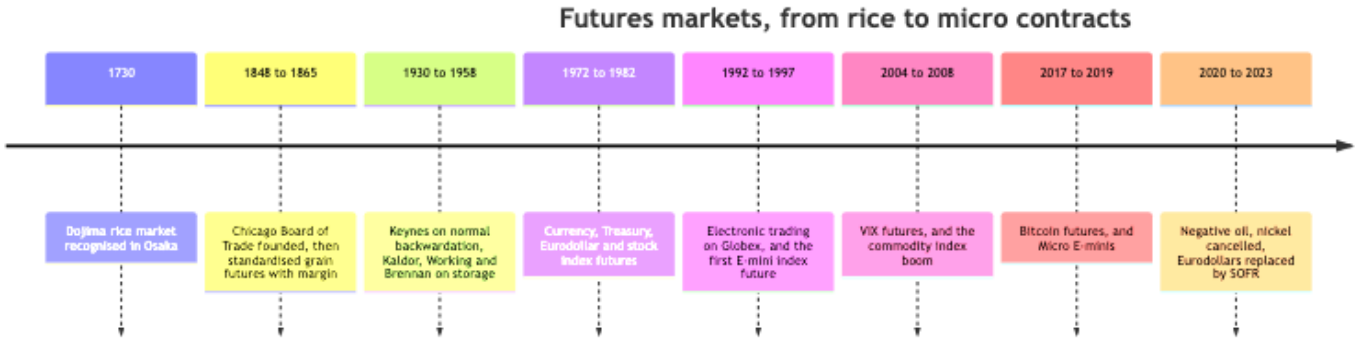
Both costs are small. A round trip in either contract costs about what a fund charging 0.10% a year costs its holders in three weeks (E-mini) to four weeks (micro). The micro's value is not cheapness. It is the ability to hold the intended position size (§10).

§2 Key takeaways

1. Compute mF, one contract's exposure, for every contract before trading it. It is usually far larger than the margin.
2. The month in a symbol is the delivery month. Energy contracts stop trading in the month before it, and grain and metal contracts reach first notice at the end of that earlier month.
3. Know the first notice day and last trading day of every physically delivered contract held, and roll several business days before the earlier one. Otherwise the broker will close the position.
4. Cash-settled index futures settle to an opening auction on expiry morning, which is a separate bet. Roll before it.
5. The exchange's settlement price, not the last trade, sets daily profit and loss. Use settlement prices consistently in research.
6. Price limits can lock a market. A stop is not a guaranteed exit.
7. Micro contracts cost more per dollar of exposure than their parents. Their value is granularity, not cost.

3. How futures markets evolved

Futures began as a solution to a storage problem. They became the cheapest way to hold almost any financial exposure. The history matters for practice in one respect. Almost every rule in §2 was written in response to a specific failure, and economists worked out the theory in §5 while trying to explain grain prices. The timeline marks the main steps, and the four eras below describe them.



Era I — storage and the grain trade (1730–1960s). *Thesis: a futures price is the price of storing something.*

- *Contribution.* Osaka officially recognised organised forward trading in rice in 1730. The Chicago Board of Trade, founded in 1848, standardised grain contracts and imposed margin in 1865. A separate clearing corporation followed in 1925. Keynes (1930) argued that producers hedge by selling forward, so the speculators who take the other side must be paid for doing so. Futures prices should therefore sit below expected spot prices, a pattern he called **normal backwardation**. Kaldor (1939), Working (1949) and Brennan (1958) answered with the **theory of storage**. In that theory, the gap between futures and spot is the price of storage. When inventories are scarce, holding the physical commodity yields a **convenience** that can make the gap negative.
- *What changed.* Economists stopped treating the futures price as a bet on the future. They started treating it as a relationship between today's price and the cost of carrying the commodity forward.
- *Limitations.* Neither theory was decisively tested. The data and the econometrics came decades later.
- *Lasting influence.* Both theories survive, and §5.4 sets out the argument between them.

Era II — financial futures (1972–1990). *Thesis: anything with a cost of carry can have a futures contract.*

- *Contribution.* Currency futures opened in Chicago in 1972, after the end of fixed exchange rates. Treasury bill, Treasury bond and mortgage futures followed. Eurodollar futures arrived in 1981 as the first contract to settle in cash rather than by delivery. Cash settlement made index futures possible, and Value Line and S&P 500 index futures began in 1982. Black (1976) priced options on futures, and Cox, Ingersoll & Ross (1981) showed exactly when futures and forward prices differ.
- *What changed.* The cost-of-carry relation of §5.1 stopped being only an economist's model. Index arbitrageurs and bond dealers now enforced it as an arbitrage relation. Futures became the cheapest way for institutions to adjust equity and bond exposure.
- *Limitations.* The October 1987 crash showed that the arbitrage could break. Index futures traded far below the value of the stocks, because the stock market's trading systems could not keep up.
- *Lasting influence.* Cash settlement, index futures, and the treatment of a futures price as "spot plus financing" all date from this era.

Era III — electronic markets and smaller contracts (1992–2019). *Thesis: futures become a retail instrument.*

- *Contribution.* CME's Globex electronic platform launched in 1992. The E-mini S&P 500, a fifth of the original contract's size and electronic from the start, followed in 1997. Electronic trading overtook the trading pits in the 2000s, and CME closed most of its futures pits in 2015. VIX futures listed in 2004. Commodity index funds hold rolling long positions in a basket of commodity futures, and they grew rapidly from about 2003. Bitcoin futures listed in December 2017. Micro E-mini equity index futures, a tenth of the E-mini, launched on 6 May 2019 and traded more than 310,000 contracts on their first day (CME, 2019).
- *What changed.* An individual with a few hundred thousand dollars could now hold a diversified futures portfolio sized in reasonable steps. A smaller account could hold a handful of markets in fine steps. Both could trade nearly around the clock, at institutional spreads.
- *Limitations.* Retail access brought retail margin products, such as intraday margins of a few hundred dollars per E-mini. These make it easy to take extreme leverage by accident (§9.2).
- *Lasting influence.* This is the market an individual trades today.

Era IV — the plumbing under stress (2008–present). *Thesis: the rare events are in the mechanics, not the prices.*

- *Contribution.* The 2008 commodity spike started a long argument over whether the rolling purchases of index funds had distorted futures prices (§4.4). CME raised silver margins five times in nine days in 2011, as prices collapsed. The broker MF Global failed in October 2011 with about 1.6 billion dollars of customer money missing (§14.2). On 20 April 2020 the May 2020 WTI crude contract settled at −37.63 dollars a barrel, because storage at the delivery point was nearly full (CFTC staff, 2020). The London Metal Exchange suspended nickel trading and cancelled trades in March 2022. SOFR futures replaced Eurodollar futures in 2023. In 2023–2024 CME moved its equity and energy margin to SPAN 2, a model based on historical value at risk.
- *What changed.* Practitioners learned that three common beliefs are assumptions, not facts: "the price can't go below zero", "the exchange will always let me trade", and "margin is stable".
- *Limitations.* Exchanges handled each event after the fact, by changing rules.
- *Lasting influence.* Most of §14.

§3 Key takeaways

1. The theory of futures prices is a theory of storage. The gap between futures and spot is the cost of carrying the asset, net of what holding it yields.
2. Cash settlement, introduced in 1981, made index, volatility and crypto futures possible.

3. Since 2019, micro contracts have brought diversified, reasonably sized futures portfolios within reach of accounts in the low hundreds of thousands of dollars.
4. The large failures of the last 15 years were mechanical: negative prices, margin spirals, cancelled trades and broker failures. Unusual price moves alone did not cause them.

4. Foundational references

4.1 The theory of the futures price

- **Keynes, J. M. (1930).** *A Treatise on Money*, vol. 2. Macmillan. — The origin of normal backwardation: speculators are paid to absorb hedgers' short positions. There is no authoritative free copy, but most libraries hold it.
- **Kaldor, N. (1939).** "Speculation and Economic Stability." *Review of Economic Studies* 7(1), 1–27. [paywalled] — Introduces the convenience yield of holding stocks of a commodity.
- **Working, H. (1949).** "The Theory of Price of Storage." *American Economic Review* 39(6), 1254–1262. — Treats the spread between futures months as the market price of storage. It is short and still the clearest statement.
- **Brennan, M. J. (1958).** "The Supply of Storage." *American Economic Review* 48(1), 50–72. — Formalises the supply of storage and the convenience yield that rises as inventories fall.
- **Samuelson, P. A. (1965).** "Proof That Properly Anticipated Prices Fluctuate Randomly." *Industrial Management Review* 6(2), 41–49. [paywalled] — Also the source of the "Samuelson effect", in which futures prices become more volatile as expiry approaches.
- **Black, F. (1976).** "The Pricing of Commodity Contracts." *Journal of Financial Economics* 3(1–2), 167–179. [paywalled] — Treats futures as contracts that cost nothing to enter, and prices options on them.
- **Cox, J. C., Ingersoll, J. E. & Ross, S. A. (1981).** "The Relation between Forward Prices and Futures Prices." *Journal of Financial Economics* 9(4), 321–346. [paywalled] — Shows when daily settlement makes a futures price differ from a forward price (§1.4).
- **Gibson, R. & Schwartz, E. S. (1990).** "Stochastic Convenience Yield and the Pricing of Oil Contingent Claims." *Journal of Finance* 45(3), 959–976. [paywalled] — Models the convenience yield as a second, mean-reverting state variable.
- **Schwartz, E. S. (1997).** "The Stochastic Behavior of Commodity Prices: Implications for Valuation and Hedging." *Journal of Finance* 52(3), 923–973. [paywalled] — The standard one-, two- and three-factor models of the commodity term structure.
- **Litzenberger, R. H. & Rabinowitz, N. (1995).** "Backwardation in Oil Futures Markets: Theory and Empirical Evidence." *Journal of Finance* 50(5), 1517–1545. [paywalled] — Explains why oil is usually backwardated. Reserves are options, and producers defer extraction when volatility is high.
- **Pindyck, R. S. (2001).** "The Dynamics of Commodity Spot and Futures Markets: A Primer." *Energy Journal* 22(3), 1–29. — The best short introduction to how inventories, spot prices, futures prices and volatility move together.

4.2 Futures risk premia and carry

- **Fama, E. F. & French, K. R. (1987).** "Commodity Futures Prices: Some Evidence on Forecast Power, Premiums, and the Theory of Storage." *Journal of Business* 60(1), 55–73. [paywalled] — Splits the basis into a forecast of the spot change and a premium. The mix differs by commodity.
- **Bessembinder, H. (1992).** "Systematic Risk, Hedging Pressure, and Risk Premiums in Futures Markets." *Review of Financial Studies* 5(4), 637–667. [paywalled] — Returns in some futures markets vary with hedgers' net positions.
- **de Roon, F. A., Nijman, T. E. & Veld, C. (2000).** "Hedging Pressure Effects in Futures Markets." *Journal of Finance* 55(3), 1437–1456. [paywalled] — Hedging pressure, in its own market and in related ones, predicts futures returns.
- **Gorton, G. & Rouwenhorst, K. G. (2006).** "Facts and Fantasies about Commodity Futures." *Financial Analysts Journal* 62(2), 47–68. — A fully collateralised commodity futures index earned an equity-like premium in 1959–2004, with negative correlation to stocks and bonds.
- **Erb, C. B. & Harvey, C. R. (2006).** "The Strategic and Tactical Value of Commodity Futures." *Financial Analysts Journal* 62(2), 69–97. — Finds that the average individual commodity future earned about nothing, and that the term structure explains which ones earned. It is the critique to read next to Gorton and Rouwenhorst.
- **Gorton, G. B., Hayashi, F. & Rouwenhorst, K. G. (2013).** "The Fundamentals of Commodity Futures Returns." *Review of Finance* 17(1), 35–105. — Shows that inventories drive the basis and the premium, the link the theory of storage predicted.
- **Hong, H. & Yogo, M. (2012).** "What Does Futures Market Interest Tell Us about the Macroeconomy and Asset Prices?" *Journal of Financial Economics* 105(3), 473–490. — Open interest predicts returns where prices do not.
- **Szymanowska, M., de Roon, F., Nijman, T. & van den Goorbergh, R. (2014).** "An Anatomy of Commodity Futures Risk Premia." *Journal of Finance* 69(1), 453–482. [paywalled] — Separates spot premia from term premia, and finds that sorting on the basis earns both.

- **Bhardwaj, G., Gorton, G. & Rouwenhorst, K. G. (2015).** "Facts and Fantasies about Commodity Futures Ten Years Later." NBER Working Paper 21243. — Finds that the 2006 conclusions largely held out of sample.
- **Koijen, R. S. J., Moskowitz, T. J., Pedersen, L. H. & Vrugt, E. B. (2018).** "Carry." *Journal of Financial Economics* 127(2), 197–225. [DOI] — Defines carry uniformly across asset classes as the return if prices do not change, and shows that it predicts returns in each (§11). The authors' firm runs carry strategies.
- **Brunnermeier, M. K., Nagel, S. & Pedersen, L. H. (2008).** "Carry Trades and Currency Crashes." *NBER Macroeconomics Annual* 23, 313–347. — Shows that currency carry has negative skewness. High-yield currencies crash together when funding liquidity dries up (§11.3).
- **Levine, A., Ooi, Y. H., Richardson, M. & Sasseville, C. (2018).** "Commodities for the Long Run." *Financial Analysts Journal* 74(2), 55–68. — Reconstructs commodity futures index returns since 1877. They are positive on average, with premia from both carry and spot. The authors include AQR employees.
- **Kang, W., Rouwenhorst, K. G. & Tang, K. (2020).** "A Tale of Two Premiums: The Role of Hedgers and Speculators in Commodity Futures Markets." *Journal of Finance* 75(1), 377–417. [paywalled] — Finds that hedgers pay an insurance premium over long horizons but earn a liquidity premium over short ones.
- **Rouwenhorst, K. G. & Tang, K. (2012).** "Commodity Investing." *Annual Review of Financial Economics* 4, 447–467. — A survey. It finds weak support for normal backwardation and stronger support for storage.

4.3 Financial futures and the basis

- **Du, W., Tepper, A. & Verdelhan, A. (2018).** "Deviations from Covered Interest Rate Parity." *Journal of Finance* 73(3), 915–957. [DOI] — Shows that since 2008, currency forwards and futures have persistently deviated from the interest-rate differential (§5.2).
- **Hazelkorn, T. M., Moskowitz, T. J. & Vasudevan, K. (2023).** "Beyond Basis Basics: Liquidity Demand and Deviations from the Law of One Price." *Journal of Finance* 78(1), 301–345. [DOI] — Shows that the equity index futures basis measures demand for leveraged exposure and predicts returns.
- **Simon, D. P. & Campasano, J. (2014).** "The VIX Futures Basis: Evidence and Trading Strategies." *Journal of Derivatives* 21(3), 54–69. [paywalled] — Finds that the VIX futures basis predicts futures returns, not changes in the VIX.
- **Eraker, B. & Wu, Y. (2017).** "Explaining the Negative Returns to Volatility Claims: An Equilibrium Approach." *Journal of Financial Economics* 125(1), 72–98. The working paper was titled "Explaining the Negative Returns to VIX Futures and ETNs". — Finds that a constant one-month VIX futures position lost about 30% a year in 2006–2013.
- **Schmeling, M., Schrimpf, A. & Todorov, K. (2023).** "Crypto Carry." BIS Working Paper / SSRN. — Finds that bitcoin futures and perpetual swaps trade at large, variable premia to spot, driven by leveraged demand.

4.4 Rolling, index flows, and the financialisation debate

- **Mou, Y. (2011).** "Limits to Arbitrage and Commodity Index Investment: Front-Running the Goldman Roll." Working paper, Columbia. — Finds that the predictable roll of the largest commodity index moved prices. Strategies that traded ahead of it earned large risk-adjusted returns in 2000–2010 (§7.4).
- **Bessembinder, H., Carrion, A., Tuttle, L. & Venkataraman, K. (2016).** "Liquidity, Resiliency and Market Quality around Predictable Trades: Theory and Evidence." *Journal of Financial Economics* 121(1), 142–166. — Studies the US Oil Fund's monthly rolls in 2008–2009. The rolls were predictable, but resilient markets absorbed them cheaply.
- **Tang, K. & Xiong, W. (2012).** "Index Investment and the Financialization of Commodities." *Financial Analysts Journal* 68(6), 54–74. [paywalled] — Finds that commodities in the major indices became more correlated with oil and with each other.
- **Irwin, S. H. & Sanders, D. R. (2011).** "Index Funds, Financialization, and Commodity Futures Markets." *Applied Economic Perspectives and Policy* 33(1), 1–31. [paywalled] — Finds no direct link between index fund positions and futures price levels. This is the opposing view.

4.5 Exchange and regulatory documents

- **CME Group.** Contract specification pages, one per product, and the equity index price limits FAQ. The authoritative source for every number in §2.
- **CME Group.** "SPAN 2 framework rollout." — How exchange margin is computed (§9.2).
- **US Code of Federal Regulations, 17 CFR 39.13.** "Risk management." — The rule that requires clearing houses to set futures margin at no less than 99% confidence over a one-day liquidation period.
- **CFTC staff (2020).** "Interim Staff Report on Trading in NYMEX WTI Crude Oil Futures Contract Leading up to, on, and around April 20, 2020." — The official account of the negative oil price (§5.6, §14.2).

4.6 Books

- **Hull, J. C. (2021).** "Options, Futures, and Other Derivatives." 11th ed., Pearson. — The standard textbook for contract mechanics, cost of carry and hedging.
- **Geman, H. (2005).** *Commodities and Commodity Derivatives: Modeling and Pricing for Agriculturals, Metals and Energy.* Wiley. — Market-by-market commodity term structures.
- **Burghardt, G. & Belton, T. (2005).** *The Treasury Bond Basis*, 3rd ed. McGraw-Hill. — The reference on Treasury futures delivery, the cheapest-to-deliver bond and the basis.
- **Schwager, J. D. & Etzkorn, M. (2017).** *A Complete Guide to the Futures Market*, 2nd ed. Wiley. — Includes the classic practitioner treatment of continuous-contract construction.
- **Carver, R. (2015).** *Systematic Trading.* Harriman House. — Volatility targeting, position rounding and buffering for futures portfolios.
- **Carver, R. (2019).** *Leveraged Trading.* Harriman House. — Written for individuals using futures and other leveraged products with small accounts.
- **Carver, R. (2023).** *Advanced Futures Trading Strategies.* Harriman House. — Covers 30 strategies on more than 100 futures, with explicit treatment of rolling, carry measurement, minimum capital and contract selection. For an individual trading futures systematically, it is the most practically useful single book.

4.7 Critiques

Three arguments in this literature are live. For each, the reader should know both sides.

- **Is there a commodity futures premium at all?** Gorton and Rouwenhorst find that an index of commodity futures earned an equity-like premium. Erb and Harvey find that the average individual commodity future earned about nothing. In their reading, the index's return came from rebalancing and from the term structure. Bhardwaj, Gorton and Rouwenhorst's out-of-sample update and Levine et al.'s long history both support a positive but noisy premium. §6.4 gives this chapter's assessment.
- **Normal backwardation or storage?** Keynes's hedging-pressure story has weak direct support (Rouwenhorst and Tang). The storage story has stronger support (Gorton, Hayashi and Rouwenhorst). Kang, Rouwenhorst and Tang reconcile the two by horizon.
- **Did index investors move prices?** Tang and Xiong find the investors' footprint in correlations, and Mou finds it in prices around the roll. Irwin and Sanders find no link to price levels.

4.8 If you only read six things

1. **Pindyck (2001)**, for the economics of spot, storage and futures prices.
2. **Koijen, Moskowitz, Pedersen and Vrugt (2018)**, for carry as one idea across every asset class.
3. **Gorton and Rouwenhorst (2006)** with **Erb and Harvey (2006)**, read as a pair, for what a futures position in commodities earns.
4. **Hull (2021)**, the futures chapters, for the mechanics in standard form.
5. **Carver (2023)**, for the practical implementation an individual needs.
6. **The CFTC staff report on 20 April 2020**, for what happens when the mechanics in §2.4 meet a full storage tank.

5. The futures price: cost of carry and the term structure

5.1 The cash-and-carry argument

Start with an asset that can be bought today and held. Let its spot price be S. Holding it until expiry costs U dollars of storage and pays I dollars of income. Both amounts are measured as their value at expiry, interest included. Money can be borrowed or lent at a simple rate r for the τ years to expiry.

There are two ways to own the asset at expiry. One is to buy it now and carry it. The other is to agree today, through a futures contract, to buy it at expiry for F. Each of the two trades below buys one of these routes and sells the other.

Cash and carry. Borrow S, buy the asset, store it, collect its income, and sell a futures contract at F. At expiry, deliver the asset into the futures contract and repay the loan. The profit is known today:

$$F - S(1 + r\tau) - U + I.$$

If this profit were positive, anyone who could borrow, buy and store would make the trade. Their selling of futures and buying of spot would close the gap.

Reverse cash and carry. Sell the asset now, either short or from inventory already held. Invest the proceeds, save the storage cost, forgo the income, and buy a futures contract to get the asset back at expiry. The profit is the negative of the one above. If it were positive, holders of the asset would sell spot and buy futures.

When both trades are available, neither can be profitable. The futures price must then equal the cost of buying the asset with borrowed money and carrying it to expiry, net of income:

$$F = S(1 + r\tau) + U - I.$$

The rest of this chapter uses the same argument with costs and income written as continuously compounded yields. Suppose the income is a yield q reinvested in the asset, and storage is a yield u paid by selling a little of the asset. One unit held for τ years then becomes $e^{(q-u)\tau}$ units. To own exactly one unit at expiry, buy $e^{-(q-u)\tau}$ units today. Pay for them with $Se^{-(q-u)\tau}$ borrowed at the continuously compounded rate r, and repay $Se^{-(q-u)\tau}e^{r\tau}$ at expiry. The futures price must equal that repayment:

$$F_t(T) = S_t\, e^{(r+u-q)\tau}.$$

From here on r, like u and q, is a continuously compounded rate, as in the notation table. For small rates the two forms agree, because $e^{r\tau} \approx 1 + r\tau$.

The futures price is therefore the spot price plus the cost of carrying the asset to expiry. Nothing in the argument mentions expectations of the future spot price. The cost of replicating a futures contract today sets its price, just as the cost of replicating an option sets the option's price (Dealer Hedging and Gamma Exposure develops the option case).

Where the argument breaks: physical commodities. The reverse trade requires someone to sell the physical commodity today and buy it back later through futures. For a financial asset, someone always can. Index arbitrageurs short baskets of stocks, banks lend and borrow currencies, and dealers lend and borrow bonds through repo. For crude oil, copper or wheat, the holders of inventory hold it because they need it. A refinery that sells its crude to capture a futures discount has to shut down.

The reverse trade therefore does not enforce the lower bound. The futures price can fall below $Se^{(r+u)\tau}$ by any amount. (That bound is the price above with $q = 0$, because most commodities pay no income. Gold's lease rate, §5.2, is the exception.) Only the upper bound survives, because the cash-and-carry trade still enforces it:

$$F_t(T) \;\le\; S_t\, e^{(r+u)\tau} \qquad \text{(storable commodity, with spare storage).}$$

The **convenience yield** y is defined to close the gap between the futures price and that bound:

$$F_t(T) = S_t\, e^{(r+u-q-y)\tau}, \qquad c = r + u - q - y.$$

The convenience yield is the implicit return to having the physical commodity on hand. It is the value of not running out (Kaldor, 1939; Brennan, 1958). Nobody observes y directly. It is backed out of the futures price, so the equation is true by construction. The economic content lies in what moves y: it rises as inventories fall.

5.2 Asset by asset

Every futures market uses the same equation, with different ingredients switched on. The table lists, for each kind of underlying, which ingredients apply, the usual shape of the curve, and the trade that enforces the price.

Underlying	Financing r	Income q	Storage u	Convenience y	Usual shape	What enforces the price
Equity index	Repo or funding rate	Dividend yield	—	—	Gentle contango when rates exceed dividends	Index arbitrage
Currency	Domestic rate	Foreign rate	—	—	Contango when the domestic rate is higher	Covered interest parity
Government bond	Repo rate	Coupon yield	—	Delivery options	Below the cash bond when coupon exceeds repo	The cash–futures basis trade
Gold	Funding rate	Lease rate	Small	≈ 0	Close to full-carry contango	Lending and borrowing of bullion
Crude oil, metals, grains	Funding rate	—	Yes	Varies with inventories	Either	Storage owners, one way only
Natural gas, power	—	—	Limited or none	—	Seasonal	Expectations, within storage limits
VIX	—	—	—	—	Contango in calm markets, backwardation in stress	None: expectation plus a premium paid by longs
Bitcoin	Funding rate	—	—	—	Contango of variable size	Cash-and-carry funds, imperfectly

Equity indices. For an index, $c = r - q$. An index future therefore trades above the index when financing rates exceed the dividend yield, and below it when dividends exceed rates. **[Fact]** Euro Stoxx 50 futures traded below the index for years while euro rates were negative and dividend yields were around 3%. The curve was in backwardation, and a long position earned positive carry. Nobody needed to expect European stocks to rise for that to happen.

The useful quantity is the **implied financing rate**. It is the value of r that makes $F = Se^{(r-q)\tau}$ hold at the observed prices:

$$r^{\text{impl}} = q + \frac{\ln(F/S)}{\tau}.$$

The implied financing rate is the rate at which the futures market finances a long index position. Compare it with a risk-free rate such as SOFR. The spread measures how much more, or less, it costs to hold the index through futures than to buy the stocks with money borrowed at that rate. **[Fact]** The spread is neither zero nor constant. It widens at quarter-ends, when bank balance sheets are expensive, and it moves with demand for leveraged equity exposure. Hazelkorn, Moskowitz & Vasudevan (2023) study index futures across global markets. They show that a rich basis coincides with heavy long demand from investors and predicts lower subsequent returns. **[Contested]** How much of the spread is a balance-sheet cost, and how much is a demand premium, is not settled. For an individual the practical point is that the financing spread embedded in an index future is the instrument's true cost. It can be larger than any commission.

Currencies. Take a currency quoted as dollars per unit of foreign currency. **Covered interest parity** says that $F = Se^{(r_{\$} - r_{\text{foreign}})\tau}$, where $r_{\$}$ is the dollar interest rate and r_{foreign} the foreign one. The foreign rate plays the part of the income yield q. The forward rate is the spot rate adjusted by the interest differential, because borrowing in one currency and lending in the other replicates the future. A long euro futures position therefore earns carry of $r_{\text{EUR}} - r_{\$}$. This is exactly the currency carry trade (see the carry section of Systematic Trading Strategies). **[Fact]** Since 2008 the parity has held only approximately. For the major currencies, the deviation, called the **cross-currency basis**, has often been tens of basis points

a year. It reflects banks' balance-sheet costs (Du, Tepper & Verdelhan, 2018). **[Practice]** For an individual the basis is a small adjustment to the carry, not a reason to change a position.

Government bonds. A Treasury future is a contract to deliver any bond from a basket, with each bond scaled by a conversion factor. The short chooses which bond to deliver, so the future tracks the bond that is **cheapest to deliver** (CTD). Its carry is the CTD's coupon income minus its repo financing cost. When the yield curve slopes up, the coupon yield exceeds the repo rate and carry is positive. The futures price, multiplied by the CTD's conversion factor, then sits below the cash bond's price, as the table shows. The short's choice of which bond to deliver, and when, has value to the short. These delivery options pull the futures price a little lower still, and they are what the table lists in the convenience column.

[Fact] With yields below the contracts' 6% notional coupon, the CTD tends to be the shortest-duration issue in the basket. A 10-year note future therefore behaves like a note with a maturity of roughly 7 years, not 10. For duration matching, use the contract's DV01, either as reported by the exchange or as computed from the CTD, and ignore the "10-year" label. Bonds and Bond Markets treats the deliverable basket, the CTD and the cash–futures basis trade in detail.

Storable commodities. The theory of storage applies here. When inventories are ample, owning the commodity confers little convenience. The convenience yield y is then near zero, and the curve sits near **full carry**: each later month sits above the previous one by financing plus storage. When inventories are scarce, the convenience yield rises and the curve inverts. **[Fact]** Across commodities, inventories and the slope of the futures curve move together, as the theory predicts (Gorton, Hayashi & Rouwenhorst, 2013).

Crude oil has spent long periods in backwardation. Litzenberger & Rabinowitz (1995) explain why with an option argument. A producer with oil in the ground can always pump it later. It pumps now only if the discounted futures price is below spot. The higher the volatility, the more valuable waiting becomes. Gold is held overwhelmingly as an investment and is plentiful in vaults, so it sits close to full carry. The **lease rate**, at which bullion can be lent, plays the part of a small income yield.

Non-storables and the VIX. Natural gas can be stored only within the limits of injection and withdrawal capacity, so its curve is seasonal. Winter months trade at a premium to summer months every year. Storage links the months only within those capacity limits. Electricity cannot be stored at all, and each delivery month is a separate market.

The VIX cannot be bought at all, because it is an index computed from option prices. With no way to buy and hold the underlying, no cash-and-carry argument exists. A VIX future is therefore priced as an expectation of where the VIX will be at expiry, plus a risk premium. The curve's usual upward slope is mostly that premium (§5.4).

Bitcoin. Cash and carry is possible here: buy bitcoin or a spot fund, and sell futures. But the trade ties up capital and carries custody and exchange risk. The basis therefore floats with demand for leveraged long exposure. **[Fact]** The basis has ranged from near zero to well above the risk-free rate. The annualised basis between perpetual swaps and spot has been large and volatile (Schmeling, Schrimpf & Todorov, 2023).

5.3 Contango and backwardation, precisely

"Contango" and "backwardation" each have two meanings, and the literature uses both without warning. The table sets them side by side.

Term	Curve meaning (observable)	Keynes's meaning (a statement about expected returns)
Contango	Later contracts trade above earlier ones, or the future above spot: $c > 0$	The futures price is above the expected future spot price: shorts are paid
Backwardation	Later contracts trade below earlier ones, or the future below spot: $c < 0$	The futures price is below the expected future spot price: longs are paid ("normal backwardation")

The two meanings are independent. Gold is almost always in contango in the curve sense. It could at the same time be in normal backwardation in Keynes's sense, if holders of gold futures earn a premium. The curve meaning is a fact about today's prices. The Keynes meaning is a claim about expected returns, which can only be estimated. In this chapter, "contango" and "backwardation" without a qualifier always refer to the curve.

Full carry as an inventory gauge. For a storable commodity with spare storage, the curve cannot be steeper than full carry, which is financing plus storage. A steeper curve would let traders buy spot, store it and sell futures at a profit. **[Practice]** Traders therefore read a commodity spread as a percentage of full carry. A calendar spread at 90% of full carry says that inventories are comfortable. A spread in backwardation says that they are tight, and the depth of the backwardation says

how tight. The upper bound has one exception, which §5.6 describes. When storage itself runs out, its price can explode, and the curve can become steeper than any normal full-carry calculation. Traders call this state **super-contango**.

5.4 Is the futures price a forecast?

To ask whether the futures price forecasts the spot price, write it as an expectation adjusted by a premium:

$$F_t(T) = E_t[S_T] - \pi_t(T),$$

where $E_t[S_T]$ is the spot price expected at expiry, given what is known at t. The **risk premium** $\pi_t(T)$, in price units, is the profit a long position expects to make per unit of the underlying for bearing the risk. If $\pi = 0$, the futures price is an unbiased forecast.

For a storable asset, the cost-of-carry relation pins $F_t(T)$ to today's spot price and today's carry. It says nothing about π. Any premium therefore shows up in the expected spot price instead. If longs are paid, $E_t[S_T]$ sits above the futures price by π. Subtracting S_t from both sides splits the basis into two parts:

$$F_t(T) - S_t = (E_t[S_T] - S_t) - \pi_t(T).$$

The basis is the expected change in spot minus the premium. Two theories have competed for 90 years to explain π.

- **Normal backwardation (hedging pressure).** Producers hedge by selling futures, and they hedge more than consumers, who hedge by buying. Speculators must therefore be paid to take the net long side, so $\pi > 0$. **[Contested]** Hedgers' net positions do predict returns in some markets (Bessembinder, 1992; de Roon, Nijman & Veld, 2000). Direct support for the theory as the general explanation of the commodity premium is weak, however (Rouwenhorst & Tang, 2012).
- **Storage.** The basis reflects inventories. Premia are higher when inventories are low and prices are more volatile. **[Fact]** The link between basis and inventories is robust (Gorton, Hayashi & Rouwenhorst, 2013).

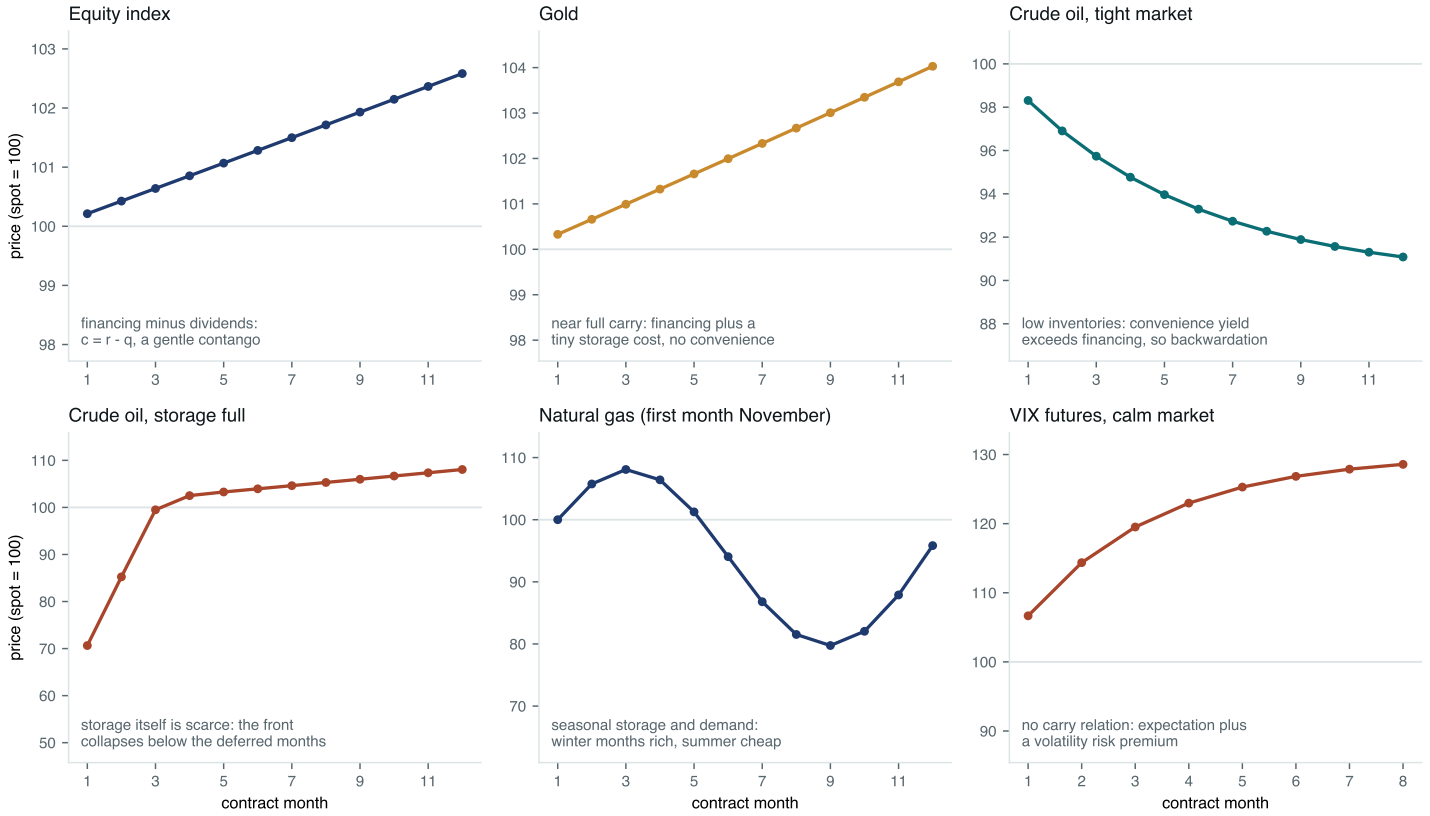
Fama & French (1987) estimated the two parts of that split: the part of the basis that forecasts the spot change, and the part that is a premium. They found both, in different proportions for different commodities. Kang, Rouwenhorst & Tang (2020) reconcile the two theories by horizon. Over long horizons, hedgers pay an insurance premium. Over days and weeks, they earn a liquidity premium from speculators who trade with momentum.

The VIX is the clean case of a premium. **[Fact]** Simon & Campasano (2014) find that the VIX futures basis predicts the futures' own returns, not changes in the VIX. When the curve is in contango, the futures price falls toward the VIX. The VIX does not rise to meet it. **[Fact]** A constant one-month position in VIX futures lost about 30% a year from 2006 to 2013 (Eraker & Wu, 2017).

[Practice] Recommendation: for financial and storable underlyings, read the slope of the curve as carry, not as the market's view of direction. The slope is a cost or a credit known in advance. For non-storables and the VIX, read the slope as mostly a premium that the long pays. In neither case is contango, by itself, a reason to expect the spot price to rise.

5.5 Reading a term structure

The figure shows the six curve shapes an individual meets most often. Five are drawn from the cost-of-carry relation, using the ingredient that dominates in each market. The VIX has no such relation, so its curve is drawn as an expectation plus a premium paid by longs.



Stylised shapes, not market quotes. In every storable market the slope is the net cost of carry: financing and storage push the curve up, income and convenience pull it down. The VIX has no such link, and its slope is mostly a premium paid to whoever is short.

Five features are worth reading off any curve.

The front spread. Take the log price of the first contract minus that of the second, and divide by the years between their expiries. The result is the annualised roll-down of the second contract into the first. It is also approximately the front contract's own carry (§6.2 derives this). It is the most useful single number on the curve (§11.1).

The shape further out. Carry is not uniform along the curve. Crude oil in backwardation is often steepest at the front. VIX futures in contango are always steepest at the front, and nearly flat beyond the fourth month. The position a trader holds on the curve therefore decides how much carry the trader earns (§7.3).

Seasonality. In natural gas, grains, heating oil and gasoline, contracts for different calendar months are effectively different commodities. December corn is the new crop and July corn the old one, and January gas is winter gas. The spread between adjacent months therefore mixes carry with seasonality. Carry must be measured between contracts a year apart, or against the usual seasonal shape (§11.2).

Volatility by maturity. [Fact] In most commodities, front contracts move more than deferred ones. This pattern is the **Samuelson effect**. Shocks to supply and demand hit today's scarcity hardest, and the market expects them to fade by later delivery dates. In equity, bond and currency futures, the curve is a financing relationship, so all contracts move almost one for one with spot.

Calendar spreads are the curve's own market. A **calendar spread** is a long position in one month against a short position in another. It has almost no exposure to the level of prices. It is a pure position on the slope, and in storable commodities, therefore, on inventories. Exchanges list spreads as instruments in their own right, and margin them far below the outright contracts (§9.6).

5.6 When storage runs out: super-contango and negative prices

The upper bound of §5.1 assumes spare storage. In April 2020 that assumption failed at the delivery point of the WTI crude contract. **[Fact]** The pandemic had cut oil demand, and storage at Cushing, Oklahoma, was nearly full. The May contract, physically delivered at Cushing, was about to expire. Anyone long the May contract at expiry had to take barrels at Cushing and put them somewhere. On 20 April 2020, the day before the last trading day, the May contract fell from about 18 dollars to settle at −37.63. Holders paid nearly 38 dollars a barrel to be relieved of the obligation (CFTC staff, 2020). The June contract, a month further from delivery, stayed well above zero.

Three lessons generalise.

- **A futures price can be negative when the contract obliges the holder to take something costly to hold.** The cost-of-carry relation still held, in its dollar form. Read $F = S(1 + r\tau) + U - I$ from §5.1 with June as the futures and May as the spot. A trader could take May barrels, store them for a month and deliver them into June. May could therefore sit below June by as much as the dollar cost U of that month's storage. For the few days left on the contract, that cost had become enormous, because storage was exactly what had run out. The exponential form cannot show this. The bound $Se^{(r+u)\tau}$ is positive for any storage yield u. Only a dollar cost, which does not shrink with the price, can push a price below zero.
- **The front of a curve can detach from the rest.** A curve is a set of separate contracts linked by arbitrage, and the arbitrage needs a physical link. Without that link, the front month trades on its own delivery problem.
- **Research tools fail at zero.** Log returns, ratio-adjusted series and percentage-based position sizing are undefined at a negative price. A data pipeline must either handle negative prices with additive methods or exclude the episode deliberately and document the exclusion (§8, §13).

§5 Key takeaways

1. For anything that can be bought and held, replication sets the futures price: spot plus financing plus storage, minus income. Expectations do not enter.
2. For physical commodities, only the upper bound is enforced. The convenience yield, backed out of prices, measures how scarce inventories are.
3. The same equation covers every market. The income term is the dividend yield for indices, the foreign interest rate for currencies, the coupon for bonds, and the lease rate for gold.
4. "Contango" and "backwardation" describe the curve. Keynes's normal backwardation is a different, unobservable claim about premia. Keep the two apart.
5. For financial and storable underlyings, read the slope as carry, not as a forecast. For the VIX and other non-storables, the slope is mostly a premium paid by longs.
6. The annualised front spread is the most useful number on a curve. Measure seasonal markets against the same month a year apart.
7. Index futures embed a financing spread over the risk-free rate, and the spread varies with demand for leverage. It is the instrument's true holding cost.
8. When storage runs out, the front of a physically delivered curve can go anywhere, including below zero. Pipelines built on log returns break there.

6. Where a futures return comes from

6.1 Excess return and total return

A futures position costs nothing to enter, so its profit and loss is not a return on any investment. To express it as a return, divide by the notional. The **futures excess return** over a period is $\Delta F/F$: the change in the price of the contract held, as a fraction of its price at the start of the period. It is called an excess return because it is earned on top of whatever the collateral earns. A position backed by cash equal to its full notional, with the cash invested in Treasury bills, earns the bill rate plus the excess return:

$$\text{total return} = r^{\text{bill}}\,\Delta + \frac{\Delta F}{F}.$$

Commodity index providers publish both versions. An **excess-return index** compounds the futures returns alone. A **total-return index** adds the interest earned on Treasury bills that fully collateralise the position. **[Fact]** The S&P GSCI and Bloomberg commodity indices both come in these two forms.

Two practical consequences follow.

- **A futures return series is already an excess return.** Its Sharpe ratio is its mean divided by its standard deviation. Subtracting the risk-free rate a second time understates the Sharpe ratio. This error is common in backtests that mix futures with funds.
- **Comparing futures with an asset requires adding the cash back.** An index future and an index fund differ by the cash rate before any question of cost or tracking arises.

6.2 The decomposition: spot, carry, and the curve

The return decomposition starts from a definition. The futures contract's **annualised log basis** is the log gap between the futures price and spot, per year to expiry:

$$c_t(T) = \frac{\ln F_t(T) - \ln S_t}{\tau},$$

so that $\ln F_t(T) = \ln S_t + c_t(T)\,\tau$ holds by construction. Under the cost-of-carry model, this c is the net cost of carry of §5.1. In general it is simply the slope of the curve from spot to this contract.

Now hold the same contract for a short period Δ. During that period its time to expiry shrinks from τ to $\tau - \Delta$. Write the definition at both dates and subtract. Then add and subtract $c_t(T)\,(\tau - \Delta)$:

$$\begin{aligned}
\Delta \ln F &= \big[\ln S_{t+\Delta} + c_{t+\Delta}(T)\,(\tau - \Delta)\big] - \big[\ln S_t + c_t(T)\,\tau\big] \\
&= \Delta \ln S + \big[c_t(T)\,(\tau - \Delta) - c_t(T)\,\tau\big] + \big[c_{t+\Delta}(T) - c_t(T)\big](\tau - \Delta) \\
&= \underbrace{\Delta \ln S}_{\text{spot return}} \;\; \underbrace{-\; c_t\,\Delta}_{\text{carry}} \;+\; \underbrace{(c_{t+\Delta} - c_t)(\tau - \Delta)}_{\text{curve shock}},
\end{aligned}$$

where the last line drops the (T), since every c refers to the same contract. This equation is an identity, not a model. It says that a futures position earns three things:

1. **The spot return.** This is what the underlying did.
2. **Carry**, $\kappa_t \Delta$ with $\kappa_t = -c_t$. Carry is known at the start of the period. In contango ($c > 0$) it is negative, and the contract drifts down toward spot. In backwardation it is positive.
3. **A curve shock.** If the slope changes, the contract's price moves by the change in slope times the time left to expiry. This term is an exposure to the slope of the curve, similar to duration. It is negligible for a front-month contract with a few weeks left, and material for a contract a year out.

The identity is the same in every market. What differs is the content of c.

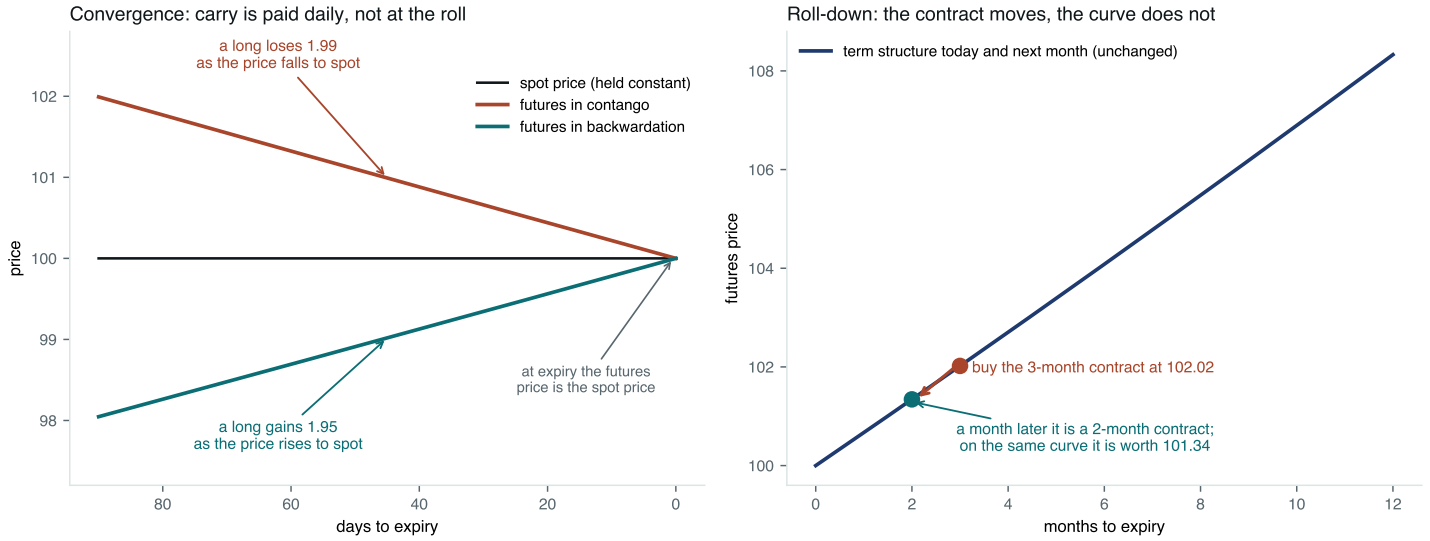
- **Financial futures at fair value.** Take $c = r - q$, with both rates constant so that there is no curve shock, and let the collateral earn r. The total return is then $\Delta \ln S - (r - q)\,\Delta + r\,\Delta = \Delta \ln S + q\,\Delta$, which is the asset's price return plus its income. **Futures plus cash is the asset.** The Micro E-mini quarter in §1.6 showed this to within a dollar. The equality holds up to the financing spread of §5.2.
- **Storable commodities.** Take $c = r + u - y$, again constant, and let the collateral earn r. The total return is then $\Delta \ln S + (y - u)\,\Delta$. This is the return from buying the commodity and storing it, plus the convenience of having it. A financial investor receives that return without needing a warehouse. When convenience exceeds storage cost, futures beat spot. When inventories are ample, futures lag spot.
- **The VIX.** The identity holds with S as the VIX index, but nothing ties c to a cost. The carry term is the roll-down of the futures toward the index. As §5.4 showed, the index has not, on average, risen to meet the futures.

Measuring carry without a spot price. Many commodities have no clean spot price, so the first contract serves as the practical anchor. Hold the second contract for the $T_2 - T_1$ years until it has the first contract's current time to expiry. Suppose the curve is unchanged in time-to-expiry terms, so that a contract with a given time to expiry trades at the same price then as now. The second contract will then trade where the first contract trades today. The roll-down return is therefore $\ln F^{(1)} - \ln F^{(2)}$ over $T_2 - T_1$ years. Annualised, it is

$$\kappa_t \;\approx\; \frac{\ln F_t^{(1)} - \ln F_t^{(2)}}{T_2 - T_1}.$$

Practitioners use this carry measure for commodities, and Koijen et al. (2018) use it in simple rather than log form. It equals the spot-based carry exactly when the log curve is a straight line in maturity. When the curve bends, the two measures differ, and that difference is why the choice of contract matters (§7.3).

The figure illustrates both mechanisms. The left panel shows convergence to a fixed spot price, and the right panel shows roll-down along an unchanged curve.



Both panels show the same 8%-a-year carry. With spot unchanged, a futures position earns or pays it continuously as the contract ages toward expiry. Rolling into the next contract resets where on the curve you sit; it does not itself make or lose money.

6.3 Nothing happens at the roll

The most common explanation of contango's cost is that a long position must "sell the cheap expiring contract and buy the expensive next one" at every roll. This explanation gets the total right over a full cycle. It gets wrong where the money goes, and that error leads to bad decisions.

Take crude oil with spot pinned and the curve unchanged. The expiring contract trades at 70.00 and the next month at 70.80. **[Computed]**

- **Over the month**, the contract held was priced at 70.80 when it was bought. As it becomes the front month and approaches expiry, it converges to 70.00. The position loses 80 cents a barrel, or 800 dollars a contract, a little each day through daily settlement. This loss is carry: $\ln(70.00/70.80) = -1.14\%$ a month, or about −13.6% a year.
- **At the roll**, the trader sells the expiring contract at its market price, 70.00, and buys the next at its market price, 70.80. One zero-value position is exchanged for another. The roll itself makes no profit or loss beyond its trading costs.

Over a full cycle the two accounts agree: the position loses 80 cents a month. The folk version, however, places the loss at the roll, and three mistakes follow from that:

1. **It suggests that rolling less often, or at a better moment, avoids the cost.** It does not. A contract held longer converges for longer. Holding a deferred contract changes the cost only if the curve is not a straight line (§7.3).
2. **It double counts in backtests.** The daily profit and loss of the contract held already contains the convergence. Subtracting the gap at each roll as a "roll cost" charges it a second time.
3. **It hides the cost when spot moves.** In a rising market the contract may gain every day, and nothing looks like a loss. The carry is still there: the position gains 13.6% a year less than spot.

One second-order effect does occur at the roll. Rolling a fixed number of contracts in contango increases the notional exposure. Each new contract has a higher price than the one it replaces, and therefore a larger notional. To hold exposure constant, the trader buys $F^{(1)}/F^{(2)} = 0.989$ new contracts per old one in this example. For a handful of contracts the difference rounds away. For a large position it is a small rebalancing trade.

6.4 What futures positions have earned

The decomposition says what to expect. The historical record says what happened.

Equity index futures earn the index's total return minus the financing rate embedded in the basis. That rate is the risk-free rate plus a spread that varies with demand for leverage (§5.2). For an individual, index futures are close substitutes for an index fund plus cash.

Currency futures earn the spot change plus the interest differential. The carry trade goes long high-rate currencies against low-rate ones. It has earned a positive premium, with severe crash risk (see the carry section of Systematic Trading Strategies).

Commodity futures are where the disagreement lies.

- Gorton & Rouwenhorst (2006) studied an equally weighted, fully collateralised index of commodity futures from 1959 to 2004. It earned about the same excess return and Sharpe ratio as US equities. It was negatively correlated with stocks and bonds and positively correlated with inflation. **[Fact]** Bhardwaj, Gorton & Rouwenhorst (2015) found that these conclusions largely held in the following decade.
- Erb & Harvey (2006) found that the average *individual* commodity future earned roughly nothing. An index earned a premium mainly from rebalancing across volatile, weakly correlated contracts. The term structure explained the cross-section of returns: commodities in backwardation earned more than those in contango.
- Levine, Ooi, Richardson & Sasseville (2018) used data back to 1877. They found positive average returns to commodity futures indices, with premia associated with both carry and spot returns. The premia varied with inflation and the business cycle.
- **[Fact]** Szymanowska et al. (2014) split the premia earned by sorting commodities on the basis. Spot premia, which compensate for spot-price risk, were 5–14% a year. Term premia, which compensate for changes in the basis, were 1–3%.

[Contested] The view taken here is that a diversified long-only commodity futures index has earned a positive but noisy excess return over long histories. It has also had long stretches of losses while carry was negative, including much of the 2010s. Most of the difference in returns between individual commodities is carry. A long-only position that ignores the curve is a bet on the spot price. It pays whatever carry the market charges, and in contango that charge is steep.

VIX futures are the extreme case. **[Fact]** A constant one-month long position lost about 30% a year from 2006 to 2013 (Eraker & Wu, 2017). The loss is the roll-down of a curve that is usually in contango, toward an index that does not, on average, rise to meet it.

6.5 Carry as a breakeven

Take expectations of the decomposition:

$$E_t[\Delta \ln F] = E_t[\Delta \ln S] + \kappa_t \, \Delta + E_t[\text{curve shock}].$$

Of the three terms, only carry is known at the start. Carry is therefore a **breakeven**: the spot move needed to offset it. In the crude oil example of §6.3, a long position needs spot to rise 13.6% over a year just to match a position in cash. **[Practice] Recommendation: before taking any directional view in futures, compute the carry over the intended holding period, and check whether the expected move is larger.** In steep curves it often is not. VIX futures in calm markets, and natural gas in the shoulder seasons, commonly embed several percent a month of negative carry at the front. A long position there needs a large, fast move to profit.

The same number works in favour of a short position. A short VIX futures position in contango earns the roll-down every day the VIX stays put. That is why the trade is popular. It is also why the trade loses catastrophically on the day the VIX doubles (§14.2). Carry is the known part of the return, not the safe part.

§6 Key takeaways

1. A futures return is an excess return. Add the cash rate to compare it with an asset, and do not subtract the risk-free rate again when computing its Sharpe ratio.
2. Every futures return splits exactly into three terms: the spot return, carry (known in advance), and a curve-shock term that matters only for longer-dated contracts.
3. Financial futures plus cash replicate the asset, including its income. Commodity futures plus cash replicate storing the commodity, plus its convenience yield.
4. Without a spot price, carry is the annualised log spread between the first two contracts.
5. The roll swaps one contract for another with no value of its own. The cost of contango accrues daily as the contract converges. Rolling less often does not avoid it, and charging the roll gap in a backtest counts it twice.
6. Commodity futures indices have earned a positive but noisy premium over long histories. Most of the difference between individual commodities is carry.
7. Carry is a breakeven: the spot move that a directional view must exceed. In steep curves it is large.

7. Rolling

7.1 What a roll is, and how to execute it

A **roll** closes a position in an expiring contract and opens the same position in a later one. For a long position, a roll sells the near contract and buys the far one. It is therefore a trade in the **calendar spread** between the two contracts. A roll changes no exposure to the underlying. It changes only which contract carries the exposure.

A roll can be executed in two ways.

- **Two outright orders.** Sell one contract, then buy the other. This method pays the bid–ask spread twice. It also carries **legging risk**: the market can move between the two fills, and on a fast day it will.
- **One spread order.** Exchanges list calendar spreads as instruments in their own right, quoted as the price difference between the two legs. An order to trade the spread fills both legs at once. In liquid markets the spread market's bid–ask is usually one tick wide. The exchange's matching engine also links the spread book to the outright books, so liquidity in either can fill the order. **[Practice] Recommendation: roll with spread orders.**

Sign conventions often cause errors. On CME, buying a calendar spread means buying the nearer month and selling the deferred one. A long position therefore rolls by *selling* the spread, and a short position rolls by buying it. Confirm the platform's convention on a small order before the first real roll.

7.2 When liquidity moves

Liquidity migrates from the expiring contract to the next one over a window that is specific to each market. Rolling inside that window gets the tightest prices. Rolling far outside it means trading an illiquid contract on one side. The table gives the window for each major market.

Market	Expiry or notice	Where the roll happens
Equity index futures (ES, NQ, RTY, YM and micros)	Third Friday of March, June, September, December	The week or so before expiry; CME publishes a roll date after which the next contract is treated as the lead month
Treasury futures (ZT, ZF, ZN, ZB)	Delivery during the contract month; notices can start at the end of the month before	Late in February, May, August and November, before notices can begin
WTI crude oil (CL, MCL)	Trading ends about the 20th of the month before delivery; the micro a day earlier	One to two weeks before the last trading day, every month
Natural gas (NG)	Trading ends three business days before the delivery month	The days before the last trading day, every month
Gold, silver, copper	First notice on the last business day of the month before delivery	Several days before first notice, between active months only
Grains (ZC, ZS, ZW)	First notice on the last business day of the month before delivery	Before first notice
Currency futures (6E, 6J, M6E)	Trading ends two business days before the third Wednesday of the contract month	The week before expiry
VIX futures	A Wednesday, 30 days before the following month's S&P 500 option expiry	The days before expiry, every month
CME bitcoin futures	The last Friday of the contract month	The days before expiry, every month

[Practice] For physically delivered contracts, the binding date is the broker's, not the exchange's. It is the date by which the broker requires long positions to be closed. Find that date for every contract held, put it in a calendar, and roll before it.

7.3 Which contract to hold

In equity, bond and currency futures the question barely arises. The deferred contracts are illiquid, and every contract moves with spot, so the trader holds the front contract and rolls it. In commodities and volatility, the choice of contract is a real decision, because carry, liquidity and volatility all vary along the curve.

Carry varies along the curve. If the log curve were a straight line, every contract would roll down at the same annual rate, and the choice would not matter. Real curves are not straight. In crude oil backwardation, the front of the curve is usually steepest, so the front contract earns the most carry. In VIX contango, the front is always steepest, so the front contract pays the most carry. Constant-maturity products (§8.3) that hold mid-curve VIX futures have therefore lost less to roll-down than one-month products. In exchange, they track the VIX less closely.

Volatility varies along the curve. Through the Samuelson effect (§5.5), deferred commodity contracts move less than the front contract. A position sized by volatility (§10.2) will therefore hold more deferred contracts. This is neither good nor bad, but it changes the contract count and the margin.

Liquidity falls off along the curve. In most markets, spreads widen and depth thins beyond the first few contracts. **[Practice]** Holding the second or third contract is usually fine in the large energy and metal markets, and costly elsewhere.

Seasonality. In seasonal markets, contracts for different months are different commodities. A roll from a summer contract to a winter contract changes what the trader owns. **[Practice]** Some systematic traders hold only one calendar month a year in seasonal markets, for example rolling from December to December. This keeps the exposure and the carry measurement consistent, and it cuts the number of rolls.

Optimised rolling. At each roll, some commodity indices choose the contract with the most favourable implied roll yield among those available. The implied roll yield is the carry of §6.2, measured for each candidate contract. **[Fact]** The DBIQ Optimum Yield index family, tracked by the Invesco DB Commodity Index Tracking Fund, works this way. The rule is a carry strategy applied to contract selection, and it inherits carry's risks.

7.4 Predictable rolls, and who trades against them

When everyone can predict a roll, someone will trade ahead of it. **[Fact]** The S&P GSCI rolls its positions from the fifth to the ninth business day of each month, a fifth of the position each day. Mou (2011) found that this "Goldman roll" moved calendar spreads measurably. Strategies that traded ahead of it earned Sharpe ratios as high as 4.39 from 2000 to early 2010. Index investors gave up about 3.6% a year to the price impact. Bessembinder, Carrion, Tuttle & Venkataraman (2016) studied the US Oil Fund's monthly rolls in 2008–2009 and reached a gentler conclusion. Strategic traders did trade ahead of the fund. But the market was resilient enough that their activity improved liquidity and the fund's proceeds.

[Contested] How large the effect is today is not settled. The research has been published, and index funds now spread their rolls over more days. **[Practice]** The robust lesson is cheap to apply. Do not roll inside the published windows of the large index funds, and do not roll at the same minute as everyone else. Rolling a few days before the crowd, or spreading a large roll over several days, costs nothing.

7.5 Roll rules for backtests and live trading

A roll rule decides, for each date, which contract the position is in. The rule must be computable from information available at the time.

- **Fixed schedule.** For each contract, set a roll date a fixed number of business days before the earlier of its last trading day and first notice day. **[Practice]** Five business days is a common default for cash-settled contracts. For physically delivered contracts held at a retail broker, roll before the broker's own deadline. The rule is reproducible and needs only the contract calendar. It is the rule §13 implements.
- **Volume or open-interest crossover.** Roll when the next contract's volume or open interest (the number of contracts still open) overtakes the current contract's. This rule tracks real liquidity. But the exchange publishes open interest the next morning, so a rule that uses it on the same day looks ahead. Lag it by a day (Trend-Following in Financial Markets makes the same point).
- **Calendars.** Use the exchange's holiday calendar, not a national one. Exchanges close on some days when banks are open, and the reverse.
- **Prices.** In a backtest, roll at both contracts' settlement prices on the roll date, and charge the cost of the spread order: about half a tick of the calendar spread plus two commissions. Do not also charge the price gap between the contracts (§6.3).
- **Never hold into delivery in the data.** In its last days, after most traders have left, a physically delivered contract trades thinly and sometimes at extreme prices. A backtest that holds through those days reports prices that no individual could have traded at.

In live trading the same calendar drives alerts. **[Practice]** Generate the roll schedule for every contract when the position is opened. Set an alert a few days before each roll date. Before rolling, check that the new contract has the expected open interest.

§7 Key takeaways

1. A roll is a calendar spread trade. Execute it as one spread order, which is cheaper and has no legging risk.
2. On CME, a long position rolls by selling the calendar spread. Confirm the sign convention on the trading platform before the first roll.
3. Each market has its own roll window. For physically delivered contracts, the broker's deadline binds before the exchange's.
4. In commodities and volatility, the contract held on the curve decides how much carry the position earns or pays, because curves are not straight lines.
5. Traders have profited from trading against predictable index rolls. Roll before the crowd's window, or spread the roll out.
6. A backtest needs a roll rule that is computable at the time: a fixed schedule from the contract calendar, or lagged volume and open interest.
7. In a backtest, charge the cost of the spread trade at the roll, never the price gap between the contracts.

8. Continuous price series

8.1 The problem

A futures "price history" does not exist. What exists is a sequence of contracts. Each one trades for a few months or years and then expires. At any moment, their prices differ from each other by the calendar spreads between them. Signals, volatility estimates, backtests and charts, however, all need one series. Building that series means choosing which property of the underlying contracts to keep, because no single series keeps them all. Market price levels, percentage returns, and dollar profit and loss per contract cannot all be correct at once. Trend-Following in Financial Markets summarises the choice, and this section works through it.

8.2 A worked example

The example uses two contracts over eight days, with one roll at the close of day 4. On day 4 the expiring contract settles at 102.00 and the new one at 104.00. The market is in contango, and the gap is 2.00. The table gives the settlement prices.

Day	Old (expiring)	New (next)	Held over the day
1	100.00		Old
2	101.00		Old
3	100.50		Old
4	102.00	104.00	Old (roll at the close)
5		103.00	New
6		105.00	New
7		106.00	New
8		105.00	New

The next table shows four constructions of the same history, with the true return of the position held. **[Computed]**

Day	Unadjusted	Back-adjusted	Ratio-adjusted	Return index (start 100)	True return
1	100.00	102.00	101.96	100.00	
2	101.00	103.00	102.98	101.00	+1.00%
3	100.50	102.50	102.47	100.50	−0.50%
4	102.00	104.00	104.00	102.00	+1.49%
5	103.00	103.00	103.00	101.02	−0.96%
6	105.00	105.00	105.00	102.98	+1.94%
7	106.00	106.00	106.00	103.96	+0.95%
8	105.00	105.00	105.00	102.98	−0.94%

Read across day 5. A trader holding the new contract lost 1.00 point, or 0.96%, from 104 to 103. The unadjusted series shows a gain of 0.98%, from 102 to 103. It has booked the 2.00-point gap between the contracts as if it were a price move.

The back-adjusted series added the gap to the earlier history, so its day-5 change is the true −1.00 point. The ratio-adjusted series scaled the earlier history by 104/102, so its day-5 change is the true −0.96%. The return index compounds the true returns. It equals the ratio-adjusted series divided by 1.0196.

Over the eight days, one contract with a multiplier of one made $1.0-0.5+1.5-1.0+2.0+1.0-1.0=3.00$ points. The back-adjusted series rises by exactly 3.00, from 102 to 105. The unadjusted series rises by 5.00, overstating the gain by the gap. The ratio-adjusted series rises by 3.04, because it scales up the point changes before the roll by 104/102. It measures a position of 1.0196 old contracts that rolls into one new contract, which keeps the notional unchanged at the roll (§6.3). The old contract rose 2.00 before the roll and the new one rose 1.00 after it, so this position made $1.0196\times 2.00+1.00=3.04$. The ratio-adjusted series therefore tracks a constant-notional position, not a constant contract count.

8.3 The constructions

Each construction below is described with the same four fields: what it does, what it keeps, what it breaks, and what it is for. A comparison table follows the six descriptions.

Unadjusted (spliced nearest contract)

- *Construction.* On each date, the series takes the price of the contract held. At a roll, the series jumps from one contract to the next.
- *Keeps.* Every level is a price that actually traded on that date.
- *Breaks.* Returns across rolls. Each roll books the price gap between the contracts as a return: a fake gain in contango and a fake loss in backwardation. Over years, these fake returns add up to roughly the cumulative carry with the sign reversed. They therefore cancel the carry, and the series tracks spot rather than the position.
- *Use for.* Charts that must show the prices traders saw, and nothing quantitative. Never use it for returns or volatility.

Back-adjusted (difference-adjusted, "panama")

- *Construction.* At each roll, shift all earlier history by the price gap between the new and old contracts, so the series has no jump. The latest segment is the current contract's actual price.
- *Keeps.* Price differences. The change over any period equals the dollar profit and loss, per unit of multiplier, of holding one contract throughout and rolling it.
- *Breaks.* Levels before the last roll, and therefore percentage returns. The level on a past date is not a price anyone saw, and it can be negative (§8.4). Every roll restates the entire history.
- *Use for.* Rules and accounting in price points with a constant contract count. It is the classic choice for systems specified in points. Do not use it for anything in percentages.

Ratio-adjusted (proportionally adjusted)

- *Construction.* At each roll, multiply all earlier history by the ratio of the new contract's price to the old contract's price.
- *Keeps.* Percentage returns, exactly. Levels stay positive. Relative distances, such as "5% below the 52-week high", are preserved.
- *Breaks.* Dollar differences, and levels before the last roll. Every roll restates the history. The construction is undefined if any price is zero or negative.
- *Use for.* Signals and volatility, when a rule needs a price level rather than returns.

Return-stitched index

- *Construction.* Compute each day's return within the contract held, and compound the returns from an arbitrary starting value. The result is the futures excess-return index.
- *Keeps.* Percentage returns, exactly. The history never changes when a new roll happens, because nothing is rescaled.
- *Breaks.* Levels have no market meaning at all.
- *Use for.* Research: signals, volatility, correlations, and the performance of a constant-notional position. If a level is needed, rescale the index to today's price. It then becomes the ratio-adjusted series.

Constant-maturity (interpolated)

- *Construction.* Each day, take a weighted average of two adjacent contracts. The weights are chosen so the average always has the same time to expiry, for example 30 days.
- *Keeps.* A fixed position on the curve, without jumps at rolls. This suits markets where carry varies steeply along the curve, such as the VIX.
- *Breaks.* The series is not a tradable price on any day. Replicating it requires trading both contracts daily, and its "returns" ignore the cost of that daily rebalancing.
- *Use for.* Describing the curve and measuring carry at a fixed maturity. Some exchange-traded products hold exactly this mixture.

Forward-adjusted

- *Construction.* Like back-adjustment, but shift each new segment to join the old history. The series is anchored at the first date instead of today.
- *Keeps.* The history never changes.
- *Breaks.* Today's level is not today's price. Any order or stop placed from it is therefore wrong.
- *Use for.* Rarely anything. It is mentioned because some vendors offer it.

The table compares the six constructions on the properties that separate them.

Construction	Levels are traded prices	Percentage returns correct	Dollar P&L per contract correct	History restated at each roll	Can go negative
Unadjusted	Yes	No	No	No	Only if prices do
Back-adjusted	Latest segment only	No	Yes	Yes	Yes
Ratio-adjusted	Latest segment only	Yes	No	Yes	No
Return-stitched	No	Yes	No	No	No
Constant-maturity	No	Approximately	No	No	Only if prices do
Forward-adjusted	First segment only	No	Yes	No	Yes

8.4 What an adjusted level means

The two adjusted series have exact interpretations. They are worth knowing, because they explain every strange thing an adjusted chart does.

Let ΔF_j be the day-j price change of the contract held over day j. Let $R_j = \Delta F_j / F_{j-1}$ be its return, where F_{j-1} is the same contract's price the day before. (On day 5 of §8.2, F_{j-1} is the new contract's 104, not the old contract's 102.) Write F_{today} for the current contract's latest price. Write P_t^{back} and P_t^{ratio} for the back-adjusted and ratio-adjusted levels on date t. Each level equals today's price, adjusted for what the position has earned since date t. For any past date t,

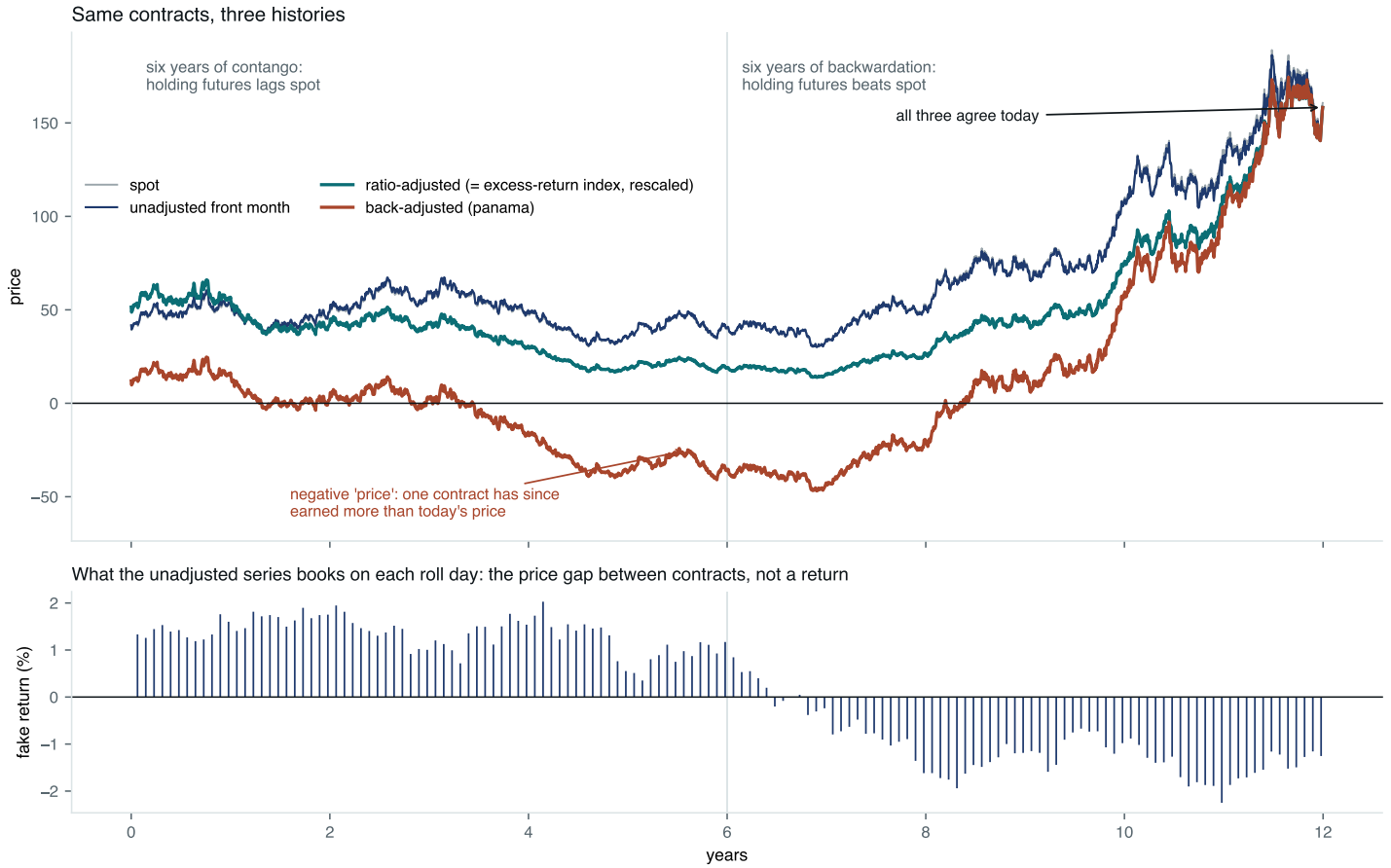
$$P_t^{\text{back}} = F_{\text{today}} - \sum_{t<j\le \text{today}} \Delta F_j, \qquad P_t^{\text{ratio}} = \frac{F_{\text{today}}}{\prod_{t<j\le \text{today}} (1+R_j)}.$$

In §8.2, for day 1, these give $105 - 3.00 = 102.00$ and $105/1.0298 = 101.96$.

The back-adjusted level on a past date is today's price minus the dollars one contract has earned since then (per unit of multiplier). If a contract has earned more than today's price since some date, the back-adjusted series is negative on that date. This can happen in any market that rallied strongly in backwardation and stayed high. The contract earned the spot rally *and* the positive carry, and the sum can exceed the current price.

The ratio-adjusted level on a past date is today's price divided by the growth of a constant-notional position since then. It is never negative. In a market with persistent negative carry, however, such as VIX futures, the early history is enormous, because the position has shrunk so much since.

The simulation below shows both effects. A market spends six years in contango with a flat spot price, then six years in backwardation while spot quadruples.



The simulation supports three readings, all **[Computed]**:

- **All three series agree today and disagree about the past.** The back-adjusted series is below zero almost continuously from about year 3.5 to year 8, and intermittently from year 1.3. It reaches about −47 shortly after the regime change. That “price” never existed. It is negative because a contract held from that date earned more than its price today.
- **The unadjusted series hides the carry.** It sits on top of spot throughout, and the two lines are almost indistinguishable. The reason is that it books the price gap between contracts as a return at every roll. That fake gain averages about 1.3% a roll in the contango years, and it turns into a fake loss in the backwardation years. Over the six contango years the fake gains add up to about 97 percentage points, which is roughly the carry the position actually paid.
- **The ratio-adjusted series is the position’s real performance.** It falls during the contango years while spot is flat. It rises faster than spot during the backwardation years.

8.5 Which series for which job

The table matches each common task to the series that serves it.

Job	Use	Why
Trend, momentum and other return-based signals	Return-stitched or ratio-adjusted	Percentage returns are exact
Volatility and correlation estimates	Return-stitched	Same; and no restatement when a new roll arrives
Rules that compare price with its own past levels, such as breakouts	Ratio-adjusted	Relative distances survive; back-adjusted levels distort them, unadjusted ones jump
Dollar risk of one contract, for sizing	Current contract’s price × multiplier × volatility of returns	No adjusted level is a current price except the latest segment
Backtest profit and loss with whole contracts	Simulate the actual contracts and roll trades	Captures contract counts, rounding, and roll costs; back-adjusted differences are the shortcut for a constant count

Job	Use	Why
Carry	Raw prices of two contracts on the same date	Carry is a cross-section of today's curve, not a time series
Charts of the prices traders watched, such as support and resistance	Each contract's actual prices	Only these were seen by the market
Periods with zero or negative prices	Back-adjusted, or contract-level data	Ratios and logarithms are undefined

[Practice] Recommendation: store every contract's raw daily settlement prices, and build a return-stitched series for research from them. Rebuild the ratio-adjusted series on demand when a rule needs levels. Size every position from the current contract's actual price.

8.6 Data pitfalls

- **Vendors differ.** Continuous series from different vendors use different roll rules and adjustments, often without saying which. Two "continuous crude oil" series can show different returns on the same day. Know how the series was built, or build it from contract prices.
- **Adjusted histories change.** A back-adjusted or ratio-adjusted series is restated at every roll. A database that stores adjusted levels and appends new ones therefore mixes adjustments. A signal computed last month will not reproduce from this month's download. Store raw contract prices and rebuild.
- **Settlement versus last trade.** Mixing the two across markets, or across time, mixes clocks (§2.1). Old histories may come from floor sessions with different closing times.
- **The last days of a contract.** Thin trading in an expiring contract produces outliers. A roll rule that leaves the contract before those days removes the outliers from the series.
- **Specification changes.** Exchanges resize and re-tick contracts over time. For example, the multiplier of the original S&P 500 futures contract was cut from 500 to 250 dollars in 1997. A history in dollars must use the units of the contract actually traded.
- **Calendars and closes.** Align markets on exchange calendars, and never forward-fill across a roll. Markets in different time zones close hours apart, which biases same-date correlations downward.
- **Survivorship.** Exchanges delist markets and discontinue contracts. A universe built from markets that trade today is selected for having survived.

§8 Key takeaways

1. No continuous series keeps traded price levels, percentage returns and dollar profit and loss at once. Choose the series by the job.
2. The unadjusted series books the price gap between contracts as a return at every roll, which hides the carry. Never compute returns from it.
3. The back-adjusted level on a past date is today's price minus what one contract has earned since. It is exact for dollar profit and loss, and it can be negative.
4. The ratio-adjusted level is today's price divided by a constant-notional position's growth since. It is exact for percentage returns, and it is the excess-return index rescaled.
5. Use return-stitched or ratio-adjusted series for signals and volatility, and the current contract's actual price for sizing.
6. Adjusted histories are restated at every roll. Store raw contract prices and rebuild; never store adjusted levels.

9. Margin and leverage

9.1 How the account works

A futures account has two numbers that matter. The cash balance changes every evening through daily settlement. The margin requirement depends on the positions held. The difference between them is the **excess**: the cash free to absorb losses or open new positions.

- **Initial margin** is the deposit required per contract to open a position.

- **Maintenance margin** is a lower level, below which the account may not fall. At the exchange it is typically about nine-tenths of the initial level.
- If equity falls below maintenance, the exchange's convention is a **margin call**, which asks the trader to restore equity to the initial level. **[Practice]** Retail brokers mostly skip the call. Their systems liquidate positions automatically, in real time and at market, as soon as equity breaches the requirement. This includes the thin overnight session.

Losses are not limited to the deposit. A position whose price gaps through its liquidation level leaves a negative balance, and the trader owes that balance to the broker.

The most useful single number about a futures position is how far the price can move against it before liquidation. To derive it, write $L = |N|mF/A$ for the position's notional leverage. Write $\mu = M^{\mathsf{M}}/(mF)$ for the maintenance margin as a fraction of one contract's notional. An adverse move of x, as a fraction of the price, loses $x\,|N|mF$ and leaves equity of $A - x\,|N|mF$. Liquidation comes when that equity falls to the maintenance requirement, $|N|M^{\mathsf{M}}$. Set the two equal and divide through by the notional $|N|mF$:

$$x^{\star} = \frac{1}{L} - \mu.$$

The distance to liquidation is one over leverage, minus the maintenance margin rate. Margin rates are a few percent of notional. At low leverage, $1/L$ is large and μ hardly matters. As leverage rises, $1/L$ shrinks toward μ, and the distance shrinks toward zero. The formula holds the maintenance margin fixed in dollars. In a fast market, maintenance margin tends to rise instead (§9.4), which brings liquidation closer.

[Computed] Take the three Micro E-mini contracts of §1.6, about 98,000 dollars of exposure, but now held against 10,000 dollars instead of 100,000. Leverage is 9.8, and maintenance margin is about 7% of notional. A fall of $1/9.8 - 7\% \approx 3.2\%$ therefore triggers liquidation. **[Fact]** In most years the S&P 500 falls 3.2% within a week at some point. Ordinary market noise will therefore eventually liquidate this position.

9.2 How margin is set

Exchange margin. The clearing house sets minimum margins. For futures, US rules require these margins to cover at least 99% of price moves over a liquidation period of at least one day (17 CFR 39.13). For decades CME computed margins with SPAN, a grid of price and volatility scenarios introduced in 1988. It has since moved its equity and energy products to SPAN 2, a historical value-at-risk model with add-ons for stress and liquidity (CME, SPAN 2). Margin is therefore a value-at-risk number of the kind Value at Risk describes. The exchange computes it for its own protection, not the trader's.

How big it is. In mid-2026 CME's initial margin for one E-mini S&P 500 contract was about 25,000 dollars, against a notional of about 325,000 to 350,000. That is 7–8% of notional. For an index with 1% daily volatility, a normal-distribution one-day 99% loss is about 2.3% of notional (2.33 daily standard deviations). The margin is therefore roughly three times a calm day's tail loss. **[Practice]** Exchange margin is deliberately that large, because the models draw on long histories that include stress periods. Two consequences follow. In calm markets margin looks generous, and after a shock it rises quickly.

Broker margin. Brokers add their own requirements on top of the exchange's. They change these requirements without notice, and they often raise them before elections, central-bank meetings and holidays.

Intraday margin. [Fact] Many retail brokers offer **day-trade margin** for positions closed before the session ends. It can be as low as a few hundred dollars per E-mini contract. At 400 dollars against 325,000 dollars of exposure, the deposit is 0.12% of notional. That is leverage of over 800, and a move of about eight index points consumes it. **[Practice]** Treat day-trade margin as a broker's convenience for very short-term traders, not as a guide to how much to hold.

9.3 Leverage is chosen, not given

Margin caps how much exposure a trader can take. It does not say how much the trader should take. The same 100,000-dollar account can hold the S&P 500 at very different leverage, as the table shows. Margin used and losses are percentages of the account's equity. Initial margin is taken as 2,500 dollars a micro (§1.6) and 25,000 an E-mini (§9.2). The fall to liquidation is $1/L - \mu$ with $\mu \approx 7\%$. **[Computed]**

Position (index at 6,500)	Notional	Leverage L	Margin used	Loss on a 5% fall	Loss on a 12% fall	Fall to liquidation
3 Micro E-mini	97,500	0.98	7.5%	4.9%	11.7%	96%
1 E-mini	325,000	3.25	25%	16.3%	39%	23.8%
3 E-mini	975,000	9.75	75%	48.8%	117%	3.3%

The 12% fall is not hypothetical: the S&P 500 fell about 12% on 16 March 2020. The first row is an unlevered index position, futures plus cash as in §1.6. On the worst day it lost what the index lost. The third row loses more than the whole account on that day, and the trader owes the broker the rest. In practice the broker would have liquidated it days earlier, at the bottom of a smaller move.

Margin-to-equity. Futures funds report the ratio of margin required to account equity. **[Practice]** Diversified managed-futures funds commonly run at roughly 10–20%. An individual with margin at 50% or more of equity is either very concentrated or very levered. In a crisis margin rates can double, and that would force such a trader to cut positions at the worst moment.

[Practice] Recommendation: size every position by risk first (§10.2), then check two numbers. First, margin used should stay below about a quarter to a third of equity, so that margins can double without forcing a sale. Second, the distance to liquidation, $1/L - \mu$, should be comfortably larger than the worst historical move over a few days.

9.4 When margin moves against a position

Margins rise when prices fall. Margin models respond to volatility. The requirement therefore rises after a large move, exactly when equity has fallen. **[Fact]** Around silver's late-April 2011 peak, CME raised silver margins five times in nine days. The increases forced leveraged holders to sell into a falling market. In March 2022 the nickel price more than doubled in hours. The London Metal Exchange then suspended nickel trading and cancelled a day's trades. Short holders were left facing margin calls they could not meet.

Liquidation is at market. Automatic liquidation sells at whatever the market bids. That may be in the overnight session, or at a price limit. **[Practice]** The account most likely to be liquidated is the one that sized its positions by the margin available. The defence is the rule above. Leave enough room that the margin required in the worst week is still well inside the account's equity.

9.5 Cash management

The collateral's interest is part of the return. The identity of §6.2, futures plus cash equals the asset, assumes that the cash earns the bill rate. Some brokers pay little or nothing on small balances. Cash left idle at such a broker loses that rate entirely. In the §1.6 example the lost rate is 3.75% a year, more than the index's dividend yield. **[Practice]** Keep unneeded cash in Treasury bills or a money-market fund, or at a broker that pays close to the bill rate. Some brokers accept bills as margin collateral, with a haircut.

Variation margin is paid in cash. Bills held as collateral do not pay a day's loss; only cash does. **[Practice]** Keep a cash buffer at the broker worth several days of large losses, for example three times the position's one-day 99% loss. Sweep the excess to bills periodically rather than daily.

Foreign-currency contracts settle in their own currency. A Euro Stoxx 50 or Bund position on Eurex settles daily in euros. The euro balance that accumulates is a currency position. If the account is measured in dollars, that balance needs converting periodically.

9.6 Spread margins

Clearing houses net offsetting risks. A calendar spread, long one month and short another month of the same contract, is margined at a fraction of the two outright positions. Related contracts receive partial credits against each other. Examples are 5-year and 10-year Treasury futures, or the E-mini S&P 500 and the E-mini Nasdaq-100. These credits make spread positions cheap to margin. They therefore make it easy to lever spreads far beyond their apparent risk.

[Fact] Amaranth Advisors lost more than 6 billion dollars in 2006, most of it in September, on natural gas calendar spreads (US Senate Permanent Subcommittee on Investigations, 2007). The low margin on those positions reflected how rarely the spreads moved, not how far they could move. Size spreads by their own volatility, including how they behave in a squeeze, not by their margin.

§9 Key takeaways

1. Margin is a deposit set by a value-at-risk model for the exchange's protection. It caps leverage. It does not measure the risk a trader should take.
2. The distance to liquidation is one over leverage minus the maintenance margin rate. At leverage near 10, a routine 3% move ends the position.
3. Leverage is notional divided by equity, and the trader chooses it. Futures plus cash at a leverage of one is an unlevered position.
4. Exchange margin is several times a calm day's tail loss. It rises after shocks, exactly when equity has fallen. Leave room for it to double.
5. Retail brokers liquidate automatically, at market, without a call. Losses can exceed the deposit.
6. Day-trade margins of a few hundred dollars per E-mini imply leverage in the hundreds. They are not a sizing guide.
7. Idle cash that earns nothing breaks the futures-plus-cash identity. Keep collateral in bills, plus a cash buffer for variation margin.
8. Spread positions are cheap to margin and easy to over-lever. Size them by their own risk.

10. Micro contracts and sizing a small account

10.1 What micro contracts are

Micro contracts are copies of existing futures at a fraction of the size. For equity indices they are a tenth of the E-mini. For crude oil, gold and the main currency pairs they are a tenth of the standard contract. For bitcoin they are a fiftieth (0.1 bitcoin against 5, §2.3). **[Fact]** CME's Micro E-mini equity index futures launched on 6 May 2019. They traded more than 310,000 contracts on their first day (CME, 2019), and passed a billion contracts, futures and options together, in November 2021 (CME, 2021). Micro contracts now exist in energy, metals, currencies, crypto and Treasury yields.

Three properties matter in practice.

- **Their liquidity is borrowed.** **[Practice]** Market makers keep a micro's price tied to its parent's by arbitrage. Bid–ask spreads, measured in ticks, are therefore similar, but the depth at each price is much smaller. For an individual's order sizes this rarely matters.
- **They are not always the same contract.** The micro crude contract is cash settled, and it stops trading a day before the physically delivered parent. Micro gold, by contrast, is physically delivered. Read each specification (§2.3).
- **They cost more per dollar of exposure.** Commissions and exchange fees do not shrink by a factor of 10. A round trip in a micro therefore costs roughly half as much again as the same exposure in the parent (§2.6). What micros buy is granularity.

10.2 Sizing a position from risk

Size positions by risk, not by margin or by notional. Suppose the account targets an annual volatility $\sigma^\star$ and allocates a share w of that risk budget to one market. The position's annual dollar volatility should then be $A\sigma^\star w$. One contract's annual dollar volatility is its notional times the volatility of its percentage returns, $mF\sigma$. The ideal number of contracts is the ratio of the two:

$$N^\star = \frac{A\,\sigma^\star\,w}{m\,F\,\sigma}.$$

The formula divides the dollar risk the account wants by the dollar risk one contract brings. For a contract priced in a foreign currency, multiply the denominator by the exchange rate, in dollars per unit of that currency, so that both sides are in dollars. Use the volatility of the return-stitched series (§8.5) and the current contract's actual price. Volatility grows with the square root of time. An annual figure is therefore the daily standard deviation times $\sqrt{252}$, where 252 is the number of trading days in a year. A volatility of 1% a day is about 16% a year.

A worked case. A 100,000-dollar account wants 15% annual volatility, all of it in the S&P 500 ($w = 1$). The S&P 500's volatility is 16%. **[Computed]**

- In E-mini contracts, $N^\star = 100{,}000 \times 0.15/(50 \times 6{,}500 \times 0.16) = 0.29$. Rounding gives zero contracts and no position. One contract would carry 3.5 times the intended risk.

- In Micro E-mini contracts, $N^\star = 2.88$. Three contracts carry 15.6% volatility, 4% more than intended.

The E-mini cannot express this account's view at all. The micro expresses it to within a few percent.

10.3 How coarse is each contract?

For sizing, the useful property of a contract is its annual dollar risk, $mF\sigma$. This is the share of the risk budget that one contract consumes. The table gives it at illustrative 2026 prices and typical volatilities, ordered from finest to coarsest. **[Computed]**

Contract	Illustrative price	Multiplier	Notional (dollars)	Assumed volatility	Annual dollar risk per contract
Micro EUR/USD (M6E)	1.15	12,500 euros	14,375	8%	1,150
Micro WTI crude (MCL)	65	100 barrels	6,500	35%	2,275
Micro Russell 2000 (M2K)	2,400	5	12,000	22%	2,640
Micro bitcoin (MBT)	100,000	0.1	10,000	50%	5,000
Micro E-mini S&P 500 (MES)	6,500	5	32,500	16%	5,200
Micro gold (MGC)	3,800	10	38,000	16%	6,080
10-Year Treasury note (ZN)	112	1,000	112,000	6%	6,720
Micro Nasdaq-100 (MNQ)	23,500	2	47,000	21%	9,870
WTI crude (CL)	65	1,000 barrels	65,000	35%	22,750
E-mini S&P 500 (ES)	6,500	50	325,000	16%	52,000
Gold (GC)	3,800	100	380,000	16%	60,800

The prices are illustrative, not quotes, so recompute with current ones. The pattern is robust, though. One micro consumes between about 1,000 and 10,000 dollars a year of risk. Its standard parent consumes 10 times as much, or 50 times as much for bitcoin.

Rounding error. Rounding an ideal position $N^\star$ to a whole number changes it by up to half a contract. The relative error is therefore up to $0.5/N^\star$. Below about two contracts the error can exceed 25%, and below half a contract the position vanishes. **[Practice]** A position needs an ideal size of three or four contracts before it can be adjusted in reasonable steps as volatility and signals change. In risk terms, each market's budget $A\sigma^\star w$ must be at least three or four times one contract's annual dollar risk.

The minimum account. Spreading a risk budget across several markets divides it. With 10 markets and equal shares, each market gets $w = 0.1$. But markets that do not move together partly cancel each other. The portfolio's volatility is therefore less than the sum of the positions' volatilities. Every position can be scaled up by a **diversification multiplier** to bring the total back to the target. Take 10 equally weighted markets with an average correlation of 0.2. The portfolio's volatility is $\sqrt{0.1 + 0.9 \times 0.2} \approx 0.53$ of what it would be if the markets moved in lockstep. The multiplier is therefore $1/0.53 \approx 1.9$, and each market's budget becomes $A\sigma^\star w$ times 1.9.

Systematic Trading Strategies works through the arithmetic. Take 10 markets of roughly Micro E-mini risk and a 20% volatility target. Each market's budget is $A \times 0.20 \times 0.1 \times 1.9 \approx 0.038A$. One contract of each, at 5,200 dollars of risk, therefore needs about 140,000 dollars. The three or four contracts that allow reasonable adjustment need about half a million. The table above shows the way around this limit. A small account can diversify across the fine-grained contracts (currencies, micro crude, small-cap equity) long before it can hold several contracts of the coarse ones (micro gold, micro Nasdaq, Treasury notes).

10.4 Buffering

Suppose an ideal position drifts between 2.4 and 2.6 contracts as volatility changes. Rounded, it will flip between 2 and 3 contracts every few days, paying costs for nothing. **[Practice]** The fix is to trade only when the current position falls outside a band around the ideal. Compute the band's edges as $N^\star \pm b$, rounded to whole contracts. If the current position is outside the band, trade to the nearer edge, not to $N^\star$. With $b = 0.5$ and the ideal drifting between 2.4 and 2.6, the band runs from 2 to 3 throughout. A position of either 2 or 3 is therefore left alone. A buffer b of half a contract or more stops the flipping in small positions. For larger positions, Carver (2015) suggests about a tenth of the average position. §13.3 implements the rule in four lines.

10.5 Micros, standard contracts, or funds?

For most exposures an individual can choose between micro futures, standard futures and exchange-traded funds. The table compares them.

	Micro futures	Standard futures	Exchange-traded funds
Smallest position	One micro: about 6,000 to 50,000 dollars of exposure	About 65,000 to 400,000 dollars	One share
Holding cost	Financing spread in the basis, plus rolls	The same, with cheaper rolls per dollar	Expense ratio; commodity funds also pay the roll
Trading cost per dollar	Highest of the futures	Lowest	Spread and commission, often small
Leverage	Chosen freely; automatic liquidation; losses can exceed the deposit	The same	None without a margin account
Going short	Symmetric with going long	Symmetric	Borrow fees, or inverse funds that decay with volatility
US tax treatment	Section 1256: 60% long-term and 40% short-term gains, marked to market at year-end	The same	Ordinary rules for equity funds; commodity funds vary, some issuing partnership forms
Trading hours	Nearly 23 hours, 5 days	The same	Exchange hours
Operations	Rolls, notice dates, margin monitoring	The same	None

[Practice] Recommendation: choose by account size. The thresholds below also depend on taxes and on the strategy.

- Below roughly 50,000 dollars, use funds for most exposures, and micros only for a few positions sized with care.
- Between about 50,000 and a few hundred thousand dollars, use micros for a diversified handful of markets, and funds for the rest.
- Above that, use standard contracts where they are fine-grained enough, and micros to fill in.

For a US taxpayer running a strategy that trades often, the Section 1256 treatment alone can decide in favour of futures (the tax section of Systematic Trading Strategies has the comparison).

§10 Key takeaways

1. Micro contracts are mostly a tenth of their parents. Their value is granularity. Per dollar of exposure, they cost more to trade.
2. Size by risk: $N^\star = A\sigma^\star w/(mF\sigma)$, using the volatility of the return-stitched series and the current contract's price.
3. Rounding to whole contracts distorts positions below three or four contracts, and erases them below half a contract. A standard E-mini cannot express a 100,000-dollar account's S&P 500 position, but three micros can.
4. Even among micros, the risk that one contract consumes differs by a factor of 10. A small account diversifies first across the fine-grained ones.
5. Buffer: trade only when the position leaves a band around its ideal, and then trade only to the band's edge.

6. Funds beat futures for very small accounts, where whole contracts are too coarse. Futures win on capital efficiency, shorting and hours, and for US taxpayers often on tax.

11. Carry as a trading signal

Carry is the part of a futures return known in advance. It therefore plays two roles. It is a cost or credit to account for in any position. It has also predicted returns across asset classes, which makes it a signal in its own right. The carry section of Systematic Trading Strategies treats carry as a strategy family. This section covers how to measure carry correctly and how to use it.

11.1 Measuring carry, market by market

Koijen et al. (2018) define carry uniformly as the return of a futures position if the spot price does not change. That return is $(S-F)/F$, earned over the contract's remaining life τ, or $(S-F)/(F\tau)$ annualised. In practice each asset class needs its own recipe, as the table shows.

Asset class	Carry, annualised	Inputs	Pitfalls
Equity index	$(S-F)/(F\tau) \approx q-r$	Index level, front future, days to expiry	Dividends are seasonal, so a quarter's carry swings; average over a year, or use dividend futures
Currencies	$r_{\text{foreign}} - r_{\text{domestic}}$, or $(S-F)/(F\tau)$	Spot and future, or the two interest rates	Quoting conventions: dollars per unit of foreign currency for CME contracts, the reverse in the spot market for some pairs
Government bonds	Coupon minus repo, plus roll-down along the curve	Yield curve and repo rate, or adjacent contracts	Deferred bond futures are illiquid, and the cheapest-to-deliver bond can differ between contracts
Commodities	$(\ln F^{(1)} - \ln F^{(2)})/(T_2 - T_1)$	The two nearest liquid contracts	Seasonality (§11.2); illiquid or model-priced deferred months; the front month during delivery
VIX	$(\ln \text{VIX} - \ln F^{(1)})/\tau_1$, with $\tau_1 = T_1 - t$, or between the first two futures	Index and futures	Very large in calm markets, and mostly a premium (§5.4)
Bitcoin	$(S-F)/(F\tau)$, or the perpetual swap's funding rate	Reference rate and future	Large and volatile; exchange risk on the spot leg

The simple form $(S-F)/(F\tau)$ and the log form $(\ln S - \ln F)/\tau$ agree to first order when carry is a few percent. Where carry is large, as in the VIX, the two forms differ. Pick one and use it consistently across markets.

[Practice] Two refinements matter more than the choice of formula.

- **Divide by volatility.** Carry is measured in return units. A 10% carry on a market with 40% volatility is a weaker signal than a 5% carry on a market with 8% volatility. Carry divided by the market's volatility is the Sharpe ratio a position would earn from carry alone, if the spot price were a random walk without drift. It is the quantity to compare across markets.
- **Smooth it.** Measured carry jumps when the pair of contracts used changes at a roll, and seasonal carry swings within the year. **[Fact]** Koijen et al. find that averaging carry over the previous 12 months keeps most of the signal's performance while roughly halving turnover.

11.2 Seasonality: compare like with like

In natural gas, the grains, heating oil and gasoline, the spread between adjacent contracts mixes carry with the seasonal pattern of the curve. Consider natural gas in October. The spread from November to December is in steep contango because

December is a winter month, not because storage is expensive. A carry signal built from that spread would short natural gas every autumn, for a reason unrelated to inventory.

[Practice] Two remedies remove the seasonal pattern.

- **Measure carry between contracts a year apart,** such as this December against next December. Seasonality cancels, and what remains is the curve's underlying slope.
- **Measure carry against its own seasonal norm.** Take the adjacent-contract spread, minus the average spread between the same calendar months in previous years.

Holding only one calendar month a year in seasonal markets (§7.3) makes the first remedy automatic.

11.3 The evidence

[Fact] Carry predicts futures returns in every major asset class, both in the cross-section and over time. Consider a strategy that is long the highest-carry contracts and short the lowest-carry contracts within an asset class. Averaged across asset classes, it earned a Sharpe ratio of about 0.8. A diversified combination across asset classes earned about 1.2 (Koijen et al., 2018; the authors' firm runs carry strategies). Commodities show the same pattern. Sorting on the basis earns sizeable spot and term premia (Szymanowska et al., 2014). The basis tracks inventories (Gorton, Hayashi & Rouwenhorst, 2013). And the relation between basis and returns held up out of sample (Bhardwaj, Gorton & Rouwenhorst, 2015).

[Fact] Carry strategies crash. Traders say that currency carry "goes up by the stairs and down by the elevator". Long positions in high-yield currencies unwind together when risk appetite collapses, as in 1998 and 2008 (Brunnermeier, Nagel & Pedersen, 2008). In commodities, high carry means scarce inventories, and scarce inventories mean higher volatility. Carry is therefore partly a payment for bearing exactly the risks that arrive together in a crisis.

[Contested] How much of the in-sample Sharpe ratio survives costs, publication and capacity is unsettled. The view taken here is that carry is a positive, modest and crash-prone premium. It is most valuable as one component of a diversified futures portfolio.

11.4 Five ways to use carry

1. **As a strategy.** Go long the highest risk-adjusted carry and short the lowest, within an asset class. Alternatively, go long or short each market on the sign of its own carry. The carry section of Systematic Trading Strategies covers the variants and their verdicts.
2. **As a tilt on another signal. [Practice]** Many futures systems combine trend and carry, because the two are weakly correlated and fail at different times. Sharp reversals hurt trend. Crises, which trend can catch, hurt carry. A trend position that pays heavy carry needs a stronger trend to justify it.
3. **As a contract-selection rule.** Where the curve bends, hold the contract with the best carry per unit of risk among the liquid ones (§7.3).
4. **As a cost line.** Before any directional futures trade, compute the carry over the intended holding period. It is the breakeven that the view must beat (§6.5).
5. **As a curve trade.** A calendar spread isolates the slope. In storable commodities it is a position on inventories: a long front contract against a short back contract profits if stocks tighten. Size the spread by its own volatility, including its behaviour in squeezes (§9.6).

A minimal monthly carry process for a futures portfolio has five steps. **[Practice]**

1. For each market, measure carry from the two nearest liquid contracts, or from contracts a year apart in seasonal markets.
2. Average carry over recent months (Koijen et al. use 12) to smooth roll jumps and seasonal swings.
3. Divide by the market's volatility.
4. Rank the markets, or scale positions in proportion to the result.
5. Size each position with the risk formula of §10.2, then round and buffer (§10.4).

11.5 How carry signals fail

- **Crashes.** The return distribution is negatively skewed. Size by what a crash does to the position, not by its volatility in quiet periods.
- **Seasonal mismeasurement.** Adjacent-month carry in a seasonal market trades the calendar, not the curve (§11.2).
- **Fictitious prices.** For deferred months that barely trade, the exchange derives settlement prices partly from spreads to the front month. Carry computed from them can show an opportunity that no one could trade.

- **Delivery-month distortions.** A front contract in its delivery period, or one squeezed near expiry, shows a carry that reflects that contract's delivery problem, not the market's (§5.6).
- **Crowding.** Carry is a well-known premium, and many traders hold it with leverage. When they unwind together, the losses are correlated across markets and across firms.

§11 Key takeaways

1. Carry is the return if spot does not move. Measure it from the index and the future for financial contracts, and from the two nearest liquid contracts for commodities.
2. Divide carry by volatility to compare markets, and smooth it to avoid trading on roll jumps.
3. In seasonal markets, measure carry between contracts a year apart, or against the seasonal norm.
4. Carry has predicted returns in every major asset class, with a diversified in-sample Sharpe ratio near 1.2. It also crashes in crises.
5. Carry can serve as a strategy, a tilt on trend, a contract-selection rule, a cost line for directional views, or, through calendar spreads, a curve trade.
6. Size carry positions by their crash risk, not by their quiet-period volatility.

12. Taxonomy and equivalences

12.1 The master form

Every futures position, in any market, is the same calculation with a handful of slots. Over a period, the profit and loss of a futures portfolio is the sum, over markets, of contracts times multiplier times price change. The number of contracts in each market comes from the risk formula of §10.2, scaled by a signal and rounded with a buffer:

$$\text{P\&L} = \sum_i N_i \, m_i \, \Delta F_i, \qquad N_i = \text{round}_b\left(\frac{A \, \sigma^\star \, w_i \, s_i}{m_i \, F_i \, \sigma_i} \right).$$

Here the sizing symbols are those of §10.2, for market i. The signal s_i is scaled to lie between −1 and 1, and round_b is rounding with a buffer b around the ideal position (§10.4). Each price change, written as $\Delta F_i \approx F_i \, \Delta \ln F_i$, splits into spot return, carry and curve shock (§6.2). The table lists the slots.

Slot	What it decides	Choices	Where
Underlying	Which carry ingredients are switched on	Index, currency, bond, storable commodity, non-storable, volatility	§5.2
Contract	Where on the curve the position sits	Front, deferred, one seasonal month, best carry	§7.3
Roll rule	When the position changes contract	Fixed schedule; lagged volume or open-interest crossover	§7.5
Series construction	What history the signal and volatility estimate see	Return-stitched, ratio-adjusted, back-adjusted	§8
Signal	Direction and strength	Trend, carry, combinations	§11; the other notes
Sizing	How many contracts	Risk formula, rounding, buffer	§10
Collateral and margin	What the cash earns; how much room there is	Bills, cash buffer, margin-to-equity limit	§9

[Practice] Which slots matter depends on what the trader is trying to protect.

- **For returns**, the underlying and the signal dominate.
- **For survival**, sizing and margin dominate, with the delivery calendar close behind. None of the losses in §14 came from a bad signal. They came from sizing, margin, delivery, exchange rules or a broker.

- **For a backtest's validity**, the series construction and the roll rule dominate. The wrong series silently adds or removes the carry.
- **For cost**, the contract choice dominates in steep, bent curves such as the VIX. It barely matters in equity, bond and currency futures.

12.2 Equivalences

Several objects in this chapter look different but are the same, exactly or approximately. The table lists them, with the conditions under which each equivalence holds.

These	are the same as	Status
Futures on a financial asset, plus cash earning the bill rate	The underlying asset, income included (for a commodity: storing it, plus its convenience yield)	Exact under cost of carry with known income; otherwise up to the financing spread
A ratio-adjusted series	The excess-return index, rescaled to today's price	Exact, for the same roll schedule
The change in a back-adjusted series	The dollar profit and loss of one contract held and rolled, per unit of multiplier	Exact
Carry, κ	Minus the net cost of carry: income plus convenience, minus financing and storage	Exact by definition, in log form
Commodity carry from the first two contracts	Carry measured from spot	Exact when the log curve, from spot out to the second contract, is a straight line in maturity; approximate otherwise
A long currency future	The carry trade: a deposit in the foreign currency financed in the domestic one	Exact under covered interest parity; the cross-currency basis is the gap
An equity index future	The index, financed at the implied financing rate	Exact, by definition of the implied rate
Roll yield, read as convergence	Carry, realised	Exact when the contract's annualised log basis c is unchanged, whatever spot does; otherwise the curve shock of §6.2 is the gap
A calendar spread	A position on the curve's slope, nearly immune to its level	Approximate; immune to a parallel shift in log prices only with legs weighted by the price ratio (§6.3)
A forward price	The futures price	Exact with non-random interest rates; a convexity adjustment otherwise
Exchange margin	A one-day 99% value at risk of the position, plus add-ons	Approximate, by regulatory design

12.3 Same name, different thing

The reverse problem also occurs: one name covers several different objects. The table lists the terms most often confused.

Term	One meaning	Another meaning	A third
Basis	$F - S$, as in this chapter and most financial-futures writing	$S - F$, "cash minus futures", in grain and physical trade	For Treasury futures, the cash bond's price minus the futures price multiplied by that bond's conversion factor
Contango, backwardation	The slope of the curve	Keynes: the futures price against the *expected* spot price	
Carry	The return if spot does not move (Koijen et al.; this chapter)	The *cost* of carry, $r + u - q - y$: the opposite sign	In bonds, coupon minus financing, sometimes excluding roll-down

Term	One meaning	Another meaning	A third
Roll yield	The return from convergence toward spot: realised carry	The price gap between contracts at the roll: not a return at all	In index reporting, the excess return minus the spot return: a residual that includes curve shocks
Margin	Futures: a performance bond that can earn interest	Securities: a loan that charges it	
Front month	The nearest expiry	The most liquid contract, which after the roll date is the second	
Excess return	A futures return, earned on top of the collateral	An asset's return minus the risk-free rate	For financial futures at fair value the two coincide (§6.2)

§12 Key takeaways

1. Every futures position is one formula: contracts times multiplier times price change. A risk budget, a signal and a rounding buffer set the number of contracts.
2. Returns come from the underlying and the signal. Survival comes from sizing and margin. Backtest validity comes from the series and the roll rule.
3. Four equivalences do most of the work. Futures plus cash is the asset. The ratio-adjusted series is the excess-return index. The back-adjusted change is dollar profit and loss. Carry is minus the net cost of carry.
4. "Basis", "carry" and "roll yield" each have several meanings. For basis and carry, two of the meanings have opposite signs. Check which one a source means before using its numbers.

13. Building it: data, calendars and code

13.1 Where the effort should go

[Practice] The six layers of a futures system are listed below in order of the damage an error in each one does.

1. **The contract master and its calendars.** This layer holds multipliers, ticks, settlement type, last trading day, first notice day, the broker's close-out deadline, and the exchange's holiday calendar. An error here produces a delivery notice, a forced liquidation, or a position 10 times the intended size.
2. **Raw contract prices.** These are daily settlement prices for every contract, stored per contract and validated. An error here corrupts every series built from them.
3. **The roll schedule and the stitched series.** A wrong roll rule, or the wrong series, adds or removes the carry from every backtest.
4. **Sizing and margin checks.** This layer covers the risk formula, rounding, buffering, margin used and distance to liquidation. Errors here are how accounts end.
5. **Execution.** This layer covers spread orders for rolls, and alerts before every roll date.
6. **The signal.** The signal comes last. It is not unimportant, but none of it matters if the five layers before it are wrong.

13.2 The contract master

The contract master has one row per product. It is maintained by hand from the exchange's specification page and the broker's rules. It is also versioned, because specifications change. The table shows the fields, with the Micro E-mini S&P 500 as an example.

Field	Example (Micro E-mini S&P 500)
Root, exchange, currency	MES, CME, USD
Multiplier; tick size; tick value	5; 0.25; 1.25
Listed months; months to trade	H M U Z; all
Settlement	Cash, special opening quotation
Last trading day rule	Third Friday of the contract month, 9:30 a.m. Eastern

Field	Example (Micro E-mini S&P 500)
First notice day rule	None (cash settled)
Broker close-out deadline	Broker-specific; record it
Roll offset	5 exchange business days before the earliest of the above
Holiday calendar	CME equity calendar
Broker identifiers	Contract identifier and trading class at the broker
Specification history	Dated changes to multiplier, tick, months

From the master, generate a table with one row per contract. Each row holds the contract's identifier, its expiry, and its **last safe day**. The last safe day is the earliest of the last trading day, the first notice day and the broker's deadline. The code below rolls a fixed number of exchange business days before it.

13.3 The code

The core fits in about 60 lines of Python with pandas. Its inputs are three objects. `closes` is a table of daily settlement prices, with one column per contract and missing values outside each contract's life. `last_day` (each contract's last safe day, §13.2) and `expiry` are series indexed by contract. **[Practice]** The functions assume two things: that each contract has a price on every day it is held and on the roll date, and that the next contract is listed. Check both before trusting the output.

```python
import numpy as np
import pandas as pd


def roll_dates(last_day: pd.Series, offset: int, calendar: pd.offsets.CustomBusinessDay) -> pd.Series:
    """Leave each contract `offset` exchange business days before its last safe day."""
    return last_day.apply(lambda d: d - offset * calendar)


def held_contract(dates: pd.DatetimeIndex, rolls: pd.Series) -> pd.Series:
    """Contract held from each close: the first whose roll date is after it."""
    order = rolls.sort_values()
    k = np.searchsorted(order.to_numpy(), dates.to_numpy(), side="right")
    return pd.Series(order.index[k], index=dates)


def stitch(closes: pd.DataFrame, held: pd.Series) -> pd.DataFrame:
    """Within-contract returns and P&L, and the continuous series built from them."""
    px = closes.to_numpy()
    j = closes.columns.get_indexer(held)
    assert (j >= 0).all(), "a held contract has no price column"
    i = np.arange(1, len(closes))
    p0, p1 = px[i - 1, j[i - 1]], px[i, j[i - 1]]  # yesterday's contract, both days
    ret = np.r_[0.0, p1 / p0 - 1.0]
    pnl = np.r_[0.0, p1 - p0]
    front = px[np.arange(len(closes)), j]
    growth = np.cumprod(1.0 + ret)
    return pd.DataFrame({
        "front": front,                                  # unadjusted: levels only
        "ret": ret,                                      # signals and volatility
        "ratio": front[-1] * growth / growth[-1],        # ratio-adjusted
        "back": front[-1] - (pnl[::-1].cumsum()[::-1] - pnl),   # back-adjusted
    }, index=closes.index)


def carry(closes: pd.DataFrame, held: pd.Series, expiry: pd.Series) -> pd.Series:
    """Annualised log slope from the held contract to the next one."""
    order = list(expiry.sort_values().index)
    nxt = held.map(lambda c: order[order.index(c) + 1])
    n = np.arange(len(closes))
    f1 = closes.to_numpy()[n, closes.columns.get_indexer(held)]
    f2 = closes.to_numpy()[n, closes.columns.get_indexer(nxt)]
    years = (expiry[nxt].to_numpy() - expiry[held].to_numpy()) / np.timedelta64(365, "D")
    return pd.Series(np.log(f1 / f2) / years, index=closes.index)


def volatility(ret: pd.Series, span: int = 35) -> pd.Series:
    """Annualised exponentially weighted volatility of within-contract returns."""
    return ret.ewm(span=span).std() * np.sqrt(252)


def contracts(equity: float, sigma_star: float, weight: float,
              multiplier: float, price: float, sigma: float) -> float:
    """Unrounded position: the account's risk budget over one contract's dollar risk."""
    return equity * sigma_star * weight / (multiplier * price * sigma)


def rebalance(current: int, ideal: float, buffer: float) -> int:
    """Trade only when outside the rounded band around the ideal, and then to its edge."""
    lo, hi = int(np.floor(ideal - buffer + 0.5)), int(np.floor(ideal + buffer + 0.5))
```

```
    return min(max(current, lo), hi)
```

For one market, the functions combine as follows:

```
cal = pd.offsets.CustomBusinessDay(holidays=exchange_holidays)
held = held_contract(closes.index, roll_dates(last_day, 5, cal))
series = stitch(closes, held)
kappa = carry(closes, held, expiry)          # carry, for §11
sigma = volatility(series["ret"])
ideal = contracts(100_000, 0.15, 1.0, 5, series["front"].iloc[-1], sigma.iloc[-1])
target = rebalance(current_position, ideal, 0.5)
```

Five details in the code carry the arguments of this chapter.

- `held_contract` decides, at each close, which contract is held *from* that close. A day's return is therefore always computed within one contract (§8.2). For the same reason, `front` already shows the new contract on a roll date. It switches one day earlier than the unadjusted column of §8.2.
- `stitch` builds every series from within-contract changes. The ratio-adjusted series is the compounded returns rescaled to today's price. The back-adjusted series is today's price minus what one contract has earned since, per unit of multiplier (§8.4). By construction, the final values of both equal the current contract's price.
- Sizing uses the current contract's actual price, `front`, and the volatility of `ret`. It never uses an adjusted level (§8.5).
- `carry` reads two contracts on the same day, because carry is a cross-section of the curve, not a time series (§6.2). It measures the slope to the next contract in `expiry`. In a seasonal market, pass it only one calendar month's contracts, so that the slope is measured a year apart (§11.2).
- `rebalance` trades only when the position leaves the band, and then only to the band's edge (§10.4).

For a backtest, compute the simulated position's profit and loss on each day as the previous day's contract count, times the multiplier, times that day's change in `back`. That change is the price change of the contract held over the day, roll days included. Charge the cost of the spread order on each roll date, and nothing for the price gap (§6.3). For periods with negative prices, use the back-adjusted series and dollar profit and loss only, because `ret`, `ratio` and `carry` are undefined there (§5.6).

13.4 The daily routine

[Practice] Run a short checklist after the settlement prices arrive.

- **Prices.** Flag any settlement price that moved more than several standard deviations, or did not move at all, before anything uses it.
- **Positions.** Compare each position with its target, after buffering.
- **Risk.** Check margin used as a share of equity, each position's distance to liquidation, $1/L - \mu$, and the cash buffer against three times the one-day 99% loss.
- **Calendar.** List every held contract's roll date, first notice day and last trading day within the next 10 business days. Add scheduled events that move these markets, such as central-bank meetings, inflation releases, the weekly US petroleum inventory report and the monthly USDA crop report.
- **Broker notices.** Check for margin changes and close-out deadlines.

13.5 Executing a roll

[Practice] A roll takes five steps.

1. Confirm the next contract. Check that its open interest and volume have overtaken the current contract's, or are approaching them.
2. Find the listed calendar spread and confirm its sign convention (§7.1).
3. During the main session, place a limit order on the spread near its mid-price. If it does not fill, improve the price in one-tick steps.
4. Confirm that both legs filled. Check that the position, contract identifiers and roll schedule in the trader's own records match the broker's.
5. For micro crude, remember that it stops trading a day before the standard contract.

§13 Key takeaways

1. Effort goes in this order: calendars, raw prices, roll rule and series, sizing and margin, execution, and only then the signal.
2. A versioned contract master, maintained from exchange pages and broker rules, prevents the expensive errors: deliveries, liquidations and mis-sized positions.
3. Store raw settlement prices per contract, and build every series from within-contract changes. About 60 lines of code are enough for the core.
4. Size from the current contract's real price and the volatility of stitched returns, then round with a buffer.
5. A daily routine checks prices, positions, margin, distance to liquidation, and every date that could force a trade.

14. Failure modes and case studies

14.1 The catalogue

The table collects the failures described in earlier sections. Each row gives the mechanism, the symptom a trader sees, and the prevention.

Failure	Mechanism	Symptom	Prevention
Wrong size	Misread multiplier; standard contract confused with micro; contract priced in another currency	Profit and loss 10 times what was expected	Compute mF before every new contract; a contract master (§13.2)
Forced close or delivery near expiry	A physically delivered contract held past the broker's deadline	The broker closes the position at a poor price, or a delivery notice arrives	Roll before the earliest of first notice, last trading day and the broker's deadline (§2.4)
Expiry settlement surprise	An index future held into its opening-auction settlement	Final price far from the previous close	Roll before expiry morning (§2.4)
Liquidation by routine noise	Position sized by the margin available	Closed out on an ordinary move	Size by risk; check $1/L - \mu$ (§9.1)
Margin spiral	Margins raised after a shock, when equity has already fallen	Forced selling near the low	Keep margin below a quarter to a third of equity (§9.3)
Locked limit	The market at its price limit, with no one trading	Cannot exit; a stop does not fill	Size for gaps; do not rely on stops (§2.5)
Carry hidden in a backtest	Returns computed from an unadjusted series	Performance overstated in contango, understated in backwardation	Return-stitched series (§8)
Roll cost counted twice	The roll gap charged on top of daily profit and loss	Performance understated in contango	Charge only the spread trade's cost (§6.3)
Look-ahead in the roll	Same-day open interest used to decide the roll	A backtest that cannot be reproduced live	Fixed schedule, or lag by a day (§7.5)
Irreproducible research	Adjusted histories restated at every roll	Signals that change between downloads	Store raw contract prices; rebuild (§8.6)
Seasonal carry misread	Adjacent-month spreads in seasonal markets	A carry signal that trades the calendar	Compare contracts a year apart (§11.2)
Pipeline failure at negative prices	Log returns and ratios undefined	Missing or infinite values; crashes	Additive methods; handle the episode explicitly (§5.6)
Idle collateral	Cash that earns nothing	Futures lag the asset by the cash rate	Bills or a money-market fund (§9.5)
Broker failure	Customer money missing from segregated accounts	Funds frozen for years	Keep only needed cash at the broker; hold the rest separately (§14.2)
Short volatility through futures	Harvesting VIX roll-down	Years of small gains, then a catastrophic day	Size by the crash (§14.2)
Spread over-leverage	Low spread margins	Losses many times the margin on a squeeze	Size spreads by their own risk (§9.6)

14.2 Case studies

Negative oil, April 2020. [Fact] §5.6 covers this episode. The May WTI contract settled at −37.63 dollars on 20 April 2020. Anyone still long at expiry had to take barrels at a nearly full storage hub (CFTC staff, 2020). The largest US oil fund had held mostly front-month contracts. In the weeks that followed, it spread its holdings across later months. *Lessons:* physical delivery is a real obligation. The front of a curve can detach from the rest. Every tool built on percentage returns breaks at zero.

The inverse VIX note, 5 February 2018. [Fact] The VelocityShares Daily Inverse VIX Short-Term note held a short position in VIX futures, rebalanced daily. It harvested the roll-down of a curve that is usually in contango. On 5 February the VIX more than doubled in a day. The note's indicative value fell by about 96%, from 115.55 to 4.22 dollars, which triggered the issuer's right to redeem it (Credit Suisse, 2018; ETF.com). **[Contested]** Such products had to buy VIX futures into the spike to rebalance. How much that buying worsened the move is debated. *Lesson:* carry is the known part of a return, not the safe part. A short-volatility position earns steadily because it is selling insurance against exactly this day.

Silver margins, 2011. [Fact] Silver peaked near 50 dollars an ounce in late April 2011. Around the peak and into May, CME raised margins five times in nine days. The increases roughly doubled the deposit required per contract (CNBC, 2011). Leveraged holders were forced to sell into a falling market, and the price fell by about a third within weeks. *Lesson:* margin is procyclical. Size positions so that a doubling of margin does not force a sale.

Nickel, March 2022. [Fact] After Russia's invasion of Ukraine, the London Metal Exchange's three-month nickel price more than doubled, from about 48,000 to above 100,000 dollars a tonne, within hours of trading on 8 March. The exchange suspended trading and cancelled all nickel trades made that day. In the later court case those trades were valued at about 12 billion dollars, and the English High Court upheld the exchange's decision (LME notice, 2022). Short holders faced margin calls they could not meet. Holders of profitable trades saw them erased. *Lesson:* an exchange can halt a market and unwind trades. Exchange rules, not just prices, are part of the risk.

MF Global, 2011. [Fact] When the broker MF Global failed in October 2011, about 1.6 billion dollars that should have been held in segregated customer accounts was missing. Customers' funds were frozen. The trustee eventually returned 100% of customer claims, under a plan that a court approved in November 2013 (NPR, 2013). *Lesson:* the clearing house guarantees trades between brokers. It does not guarantee how a broker handles its customers' cash. Keep only the needed margin and buffer at the broker, and hold the rest elsewhere.

Amaranth, 2006. [Fact] The hedge fund lost more than 6 billion dollars, mostly in September 2006, on natural gas calendar spreads. It was long winter contracts and short the months around them (US Senate Permanent Subcommittee on Investigations, 2007). Spreads look low-risk because they usually move little, and exchanges margin them accordingly. *Lesson:* size a spread by how it behaves when the seasonal relationship breaks, not by its margin or its quiet-period volatility.

§14 Key takeaways

1. Most futures disasters are mechanical: size, delivery, margin, liquidity and broker failures. Unusual prices alone rarely cause them.
2. Most research errors with futures come from the wrong series or the wrong roll accounting. Both silently add or remove the carry.
3. A short-volatility position that harvests contango is a carry trade with a known catastrophic tail. Size it by that tail.
4. Margin is procyclical, exchanges can halt markets and cancel trades, and brokers can fail. The defences are headroom, and keeping only the needed cash at the broker.

15. Synthesis

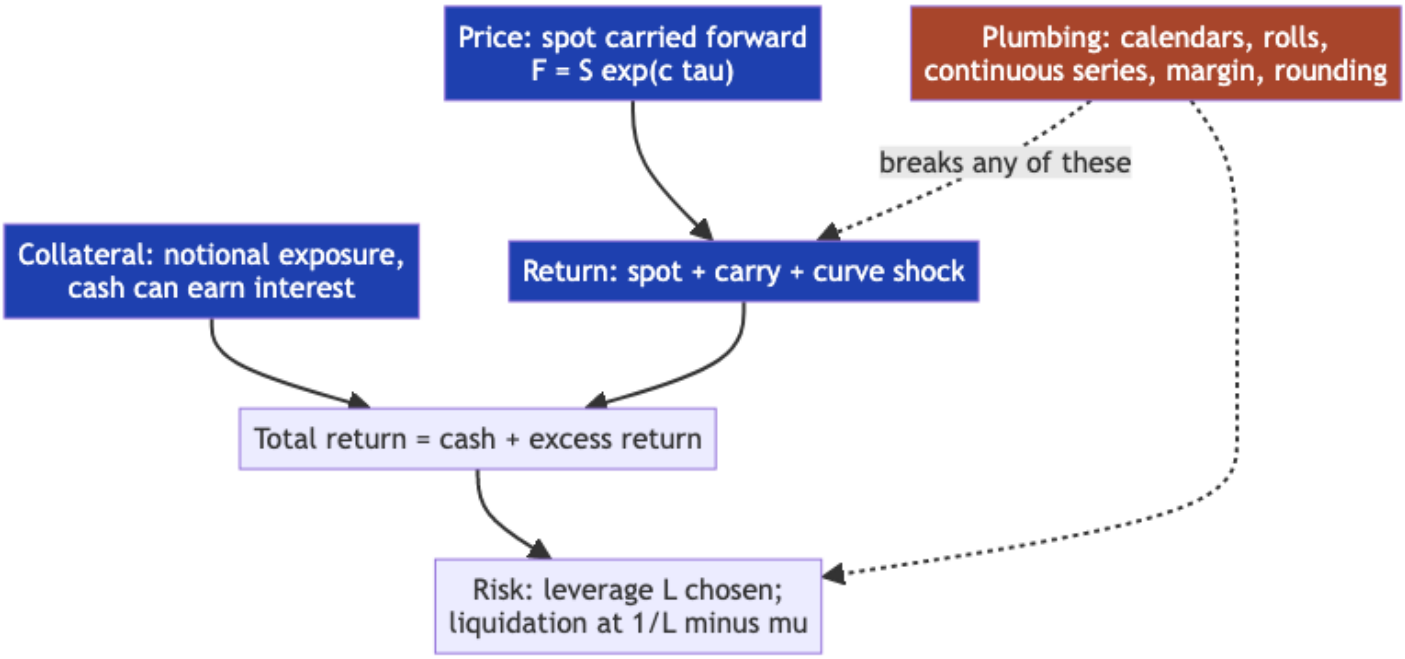
15.1 The framework on one page

Three identities generate almost everything in this chapter.

- **Price.** For anything that can be held, $F = Se^{c\tau}$ with $c = r + u - q - y$. The futures price is the spot price carried forward at the net cost of carrying the asset. The sign of c decides between contango and backwardation. Where nothing can be held, the futures price is an expectation adjusted by a risk premium. For the VIX that premium is large, and longs pay it.
- **Return.** $\Delta \ln F = \Delta \ln S + \kappa\Delta +$ curve shock, with $\kappa = -c$. Carry is the known part of the return. It is paid daily as the contract converges, never at the roll.

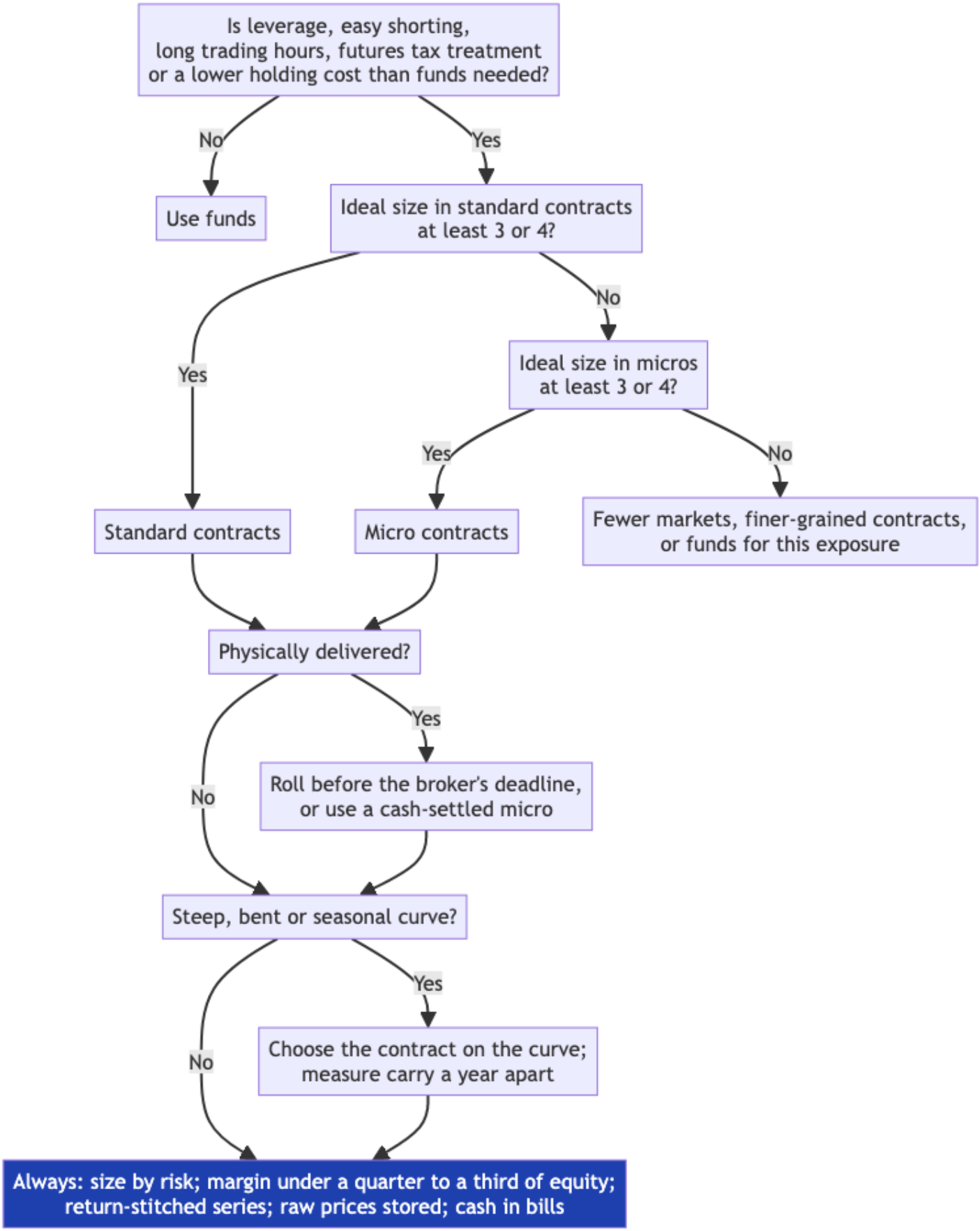
- **Collateral.** The exposure is NmF, acquired with no money. The cash can earn interest separately, and the total return is that interest plus the futures excess return. The trader chooses the leverage. Margin decides only how much leverage the trader may take, and when the trader is forced out: after an adverse move of $1/L - \mu$.

Around these three identities sits the plumbing, where most failures happen. It covers the calendar of notice and expiry dates, the construction of continuous series, margin, and the rounding of positions into whole contracts. The diagram shows how the parts connect.



15.2 A decision tree for an individual

The decision tree turns the chapter into a sequence of choices for an individual account. It ends in one of three instruments, then in the checks that apply to every futures position.


Is leverage, easy shorting, long trading hours, futures tax treatment or a lower holding cost than funds needed?
No
Use funds
Yes
Ideal size in standard contracts at least 3 or 4?
Yes
Standard contracts
No
Ideal size in micros at least 3 or 4?
Yes
Micro contracts
No
Fewer markets, finer-grained contracts, or funds for this exposure
Physically delivered?
No
Yes
Roll before the broker's deadline, or use a cash-settled micro
Steep, bent or seasonal curve?
No
Yes
Choose the contract on the curve; measure carry a year apart
Always: size by risk; margin under a quarter to a third of equity; return-stitched series; raw prices stored; cash in bills

15.3 A staged build, with gates

[Practice] The early stages are infrastructure, not trading. They are also where futures projects usually fail. Each stage below ends in a gate: a check that must pass before the next stage starts.

1. **Contract master and calendars** for three to five markets. *Gate:* the generated roll dates, first notice days and broker deadlines for the past year match the exchange's and the broker's published dates.
2. **Raw prices and stitched series.** *Gate:* the stitched S&P 500 futures excess return, plus a bill rate, tracks the index's total return to within a financing spread of a fraction of a percent a year. The continuous series also reproduce the adjusted columns of §8.2.
3. **Sizing and risk reporting.** *Gate:* a daily report shows positions against targets, margin used, distance to liquidation, the cash buffer, and the dates in the next 10 business days.
4. **One market at the smallest size**, one to three micros, held through at least two rolls. *Gate:* rolls are executed as spread orders at the planned time, and the trader's records match the broker's.
5. **The diversified set the account supports** (§10.3). *Gate:* the margin and buffer rules hold through a volatile month without a forced trade.
6. **Signals**: trend, carry, or both, from the other notes and §11.

15.4 Advice for someone starting today

These recommendations condense the chapter for an individual starting to trade futures.

1. Before trading a contract, compute one contract's exposure, multiplier times price. It is usually far larger than the margin.
2. Futures plus cash is the asset. A futures return is already an excess return.
3. Carry is paid every day as the contract converges. The roll swaps one contract for another and has no value of its own.
4. Read a curve's slope as carry, not as a forecast. The exceptions are the VIX and other things that cannot be stored, where the slope is mostly a premium.
5. Know every held contract's first notice day, last trading day and broker deadline. Roll early, with a spread order.
6. Use return-stitched series for research and the current contract's price for sizing. Store raw contract prices, never adjusted levels.
7. Size by risk, never by the margin available. Leverage is a number the trader chooses.
8. Keep margin below a quarter to a third of equity. Know the distance to liquidation: one over leverage minus the maintenance rate.
9. Micros buy granularity, not cheapness. For very small accounts, funds are better.
10. Carry is the known part of the return, not the safe part. Size carry and short-volatility positions by their crash.

15.5 What is known, and what is not

The mechanics are not in doubt. The cost-of-carry relation for financial futures, convergence at expiry, the decomposition of a futures return, the meaning of each continuous series, and the arithmetic of margin are definitions and arbitrage relations. They hold by construction, or traders who profit when they fail enforce them every day. Most of this chapter falls in that category, and a reader can rely on it.

Some empirical regularities are robust. Carry has predicted returns across asset classes. Commodity curves track inventories. The slope of the VIX futures curve is mostly a premium. And traders have profited from trading against predictable flows, such as index rolls.

Much remains open. Nobody knows how large the commodity futures premium will be. The two best-documented answers are both defensible: an equity-like premium for a diversified index, and roughly nothing for the average contract. Whether hedging pressure or storage drives the premium is resolved only partly, by horizon. Two further questions are unsettled. One is how much of the financing spread in index futures is a balance-sheet cost and how much is a demand premium. The other is how much carry strategies earn after costs, now that they are widely known.

[Practice] Recommendation: treat the mechanics as fixed, and treat carry as a modest, crash-prone premium worth holding in diversified form. Treat the commodity premium as uncertain. A long-only commodity position is then justified by its diversification and inflation properties, not by an expected return. Spend the first months on the calendar, the data and the sizing, because that is where an individual's futures trading is won or lost.

Appendix A. Concepts and prerequisites

This appendix explains the concepts that the main text uses without stopping to build them up. It is written for a reader who is at home with mathematics but does not work in finance. Such a reader meets “the repo rate” in §5.1, “the cheapest-to-deliver bond” in §5.2 or “SPAN 2” in §9.2. That reader wants the idea rather than a citation, and would rather not leave the chapter to get it.

The entries are ordered by **dependency**, not alphabetically, so later entries use earlier ones. They fall into six parts, which are also in dependency order, so the appendix reads as a build-up. The parts cover, in turn:

1. rates, returns and statistics;
2. how exchanges and accounts work;
3. the money markets, the arbitrage that ties futures to spot, and the financial underlyings;
4. commodities, volatility and premia;
5. risk and sizing;
6. the vocabulary of funds, data and backtests.

Each entry gives the idea in words first, then the formal definition, then why it appears here, then where to go deeper. Notation follows the main text’s notation table exactly. Symbols that an entry introduces are local to it. Where a field’s usual symbol would collide with the notation table, the entry uses another symbol and says so.

None of this is needed to follow the *argument* of the chapter. Much of it is needed to check the argument, or to act on it.

Index. The table lists where each concept first matters.

Concept	First used	Concept	First used
Continuous compounding, basis points	§1.5	Storage, convenience yield	§3
Log returns	§1.5	Hubs, grades and units	§1.2
Conditional expectation, martingales	§5.4	Seasonal markets	§5.2
Volatility, EWMA	§9.2	Gold lease rate	§5.2
Correlation, portfolio volatility	§4.2	Real options	§5.2
Sharpe ratio	§6.1	Normal backwardation, COT	§3
Skewness, crash risk	§6.4	Spot, term, insurance premia	§5.4
Rebalancing return	§6.4	Samuelson effect, factor models	§5.5
Exchanges, sessions, calendars	§2.1	Squeezes and corners	§9.6
Clearing houses, FCMs	§1.3	VIX, volatility risk premium	§1.1
Settlement, marking to market	§1.2	Bitcoin reference rate, perpetuals	§2.3
The margin account	§1.2	Commodity indices	§3
Orders, the order book	§2.1	Financialisation, price impact	§3
Listed calendar spreads	§5.3	Value at risk	§3
Price limits, circuit breakers	§2.1	SPAN and SPAN 2	§3
Expiry, notice, delivery	§2.1	Leverage, margin-to-equity	§9.3
Volume, open interest	§2.3	Volatility targeting	§10.2
Bills, money-market funds	§1.6	Turnover, no-trade buffers	§10.4
Repo	§5.1	Crowding, carry unwinds	§11.3
Arbitrage, replication	§5.1	ETFs, ETNs, leveraged funds	§1.7
SOFR, Eurodollar futures	§1.4	Section 1256	§10.5
Forwards, convexity, tailing	§1.4	Roll-down, roll yield, roll gap	§6.2
Duration, DV01	§5.2	Non-synchronous closes	§8.6
Treasury futures, CTD	§2.3	Backtest biases	§4.2
Covered interest parity	§5.2	Cross-sectional, time-series	§6.4
Index arbitrage, implied financing	§3	Trend, momentum, breakouts	§5.4

Concept	First used	Concept	First used
Options, Black model	§1.7	Scheduled releases	§13.4

Part I — Rates, returns and statistics. This part covers the arithmetic of rates and returns, in which every formula in the chapter is written. It also covers the few statistics the chapter uses to describe risk.

A.1 Continuous compounding, simple rates and basis points

The idea. An interest rate describes growth. The same growth gets a different number depending on how often interest is added to principal. A simple rate pays interest only on the original sum. An annually compounded rate adds the interest once a year, and that interest then earns interest. A continuously compounded rate adds interest at every instant.

Pricing theory uses continuous compounding because it turns growth into exponentials. Rates then add, and carrying for two periods is the product of carrying for each. The futures price is spot times $e^{c\tau}$ whatever the horizon. Money markets and brokers quote simple rates instead. At the rates and horizons of this chapter the two conventions differ very little. A formula must still use the convention it was derived in. Small rate differences are quoted in basis points.

Formally. One dollar invested for τ years grows to $1 + r_s\tau$ at a simple rate r_s. It grows to $(1 + r_a)^\tau$ at an annually compounded rate r_a, and to $e^{r\tau}$ at a continuously compounded rate r. The three rates describe the same growth when

$$r = \frac{\ln(1 + r_s\tau)}{\tau} = \ln(1 + r_a).$$

For small $r\tau$, $e^{r\tau} \approx 1 + r\tau$. §5.1 uses this step to pass between its two forms of the cost-of-carry relation. The discount factor $e^{-r\tau}$ is the value today of a dollar paid at T. A **basis point** (bp) is 10^{-4}, one hundredth of a percent. A rate that rises from 3.75% to 3.80% has risen 5 bp. A cost of 0.52 bp of notional is 0.0052% of notional.

Why it appears here. The notation table makes r, q, u and y continuously compounded, so $F_t(T) = S_t e^{c\tau}$ in §1.5 and §5.1 is exact. §1.6 computes the interest on cash at a simple rate. §2.6 states trading costs in basis points, and §5.2 gives the cross-currency basis in them.

Deeper. Hull (2021), the chapter on interest rates.

A.2 Log prices and log returns

The idea. A return can be measured in two ways. The simple return is a ratio minus one. The log return is the change in the logarithm of the price. Logs turn multiplication into addition. The log return over a month is therefore the sum of the daily log returns, so compounding becomes summing, and an exponential price path becomes a straight line. For the small daily moves of most markets the two measures nearly agree. They diverge for large moves. Once a price touches zero or goes negative, the log return does not exist at all.

Formally. For prices F_{t-1} and F_t, the simple return is $R = F_t/F_{t-1} - 1$, and the log return is $\Delta \ln F = \ln F_t - \ln F_{t-1} = \ln(1 + R)$. Log returns sum over time, and they match simple returns to first order:

$$\ln F_T - \ln F_0 = \sum_{t=1}^{T} \Delta \ln F_t, \qquad \Delta \ln F = R - \tfrac{1}{2}R^2 + O(R^3).$$

So $\Delta \ln F \approx \Delta F/F$ to first order, which is the approximation §12.1 uses. Log returns add over time but not across assets. A portfolio's simple return is the weighted sum of its holdings' simple returns, but its log return is not the weighted sum of theirs. The logarithm is defined only for positive arguments. Any series, ratio or signal built from $\ln F$ or F_t/F_{t-1} therefore fails at $F \leq 0$.

Why it appears here. The return identity of §1.5 and §6.2 is written in log returns, which is what makes it exact. §5.6 and §13.3 deal with the failure at negative prices. The May 2020 WTI contract made log returns, ratio-adjusted series and percentage-based sizing undefined.

Deeper. Simple and Log Returns, the note in this collection devoted to the distinction.

A.3 Conditional expectation, martingales and the random walk

The idea. A forecast is an average over the outcomes still possible, given what is known now. Conditional expectation makes that precise: $E_t[X]$ is the best guess of X that uses only information available at t. A martingale is a process whose best forecast of its own future value is its current value. Nothing known today predicts which way it will move. A random walk is the simplest example. Its steps are independent draws from one fixed distribution. Samuelson (1965) observed that a price set as the expectation of a future value must be a martingale. Suppose the futures price is today's best forecast of the spot price at expiry. Any predictable change in the futures price would then already be part of it.

Formally. Let $\mathcal{F}_t$ be the information available at t. Then $E_t[X] = E[X \mid \mathcal{F}_t]$ is the function of that information that minimises the mean squared error $E[(X - Z)^2]$ over all such functions Z. The tower property, $E_t\big[E_{t+\Delta}[X]\big] = E_t[X]$, says that today's forecast of tomorrow's forecast is today's forecast. A process X_t is a **martingale** if $E_t[X_{t+\Delta}] = X_t$ for all $\Delta > 0$. If $F_t(T) = E_t[S_T]$, the tower property gives $E_t[F_{t+\Delta}(T)] = F_t(T)$, so an unbiased futures price is a martingale. Now add the premium of §5.4, $F_t(T) = E_t[S_T] - \pi_t(T)$. The expected change in the futures price is then the premium that runs off over the period:

$$E_t[F_{t+\Delta}(T)] - F_t(T) = \pi_t(T) - E_t[\pi_{t+\Delta}(T)].$$

The futures price therefore drifts toward the expected spot price as the premium runs off. It reaches that price at expiry, where $\pi_T(T) = 0$. A **random walk** is $X_{t+1} = X_t + \varepsilon_{t+1}$, with the ε independent and identically distributed. It is a martingale when $E[\varepsilon] = 0$, and a random walk *with drift* otherwise.

Why it appears here. §5.4 writes the futures price as $E_t[S_T] - \pi_t(T)$. §6.5 takes E_t of the return identity to show that carry is a breakeven. §11.1 reads carry divided by volatility as a Sharpe ratio. That reading assumes the spot price is a random walk without drift, so that carry is the whole expected return.

Deeper. Samuelson (1965); D. Williams, *Probability with Martingales* (Cambridge University Press, 1991); Stochastic Processes in this collection.

A.4 Volatility: daily, annual and exponentially weighted

The idea. Volatility is the standard deviation of returns: the typical size of a move, in either direction. It is quoted per year so that markets and horizons can be compared. It scales with the square root of time, not with time, because independent moves partly cancel. A market that moves 1% on a typical day moves about 16% in a typical year, not 252%. Volatility also changes over time, with calm and turbulent stretches. An estimate should therefore weight recent days more than distant ones. The exponentially weighted estimate does this with a single parameter.

Formally. Suppose daily returns are uncorrelated, with standard deviation σ_d. Variance then adds over days, so the annual volatility is $\sigma = \sigma_d\sqrt{252}$, where 252 is the usual count of trading days in a year. Over n days the standard deviation is $\sigma_d\sqrt{n}$, so 16% a year is about 1% a day. The exponentially weighted moving average (EWMA) estimate of daily variance blends yesterday's estimate with today's squared deviation, once a day:

$$\hat{\sigma}^2_{d,t} = (1-\alpha)\,\hat{\sigma}^2_{d,t-1} + \alpha\,(R_t - \bar{R}_t)^2, \qquad \alpha = \frac{2}{\text{span} + 1}.$$

Here R_t is the day's return, and $\bar{R}_t$ is the matching weighted mean, often set to zero for daily returns. The weight on a return k days old is proportional to $(1-\alpha)^k$. A span of 35 gives $\alpha = 1/18 \approx 0.056$ and a half-life of $\ln 2/(-\ln(1-\alpha)) \approx 12$ trading days. The pandas method `ewm(span=...)` uses this parameterisation, with a correction for small-sample bias.

Why it appears here. In the notation table, σ is the annualised volatility of a contract's percentage returns. §9.2 turns 1% daily volatility into a one-day 99% loss, and §10.2 sizes positions from σ. The `volatility` function in §13.3 is the EWMA above, annualised with $\sqrt{252}$.

Deeper. Hull (2021), the chapter on estimating volatilities and correlations; Carver (2015) and Carver (2023), on volatility estimates for sizing.

A.5 Correlation and the volatility of a portfolio

The idea. Two markets that do not move in lockstep partly cancel each other's moves when held together. A portfolio is therefore less volatile than the sum of its parts. Correlation measures how much two markets move together. It runs from −1

(exact opposites) through 0 (unrelated) to +1 (lockstep). How much cancels depends on correlation, not just on the number of markets: 10 markets that all move together behave like one market.

Formally. For returns R_i and R_j with volatilities σ_i and σ_j, the correlation is $\rho_{ij} = \mathrm{Cov}(R_i, R_j)/(\sigma_i \sigma_j)$. Consider a portfolio that holds fractions v_i of its capital in each market. (The entry writes v because the notation table reserves w for risk shares.) Its variance sums the contributions of every pair of markets:

$$\sigma_p^2 = \sum_i \sum_j v_i v_j \, \sigma_i \sigma_j \, \rho_{ij}.$$

Suppose each of n positions contributes the same volatility $\sigma_0 = v_i \sigma_i$, and every pair has correlation $\bar{\rho}$. Then $\sigma_p = n\sigma_0 \sqrt{1/n + (1 - 1/n)\bar{\rho}}$. The square root is the fraction of the lockstep volatility $n\sigma_0$ that remains. As n grows, the fraction falls toward $\sqrt{\bar{\rho}}$. Correlation, not the count of markets, therefore sets the floor. Correlation estimated from same-day returns is biased toward zero when markets close at different times (A.50).

Why it appears here. §4.2 and §6.4 report that commodity futures were negatively correlated with stocks and bonds. §8.6 warns of the bias from markets that close at different times. §10.3 uses the square-root formula, with $n = 10$ and $\bar{\rho} = 0.2$, to get the diversification multiplier (A.44).

Deeper. H. Markowitz, "Portfolio Selection," *Journal of Finance* 7(1) (1952), 77–91; Portfolio Construction and the Covariance Matrix in this collection.

A.6 The Sharpe ratio

The idea. The Sharpe ratio is return per unit of risk. Scaling a position up multiplies its excess return and its volatility by the same factor, so their ratio does not change. This makes the Sharpe ratio the natural way to compare strategies that can be levered up or down at will, as futures positions can. It is measured on *excess* returns, because anyone can earn the cash rate without risk. A strategy is paid only for the return above the cash rate.

Formally. For excess returns R^e, $\mathrm{SR} = E[R^e] / \mathrm{sd}(R^e)$. With n periods a year and independent returns, the mean scales by n and the standard deviation by $\sqrt{n}$. An annual Sharpe ratio is therefore $\sqrt{252}$ times a daily one. For an asset, $R^e = R - r^{\text{bill}}\Delta$. A futures return $\Delta F/F$ is already an excess return (§6.1), because nothing funds the position and the collateral's interest is counted separately. Its Sharpe ratio is therefore its own mean over its own standard deviation. Subtracting the bill rate again understates it by r^{bill}/σ a year. Suppose spot is a driftless random walk (A.3) and carry κ is known in advance. A futures position's expected excess return is then κ, so κ/σ is the Sharpe ratio of carry alone.

Why it appears here. §6.1 warns against subtracting the risk-free rate twice. §7.4 and §11.3 report the Sharpe ratios of strategies, and §11.1 divides carry by volatility to compare markets.

Deeper. W. F. Sharpe, "The Sharpe Ratio," *Journal of Portfolio Management* 21(1) (1994), 49–58; A. W. Lo, "The Statistics of Sharpe Ratios," *Financial Analysts Journal* 58(4) (2002), 36–52, for annualisation when returns are autocorrelated.

A.7 Skewness and crash risk

The idea. Volatility treats a gain and a loss of the same size alike. Many strategies, however, are lopsided. They earn small, steady amounts most of the time and occasionally lose a large amount. Such a distribution is negatively skewed. If its volatility is measured in a quiet stretch, it understates the risk badly, because the loss that matters may not yet be in the sample. Selling insurance has this shape. So does any strategy that is paid for bearing a risk that arrives all at once.

Formally. The skewness of a return R with mean $\bar{R}$ is $E[(R - \bar{R})^3] / \mathrm{sd}(R)^3$. It is zero for a symmetric distribution, and negative when the left tail is longer than the right. Take a strategy that gains a small g with probability $1 - p$, and loses a much larger ℓ with a small probability p. Its skewness is strongly negative. In a sample shorter than about $1/p$ periods, the loss is often absent, so the sample Sharpe ratio overstates the strategy's prospects. In this chapter, "crash risk" means this left tail: large losses that arrive together across markets.

Why it appears here. §6.4 notes the crash risk of the currency carry trade. §11.3 and §11.5 describe carry's distribution as negatively skewed, and advise sizing by the crash. §6.5 and §14.2 describe short VIX futures positions, which have exactly the insurance-selling shape.

Deeper. Brunnermeier, Nagel & Pedersen (2008), on the skewness of currency carry.

A.8 The rebalancing return

The idea. A portfolio rebalanced to fixed weights sells what has risen and buys what has fallen. Suppose its holdings are volatile and imperfectly correlated. Rebalancing then mechanically makes the portfolio compound faster than the weighted average of its holdings' compound growth rates. No holding earns more. The gain comes from diversification, which reduces the drag that volatility puts on compounding. This is why an index of commodity futures can grow faster than the typical commodity in it.

Formally. A holding with arithmetic mean return a_i and volatility σ_i has compound (log) growth of approximately $g_i \approx a_i - \frac{1}{2}\sigma_i^2$. A portfolio rebalanced to weights v_i has arithmetic mean $\sum_i v_i a_i$ and volatility σ_p (A.5). Its growth is the weighted growth of its holdings plus a term that measures how much diversification lowers variance:

$$g_p \approx \sum_i v_i \, g_i + \tfrac{1}{2}\Big(\sum_i v_i \, \sigma_i^2 - \sigma_p^2 \Big).$$

The bracketed term is the rebalancing (or diversification) return. It is zero when all holdings are perfectly correlated with equal volatility. It grows with volatility and with lower correlation. For example, take 10 equally weighted holdings with 30% volatility and pairwise correlation 0.2. Then $\sigma_p^2 = 0.09 \times 0.28 \approx 0.025$, and the bracket is about $\frac{1}{2}(0.09 - 0.025) \approx 0.032$, or 3.2% a year.

Why it appears here. §4.7 and §6.4 report Erb and Harvey's finding. An index's commodity premium came mainly from rebalancing across volatile, weakly correlated contracts, while the average individual contract earned roughly nothing.

Deeper. Erb & Harvey (2006); D. G. Booth & E. F. Fama, "Diversification Returns and Asset Contributions," *Financial Analysts Journal* 48(3) (1992), 26–32.

Part II — Market structure and trading mechanics. This part covers how exchanges, clearing houses and brokers turn a futures contract into daily cash flows. It also covers the orders, limits and dates that a position has to get through.

A.9 Exchanges, trading sessions and holiday calendars

The idea. A futures contract belongs to one exchange, which writes its rules, lists its months and publishes its settlement prices. Most US contracts that an individual trades belong to CME Group. CME Group owns four exchanges, which kept their historical specialities when they merged:

- the Chicago Mercantile Exchange (CME): equity indices, currencies and short-term interest rates;
- the Chicago Board of Trade (CBOT): Treasuries, the Dow and grains;
- the New York Mercantile Exchange (NYMEX): energy;
- COMEX: metals.

All four trade on one electronic platform, Globex. VIX futures trade on the Cboe Futures Exchange (CFE), and European index and bond futures trade on Eurex. Base metals trade on the London Metal Exchange (LME), whose date structure differs from the rest.

Formally. A US-listed contract trades electronically for nearly 23 hours a day, Sunday evening to Friday afternoon, Central time, with a daily one-hour break. The *regular* or day session is the part that overlaps the underlying cash market's hours. The rest is the *overnight* session, with thinner books and wider spreads. A *trade date* starts the evening before, so a trade on Sunday evening belongs to Monday's trade date and Monday's settlement. Each exchange publishes a **holiday calendar** of closures and early closes. These calendars differ by product group, and they differ from national bank holidays. Business-day arithmetic for rolls and expiries must use them. The LME lists a **prompt date** for every business day out to three months, then weekly and monthly dates further out. Its benchmark "three-month" price is for delivery three months from today. That makes it a rolling, constant-maturity forward rather than a contract with a fixed expiry.

Why it appears here. §2.1 and §2.5 describe the sessions, and §2.3 lists contracts by exchange. §7.5 and §13.3 build roll dates on exchange calendars. §14.2 quotes the nickel case at the LME's three-month price.

Deeper. The exchanges' own contract-specification and holiday-calendar pages; Hull (2021), the chapter on futures markets and central counterparties.

A.10 Clearing houses, FCMs and segregated accounts

The idea. If every trader faced every counterparty directly, each would have to judge the credit of strangers. A clearing house removes that need by standing in the middle of every trade. It can do so cheaply because it nets. A member that bought and sold the same contract with many counterparties owes the clearing house only on its net position. The clearing house protects itself by collecting margin and by keeping a pool of resources that absorbs a member's default in a fixed order. An individual is not a member. They reach the clearing house through a broker, and the broker is the weak link.

Formally. In the US a clearing house for futures is a **derivatives clearing organization** (DCO), registered with the Commodity Futures Trading Commission (CFTC). After **novation**, a trade between A and B becomes two contracts: A with the DCO, and the DCO with B. The DCO's net position in every contract is therefore zero, and its risk is that a member fails to pay variation margin. Its **default waterfall** absorbs losses in this order:

1. the defaulter's margin;
2. the defaulter's contribution to the **guarantee fund**;
3. a slice of the DCO's own capital;
4. the other members' contributions.

A **futures commission merchant** (FCM) is a CFTC-registered broker that accepts customers' money for futures, and clearing FCMs are DCO members. Customer funds must be held in **segregated** accounts, apart from the FCM's own money. That protection has two limits. Customer funds are pooled, so a shortfall can be shared across an FCM's customers. And the insurance that covers US securities accounts (SIPC) does not cover futures accounts.

Why it appears here. §1.3 describes the arrangement and diagrams it. The MF Global case in §14.2 was a failure at the FCM layer, which the clearing house does not guarantee.

Deeper. Hull (2021), the chapter on futures markets and central counterparties.

A.11 Settlement prices and marking to market

The idea. Daily settlement needs one official price per contract per day. The exchange computes it by a published procedure rather than taking the last trade, so that a single late print cannot move everyone's cash. Marking to market means revaluing every open position at that price and paying the difference in cash. For a heavily traded contract, the procedure gives a price close to where it traded at the close. For a deferred month that barely trades, the exchange builds the price from the spread to a liquid month. The "settlement" can then be a price at which nothing traded.

Formally. On trading day t the exchange publishes a settlement price F_t for each listed contract. Typically it is a volume-weighted average of trades in a short window near the close. When trades are scarce, it is derived from bids, offers and spread trades. A deferred month with no trades is usually settled at an anchor month's settlement plus the settled calendar spread between the two. Marking to market pays $Nm(F_t - F_{t-1})$ as **variation margin** (§1.2), after which the position is carried at F_t. At expiry, a **final settlement price** set by the contract's own rule (A.16) makes the last payment. Data vendors' daily "close" for futures is usually the settlement price, not the last trade.

Why it appears here. §1.2 introduces daily settlement, and §2.1 and §2.5 warn that settlement is not the last trade. §11.5 notes that carry measured from spread-derived settlements of illiquid months can show an opportunity no one could trade. §8.6, §13.1 and §13.3 treat raw settlement prices as the stored data.

Deeper. The settlement-procedure documents each exchange publishes per product; Hull (2021), the chapter on futures markets and central counterparties.

A.12 The margin account: initial and maintenance margin, calls and liquidation

The idea. Futures margin is collateral, not a loan. A share bought "on margin" is partly paid for with money borrowed from the broker, who charges interest. A futures contract costs nothing to buy, so nothing is borrowed. The margin is a deposit that guarantees the trader can pay tomorrow's loss, and it is sized to cover a bad day (A.41, A.42). There are two levels, so that an account does not get a call for every small loss: a higher one to open a position, and a lower one below which the account must not fall.

Formally. Write M^{I} for the initial margin per contract (a local symbol). A is the account's equity. Daily settlement changes it by the variation margin.

- The **initial** requirement for $|N|$ contracts is $|N|M^{\mathsf{I}}$. The **maintenance** requirement is $|N|M^{\mathsf{M}}$, which is lower: at the exchange, about nine-tenths of initial (§9.1).
- The **excess** is A minus the requirement in force. Opening a position needs A to cover the initial requirement summed over all positions.

- If A falls below the maintenance requirement, the exchange convention is a **margin call** to restore A to the initial requirement. Retail brokers more often liquidate positions automatically, at market.
- If the market gaps through the liquidation level, A can go negative, and the trader owes the deficit.
- **Day-trade margin** is a broker's lower requirement for positions closed within the session. A position still open at the session's end must meet full margin, or the broker closes it.

US **securities margin** (Regulation T) works differently. A buyer of shares borrows up to half the purchase price from the broker and pays interest on the loan.

Why it appears here. §1.2 calls margin a performance bond, and §1.7 contrasts it with securities margin. §9.1, §9.2 and §9.3 build the liquidation distance $x^\star = 1/L - \mu$, day-trade margin and the sizing rules on these definitions.

Deeper. Hull (2021), the chapter on futures markets and central counterparties; the appendix of Systematic Trading Strategies, on Regulation T.

A.13 Orders and the order book

The idea. An electronic market is a queue of standing offers to buy and sell. The best price anyone will pay is the **bid**, and the best price anyone will sell at is the **offer** (or ask). The gap between them is the **bid–ask spread**. A trader who wants to trade at once pays the gap, "crossing the spread". A trader willing to wait posts an order and may be filled later, or never. **Market makers** are firms that post bids and offers continuously and earn the spread in exchange for the risk of holding inventory. **Depth** is how many contracts are offered at each price. A one-tick spread with little depth is cheap for a small order and expensive for a large one.

Formally. The **mid-price** is the average of the best bid and the best offer. A trade at the bid or the offer costs half the spread relative to the mid, and a round trip costs the full spread. The main order types are these:

- A **market order** trades immediately against the best standing orders, at whatever price that takes.
- A **limit order** trades only at its stated price or better, and otherwise rests in the book.
- A **stop order** lies dormant until the market trades at its trigger price, then becomes a market order. It guarantees execution once triggered, but not the price. If the market jumps past the trigger, the fill can be far from it. If the market is locked at a limit (A.15), the fill may not happen at all.
- A **stop-limit** order becomes a limit order instead, which caps the price but may not fill.

Why it appears here. §2.1 and §2.6 measure costs in ticks of spread. §2.5 and §14.1 warn that a stop is not a guaranteed exit. §10.1 notes that micro contracts share their parents' spreads but not their depth. §13.5 rolls with a limit order on the spread near its mid-price.

Deeper. L. Harris, *Trading and Exchanges: Market Microstructure for Practitioners* (Oxford University Press, 2003).

A.14 Calendar spreads as listed instruments

The idea. A calendar spread is long one delivery month and short another of the same contract. Both legs move with the price level, so the spread moves mainly with the slope of the curve. Exchanges list spreads as instruments in their own right, with their own order books. A trader can therefore buy or sell the difference between two months in one order instead of two. The exchange's matching engine also links each spread book to the books of its two legs, so liquidity resting in any of the three can fill an order in the others.

Formally. A spread is quoted as the near leg's price minus the far leg's price. On CME, buying the spread buys the near month and sells the far one. An order in the spread fills both legs at once, at prices that differ by exactly the traded spread. **Implied** pricing combines orders across books in two ways. A bid for the near month and an offer for the far month together imply a bid for the spread (implied-in). A resting spread order and an outright order in one leg together imply an order in the other leg (implied-out). An **outright** order is an order in a single contract. Executing a roll as two outright orders pays two spreads instead of one. It also carries **legging risk**: the price can move between the two fills. Clearing houses margin a calendar spread far below two outright positions (A.42).

Why it appears here. §5.3 reads spreads against full carry, and §5.5 calls the spread the curve's own market. §7.1 and §13.5 roll with a spread order. §9.6 and §14.2 show how cheap spread margin invites over-leverage.

Deeper. Carver (2023), on executing rolls; the exchange's documentation of its implied-matching functionality.

A.15 Price limits, locked markets and circuit breakers

The idea. Exchanges cap how far some contracts may move in a session. The cap slows a panic and gives traders time to post margin. Its cost is that the market may stop trading at the cap. If every seller wants a price below the limit and the limit forbids it, nothing trades, and a trader who needs to get out cannot. Equity index futures are tied to the stock market's own halts. Electronic markets also add brief automatic pauses when prices move too fast.

Formally. A **daily price limit** is a band around the previous settlement price outside which no trade may occur. A market is **limit up** or **limit down** when it trades at the band's edge. It is **locked** when orders exist on only one side at the limit, so nothing trades. For grains the bands are reset periodically from the price level, and widened for the next session after a limit close.

The US stock market's **market-wide circuit breakers** are set by falls in the S&P 500 from the previous close. A 7% fall (Level 1) or a 13% fall (Level 2) halts trading in all stocks for 15 minutes, if it comes before 3:25 p.m. Eastern. A 20% fall (Level 3) halts trading for the rest of the day. CME's tiered limits on equity index futures sit at the same three levels during the stock market's hours.

A **dynamic circuit breaker** briefly pauses trading when the price moves more than a set amount from a reference price over a rolling window. Trading then resumes with the band re-centred.

Why it appears here. §2.1 lists price limits among the specification's fields. §2.5 describes the equity index tiers, the overnight band and locked markets. §9.4 and §14.1 warn that liquidation can happen at a limit, or not at all.

Deeper. CME, equity index price limits FAQ; Hull (2021), the chapter on futures markets and central counterparties.

A.16 Expiry: notice, delivery and final settlement

The idea. A contract's life ends in one of two ways. A physically delivered contract ends with a short handing over the goods and a long paying for them. The clearing house runs the hand-over as a sequence of notices over several days, and anyone still long when notices begin can be chosen to receive. A cash-settled contract ends with one last variation-margin payment, against a reference price computed by a special procedure on expiry morning. Either way, special rules govern a contract's last days. A position held only for its price exposure should leave before those rules apply.

Formally. The **last trading day** (LTD) is the last day the contract trades. The **first notice day** (FND) is the first day on which a short may give notice of its intention to deliver. CBOT's grain and Treasury contracts, for example, run delivery as a three-day sequence:

1. *Position* (or *intention*) day: the short's clearing member declares its intention to deliver.
2. *Notice* day: the clearing house matches the short to a long, usually the holder of the oldest open long position, and notifies both.
3. *Delivery* day: the goods move and the long pays the invoice price.

Grains and metals have their FND before their LTD, so a long can be assigned delivery while the contract still trades. Crude oil has its LTD first, and it delivers to whoever is long when trading stops. Brokers set their own **close-out deadlines**, earlier than either date.

A cash-settled index future settles to a **special opening quotation** (SOQ): the index computed from each constituent stock's opening price on expiry morning. VIX futures settle to an SOQ of the VIX, computed from the opening prices of S&P 500 options on the futures' expiry morning.

Why it appears here. §2.1 and §2.4 describe the dates and the SOQ, and §7.2 and §7.5 set roll dates from them. §13.2 records them in the contract master. §14.1 lists the forced closes and settlement surprises they cause.

Deeper. Hull (2021), the chapter on futures markets and central counterparties; each contract's specification page (§2.3 gives CME's URL pattern).

A.17 Volume, open interest and active months

The idea. Volume counts how many contracts changed hands in a session. Open interest counts how many are still open at its end. They answer different questions. Volume says where trading is happening today. Open interest says how many positions exist, and so how many will have to be closed or rolled before expiry. Every contract has one long and one short, which gives three cases:

- two traders opening new positions add one to open interest;
- two traders closing positions remove one;
- a trade that transfers an existing position leaves it unchanged.

In most markets both measures concentrate in one or two months at a time, called the **active** months. As expiry approaches, they migrate to the next month.

Formally. Let V_t be the number of contracts traded on day t and OI_t the number open at the end of day t, counting each contract once rather than once per side. Then $\text{OI}_t - \text{OI}_{t-1}$ equals the number of trades that opened two new positions minus the number that closed two existing ones. Open interest is known only after the clearing house has processed the day's trades, so exchanges publish it the following morning, often first as a preliminary figure. A rule that rolls when the next contract's open interest exceeds the current one's can therefore use OI_{t-1} on day t, but not OI_t. **Active months** are the cycle where liquidity concentrates. For gold these are February, April, June, August, October and December. Other months trade thinly.

Why it appears here. §2.3 warns about inactive metal months. §7.2 and §7.5 time rolls by the migration of liquidity, and §7.5 and §14.1 lag open interest to avoid look-ahead (A.51). §13.5 checks open interest before rolling.

Deeper. Hong & Yogo (2012), on what open interest predicts; Hull (2021), the chapter on futures markets and central counterparties.

Part III — Rates, arbitrage and financial underlyings. This part covers the money-market rates that finance a futures position, and the arbitrage argument that ties futures to spot. It also covers the details of the interest-rate, bond, currency, equity index and option markets that the main text draws on.

A.18 The risk-free rate: Treasury bills, money-market funds and collateral

The idea. "Risk-free" in finance means free of default risk over a short horizon. In dollars the benchmark is the yield on US Treasury bills: short-term government debt that pays no coupon and is sold at a discount to the amount it repays. A futures position's collateral can sit in three places: in bills, in a money-market fund that holds bills and similar short instruments, or as cash at the broker. The choice decides whether the futures-plus-cash identity holds in practice. Brokers that accept bills as margin value them at less than their market price. This haircut protects the broker against a fall in the bills' value before they can be sold.

Formally. A bill bought at price P that repays 100 in τ years yields the simple rate $r^{\text{bill}} = (100/P - 1)/\tau$. Bills are issued with maturities from a few weeks to a year. The market quotes them on a discount basis and on a bond-equivalent basis, both close to this rate for short bills. A government **money-market fund** holds bills, repurchase agreements backed by Treasuries (A.19) and similar instruments, and pays close to the bill rate less its fee. Collateral with market value V and **haircut** η counts as $(1 - \eta)V$ towards a margin requirement. Variation margin is paid in cash, so bills held as collateral cover the requirement but not a day's loss.

Why it appears here. §1.5, §1.6 and §6.1 count the collateral's interest at the bill rate as one of the two parts of a futures position's total return. §9.5 sets out cash management. Idle cash goes into bills or a money-market fund, bills pledged as collateral carry a haircut, and a cash buffer covers variation margin.

Deeper. Hull (2021), the chapter on interest rates; B. Tuckman & A. Serrat, *Fixed Income Securities*, 3rd ed. (Wiley, 2011).

A.19 Repo and the repo rate

The idea. A repurchase agreement, or repo, is a loan secured by a security. The borrower sells a bond to the lender today and agrees to buy it back later at a slightly higher price. The difference is the interest. If the borrower fails, the lender keeps the bond, so the rate is close to risk-free. Repo is how dealers finance holdings of bonds. It is also the rate at which an arbitrageur could borrow to buy a bond and carry it. The mirror trade, a reverse repo, lends cash against a bond. A dealer uses it to borrow a bond and sell it short.

Formally. A repo sells a security for P_0 and repurchases it after d days for $P_1 > P_0$. The **repo rate** is the simple rate $r^{\text{repo}} = (P_1/P_0 - 1) \times 360/d$, on the money-market convention of a 360-day year. The cash lent is usually less than the security's value by a haircut. *General collateral* repo accepts any security from a class and pays a rate close to other secured overnight rates. A security in high demand to borrow trades *special*, at a lower repo rate, because lenders of cash accept less interest to get that particular bond. For a bond held with repo financing, carry over a period is the coupon income minus the repo interest, the "coupon minus repo" of §11.1. The Secured Overnight Financing Rate (A.21) is computed from overnight Treasury repo trades.

Why it appears here. The notation table defines r as the risk-free rate or the repo rate. In §5.1, dealers borrow and lend bonds through repo to enforce the cost-of-carry bound. §5.2 and §11.1 make a bond future's carry the coupon of the cheapest-to-deliver bond minus its repo cost.

Deeper. Burghardt & Belton (2005); Tuckman & Serrat (2011).

A.20 Arbitrage, replication and the law of one price

The idea. An arbitrage is a trade that costs nothing, cannot lose and might win. In a market with many well-funded traders an arbitrage cannot persist. Traders would repeat it until prices moved enough to remove it. Two consequences follow. If two portfolios pay the same in every future state, they must cost the same today: the law of one price. And if a contract's payoff can be reproduced by trading other instruments (a replication), the contract's price is pinned to the cost of that replication, whatever anyone expects. The argument works only in directions where the replicating trade can actually be done. Suppose one leg requires selling something that nobody can sell short. Then only one of the two inequalities survives.

Formally. A self-financing strategy with value V_t is an **arbitrage** if $V_0 \le 0$, $V_T \ge 0$ in every state, and $V_T > 0$ with positive probability. The fundamental theorem of asset pricing states that a market has no arbitrage if and only if there is a probability measure Q under which every traded price, discounted at the riskless rate, is a martingale (A.3). Q is called the **risk-neutral** or pricing measure, and prices are discounted Q-expectations of payoffs. A futures position costs nothing and pays ΔF daily, so the futures price itself is a Q-martingale: $F_t(T) = E_t^Q[S_T]$.

Short selling means borrowing an asset, selling it, and buying it back later to return it. Meanwhile the short seller pays the lender a **borrow fee**. When shorting is impossible or costly, any bound that requires it fails. That is why §5.1 keeps only the upper bound for physical commodities. *Limits to arbitrage* are the capital, margin and funding constraints that let prices stray from arbitrage values for a while.

Why it appears here. §5.1 prices futures by replication and derives the one-sided bound for commodities. §3 and §5.2 name the arbitrage that enforces each market: index arbitrage, covered interest parity, and the cash–futures basis trade. §4.3 cites deviations from the law of one price, and §10.5 contrasts shorting futures with paying stock borrow fees.

Deeper. Hull (2021), the chapter on the determination of forward and futures prices; A. Shleifer & R. W. Vishny, "The Limits of Arbitrage," *Journal of Finance* 52(1) (1997), 35–55.

A.21 SOFR, LIBOR and short-term interest-rate futures

The idea. Short-term interest-rate futures let traders lock in, or bet on, the average overnight rate over a future three-month period. The underlying is a benchmark rate rather than an asset. The contract is quoted so that it behaves like a bond. Its price is 100 minus the rate, so the price falls when rates rise. For four decades the benchmark was LIBOR, and the contract was the Eurodollar future. LIBOR was an average of the rates at which banks said they could borrow unsecured. LIBOR rested on banks' estimates rather than on trades. It was retired for the dollar in 2023, and its successor is SOFR, a rate computed from actual overnight repo trades.

Formally. The **Secured Overnight Financing Rate** (SOFR) is published each business day by the Federal Reserve Bank of New York. It is a volume-weighted median of rates on overnight repo trades secured by Treasuries (A.19). A **three-month SOFR future** settles to $100 - R$, where R is the compounded average of daily SOFR over the contract's three-month reference period, in per cent. The contract's notional is one million dollars for a quarter of a year, so a one-basis-point change in R is worth $10^6 \times 10^{-4} \times 0.25 = 25$ dollars. The **Eurodollar** future it replaced settled to 100 minus three-month dollar LIBOR, on the same 25-dollars-per-basis-point scale. Before expiry, the futures rate $100 - F$ is the market's rate for the reference period, adjusted for convexity (A.22).

Why it appears here. §1.4 uses SOFR futures several years out as the case where futures and forward prices differ measurably, and §2.3 lists the contract. §3 records Eurodollar futures as the first cash-settled contract and their replacement by SOFR futures in 2023. §5.2 measures the financing spread in index futures over SOFR.

Deeper. Hull (2021), the chapter on interest rate futures.

A.22 Forwards, the convexity adjustment and tailing the hedge

The idea. A forward is a futures contract without the daily settlement: the whole gain or loss is paid at expiry. If interest rates were fixed, the timing would not matter and the two prices would agree. When rates are random and move with the underlying, daily settlement favours one side. It hands that side its gains when they can be reinvested at high rates, and its losses when they can be financed cheaply. The futures price adjusts to absorb that advantage. For interest-rate futures, whose price is itself a function of rates, the effect is systematic and grows with maturity. Tailing is the hedger's correction for the same timing difference.

Formally. Take the risk-neutral measure Q of A.20 and the stochastic discount factor $D_T = \exp\big(-\int_t^T r_v\,dv\big)$, where r_v is the short-term rate at time v. The futures and forward prices are

$$F_t^{\text{fut}} = E_t^Q[S_T], \qquad F_t^{\text{fwd}} = \frac{E_t^Q[D_T\, S_T]}{E_t^Q[D_T]},$$

$$F_t^{\text{fut}} - F_t^{\text{fwd}} = -\frac{\text{Cov}_t^Q(D_T, S_T)}{E_t^Q[D_T]}.$$

The difference is the covariance between the discount factor and the price at expiry. If S_T tends to be high when rates are high, D_T is low at those times. The covariance is then negative, and the futures price is above the forward. A SOFR future is priced as 100 minus a rate. Its futures price is therefore below the forward price, and the futures rate above the forward rate. In the simple Ho–Lee model of rates, the forward rate equals the futures rate minus $\frac{1}{2}\sigma_r^2 t_1 t_2$. Here σ_r is the annual volatility of the short rate in rate units, and t_1 and t_2 are the times to the start and end of the reference period.

Tailing. A futures gain realised on day t earns interest until the hedge ends at T_h. A hedge that needs the exposure of N contracts at T_h therefore holds $Ne^{-r(T_h - t)}$ contracts on day t.

Why it appears here. §1.4 states the forward–futures difference, the SOFR convexity adjustment and tailing. §12.2 lists forward and futures prices as equal exactly only when rates are not random.

Deeper. Cox, Ingersoll & Ross (1981); Hull (2021), the chapters on interest rate futures and on hedging strategies using futures.

A.23 Bond prices, yields, duration and DV01

The idea. A bond is a schedule of fixed payments: coupons plus a final principal. Its price is the present value of those payments. The yield is the single discount rate that makes that present value equal the market price, so prices and yields move in opposite directions. Duration measures how far the price moves. It is the percentage change in price for a one-unit change in yield, and it equals the payment-weighted average time to the bond's cash flows. Long bonds have high duration and move a lot when yields change. Short bonds move little. DV01 is the same sensitivity in dollars per basis point, which is what a hedger needs to match two positions. The yield curve plots yields against maturity. When the curve slopes up, a bond's yield falls as it ages toward shorter maturities, and its price rises. This is the bond version of roll-down.

Formally. A bond paying C_j at times t_j has price $P = \sum_j C_j e^{-Y t_j}$ at continuously compounded yield Y. (The yield is written Y, not y, because the notation table reserves y for the convenience yield.) Its duration is the present-value-weighted average time to payment:

$$D = -\frac{1}{P}\frac{\partial P}{\partial Y} = \sum_j t_j \frac{C_j e^{-Y t_j}}{P},$$

so that, to first order, $\Delta P/P \approx -D\,\Delta Y$. The **DV01** (dollar value of a basis point) is $DP \times 10^{-4}$ per unit of face value, scaled to the position. Two positions are duration-matched when their DV01s are equal and opposite. When the yield curve slopes up, long yields sit above short ones. A bond's coupon yield then usually exceeds the overnight repo rate, so holding it with repo financing has positive carry (A.19).

Why it appears here. §5.2 uses duration and DV01 to explain why a 10-year note future behaves like a seven-year note, and the slope of the yield curve to sign a bond future's carry. §11.1 adds roll-down along the curve to bond carry.

Deeper. Bonds and Bond Markets in this collection; Tuckman & Serrat (2011).

A.24 Treasury futures: the deliverable basket, conversion factors and the cheapest to deliver

The idea. A Treasury future is not a contract on one bond. The short may deliver any Treasury issue from a list with maturities in a set range. To make different coupons and maturities roughly comparable, each bond's invoice price is the futures price times a **conversion factor**. The factors are computed as if every bond yielded 6%, so they equalise the bonds only when yields are 6%. At any other yield, one bond is the cheapest to deliver (CTD). The short will deliver that bond, so the futures price tracks it. The short's freedom to choose which bond to deliver, and when, has value. The futures price is a little lower to pay for it.

Formally. The **conversion factor** CF_i is the price per unit of face value of bond i at a 6% yield, with its maturity rounded down to whole months or quarters. On delivery the long pays the short $F\,\text{CF}_i$ plus accrued interest per unit of face. The short

delivers the bond that minimises the **gross basis** of §12.3, $P_i - F\,\mathrm{CF}_i$. Since F is common to all bonds, this is approximately the bond with the smallest P_i/CF_i. For a yield Y not too far from 6%, with duration D_i (A.23), $P_i(Y)/\mathrm{CF}_i \approx e^{D_i(0.06-Y)}$, because $\mathrm{CF}_i \approx P_i(6\%)$. Below 6% this ratio is smallest for the shortest-duration bond.

The short holds three **delivery options**:

- the *quality* option: which bond to deliver;
- the *timing* option: which day of the delivery month to deliver on;
- the *wild card* option: the short can give notice after the day's futures settlement price, which sets the invoice, is fixed, while cash bonds still trade.

Prices are quoted in points and thirty-seconds of a point per 100 of face. For example, 112-16 means $112 + 16/32$. The 10-year note's tick of half a thirty-second is worth $100{,}000 \times (0.5/32)/100 = 15.625$ dollars. The **cash–futures basis trade** buys the CTD with repo financing and sells futures. It earns the gap between two rates: the financing rate that the futures price implies (the *implied repo rate*) and the actual repo rate.

Why it appears here. §2.3 lists the contracts and their deliverable baskets. §5.2 explains the CTD, the delivery options and why the 10-year future behaves like a seven-year note. §12.3 defines the Treasury basis.

Deeper. Burghardt & Belton (2005); Bonds and Bond Markets.

A.25 Covered interest parity, the cross-currency basis and the carry trade

The idea. There are two ways to hold euros a year from now. One is to buy euros today and deposit them at the euro rate. The other is to keep dollars at the dollar rate and agree today to buy euros forward. Both start and end in the same place, so the forward rate must make them cost the same. This pins the forward exchange rate to the spot rate and the two interest rates. The relation is called covered interest parity, "covered" because the forward removes the exchange-rate risk. The carry trade does the opposite. It holds the high-rate currency uncovered, earning the rate difference and bearing the exchange-rate risk.

Formally. Take S and F in dollars per unit of foreign currency, and continuously compounded rates $r_{\$}$ and r_{foreign}. Buy $e^{-r_{\text{foreign}}\tau}$ units of foreign currency for $Se^{-r_{\text{foreign}}\tau}$ dollars borrowed at $r_{\$}$. Deposit them, so they grow to one unit at T. At T the dollar loan has grown to $Se^{(r_{\$}-r_{\text{foreign}})\tau}$. No arbitrage then gives

$$F = S\,e^{(r_{\$}-r_{\text{foreign}})\tau},$$

the cost-of-carry relation with $q = r_{\text{foreign}}$. The **cross-currency basis** is the annualised spread that must be added to one of the two interest rates to make this hold at market prices. Its sign depends on the convention. A long futures position in the foreign currency earns carry of $r_{\text{foreign}} - r_{\$}$ if spot does not move. *Uncovered* interest parity is the hypothesis that the high-rate currency is expected to depreciate by exactly the rate differential, so that the carry trade earns nothing on average. The main text records that the carry trade has in fact earned a premium, with crash risk (A.7).

Why it appears here. §5.2 states covered interest parity and the post-2008 basis. §6.4 and §12.2 equate a long currency future with the carry trade, and §11.1 measures currency carry as the rate differential.

Deeper. Du, Tepper & Verdelhan (2018); Hull (2021), the chapter on the determination of forward and futures prices; the carry section of Systematic Trading Strategies.

A.26 Index arbitrage, fair value and the implied financing rate

The idea. An equity index future can be replicated by buying every stock in the index with borrowed money and collecting the dividends. Index arbitrageurs, mostly dealers, do this whenever the future strays far enough from that cost. They sell the expensive side and buy the cheap one. The price at which the trade breaks even is the future's **fair value**. Banks do the arbitrage on their balance sheets, so its cost includes the cost of that balance sheet. This cost rises when regulation makes balance sheet scarce, notably around quarter-ends.

Formally. With dividend yield q and financing rate r, fair value is $F^{\star} = Se^{(r-q)\tau}$. With discrete dividends D_j paid at times t_j, it is $F^{\star} = \big(S - \sum_j D_j e^{-r(t_j - t)}\big)e^{r\tau}$. If F exceeds $F^{\star}$ by more than costs, arbitrageurs buy the stocks and sell futures. If F falls below $F^{\star}$ by more than costs, they sell or short the stocks and buy futures. The **implied financing rate** of §5.2, $r^{\text{impl}} = q + \ln(F/S)/\tau$, is the r that makes the observed F fair. Its spread over SOFR (A.21) measures how much more, or less, futures holders pay than a borrower at the benchmark rate.

Leverage-ratio rules require banks to hold capital against the size of their balance sheets. Where those ratios are measured on quarter-end snapshots, balance sheet is scarcest at quarter-ends. **Dividend futures**, listed on some indices, settle to the dividends the index's constituents pay over a stated period. They let a trader replace an estimated dividend yield q with a traded one.

Why it appears here. §3 credits index arbitrageurs with turning cost of carry into an enforced relation. §5.2 defines the implied financing rate and describes its quarter-end behaviour. §6.2 treats financial futures at fair value, and §11.1 suggests dividend futures for equity carry.

Deeper. Hazelkorn, Moskowitz & Vasudevan (2023); Du, Tepper & Verdelhan (2018), on quarter-end balance-sheet effects; Hull (2021), the chapter on the determination of forward and futures prices.

A.27 Options, options on futures and the Black model

The idea. An option is the right, but not the obligation, to buy (a call) or sell (a put) at a fixed price, the strike, on or before a date. The holder exercises only when exercise pays, so the payoff is one-sided. The buyer pays a premium up front and can lose no more than that premium. The seller can lose a great deal. An option's value depends on how much the underlying is likely to move, so option prices encode the market's volatility expectations. Options are compared by their implied volatility: the volatility that reproduces each option's price. An option on a futures contract delivers a futures position on exercise, and Black's 1976 model prices it.

Formally. A European call on a futures price pays $(F_T - K)^+$ at expiry for strike K, and a put pays $(K - F_T)^+$. Suppose F follows a geometric Brownian motion with volatility σ (here the model's assumed volatility, not the notation table's estimate). The Black (1976) call and put prices C and P are

$$C = e^{-r\tau}\big[F\,\Phi(d_1) - K\,\Phi(d_2)\big], \qquad P = e^{-r\tau}\big[K\,\Phi(-d_2) - F\,\Phi(-d_1)\big],$$
$$d_1 = \frac{\ln(F/K) + \tfrac{1}{2}\sigma^2\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau},$$

where Φ is the standard normal distribution function. The **implied volatility** is the σ that makes the formula equal an option's market price. A futures contract has a linear payoff, $F_T - F_t$ per unit, while an option's payoff is convex.

Why it appears here. §1.7 contrasts futures with options. §3 and §4.1 credit Black (1976) with pricing options on futures. §5.1 compares futures replication with option replication. §5.2 and §5.4 rest on the VIX, which is computed from option prices (A.37).

Deeper. Black (1976); Hull (2021), the chapter on futures options and Black's model; Implied Volatility and Dealer Hedging and Gamma Exposure in this collection.

Part IV — Commodities, volatility and premia. This part covers the economics of storable and non-storable underlyings, and the physical details of the main commodity markets. It also covers the risk premia that the literature argues over, and the index products whose flows move prices.

A.28 The theory of storage: convenience yield and full carry

The idea. For a storable commodity, the gap between a later futures month and an earlier one is the price of storing the commodity between the two dates. When inventories are ample, the gap equals the full cost of carrying the commodity: financing plus warehousing. A larger gap would let anyone buy now, store and sell forward at a profit. When inventories are scarce, holders value having the commodity on hand, for example to keep a refinery running or to meet an order. They value it more than any futures contract can compensate, so they will not sell it to capture a futures discount. That value is the convenience yield. It rises steeply as inventories run down, and it is what turns a curve from contango into backwardation. Working (1949) called the spread "the price of storage". Kaldor (1939) and Brennan (1958) supplied the convenience yield.

Formally. Let Z be the level of inventories. The theory posits a marginal convenience yield $y(Z)$ that is decreasing and convex in Z. It is near zero when stocks are ample and rises without bound as stocks approach the minimum needed to operate. With $F_t(T) = S_t e^{(r+u-y)\tau}$, the curve sits at **full carry**, $F_t(T) = S_t e^{(r+u)\tau}$, when $y \approx 0$. It slopes down when $y > r + u$. Between two contracts, the spread as a **fraction of full carry** is

$$\frac{F^{(2)} - F^{(1)}}{F^{(1)}\big(e^{(r+u)(T_2 - T_1)} - 1\big)}.$$

The fraction is near one when storage is ample, and negative in backwardation. Full carry is an upper bound only while storage has spare capacity at a stable cost. When storage fills, its marginal cost rises, and u stops being a constant. The spread can then exceed any normal full-carry figure, a state called **super-contango**.

Why it appears here. §3 introduces the theory and §5.1 defines the convenience yield. §5.2 and §5.3 use full carry, and spreads as a fraction of it, as an inventory gauge. §5.6 describes super-contango and negative prices. §11.4 calls a calendar spread a position on inventories.

Deeper. Working (1949); Pindyck (2001); Gorton, Hayashi & Rouwenhorst (2013).

A.29 Physical underlyings: grades, delivery points and units

The idea. A physically delivered contract specifies not just a quantity but exactly what may be delivered, where, and in what unit. Those details decide what its price means. A crude oil futures price is the price of a particular grade of oil at a particular tank farm. A natural gas price is the price of gas at a particular pipeline junction. When the delivery point has its own problem, as Cushing did in April 2020, the futures price reflects that problem, not the world price of the commodity.

Formally.

- **Crude oil. WTI** (West Texas Intermediate) is a light (low-density), sweet (low-sulphur) US crude. The NYMEX contract delivers 1,000 barrels of it, or of specified alternative grades, at **Cushing, Oklahoma**, a hub of pipelines and storage tanks.
- **Natural gas. Henry Hub** is a pipeline interconnection in Louisiana and the delivery point of NYMEX natural gas. Prices are quoted in dollars per **MMBtu**, a million British thermal units, which measures energy content.
- **Precious metals** are measured in **troy ounces** of about 31.1 grams. Delivered bars must meet a stated purity and bar size, and must come from an approved refiner.
- **Grains** are measured in **bushels**, a volume unit converted to a standard weight for each grain, with premiums and discounts for grade and delivery location.
- **Base metals** on the LME are measured in tonnes and delivered by warrant at approved warehouses.

Why it appears here. §1.2 and §2.1 cite delivery at Cushing, and §2.3 lists the units. §5.6 and §14.2 explain negative oil as a storage problem at Cushing. §7.2 times rolls by each market's delivery rules.

Deeper. CFTC staff (2020), for Cushing in practice; Geman (2005), for market-by-market detail.

A.30 Seasonal markets: gas storage, shoulder seasons and crop years

The idea. In some markets the calendar month is part of the commodity. Natural gas demand peaks in winter, and storage can move gas from summer to winter only as fast as it can be pumped in and out. Winter contracts therefore carry a premium every year that storage cannot arbitrage away. Grain is harvested once a year. Storage can carry old grain forward into the new crop year, but it cannot carry the coming harvest back, so the months on either side of a harvest are linked in one direction only. In both markets, the gap between adjacent months mixes carry with a seasonal shape that repeats every year.

Formally. US gas storage is filled in the *injection* season, roughly April to October, and drawn down in the *withdrawal* season, roughly November to March. Both are subject to limits on total capacity and on daily injection and withdrawal rates. The arbitrage bound of A.28 therefore links a summer month to a winter month only up to the volume that storage can move. The winter premium can then exceed the cost of storage. The **shoulder seasons**, spring and autumn, fall between heating and cooling demand, when demand is lowest. In grains, the **old crop** is the stock from the last harvest and the **new crop** is the coming one. For US corn, harvested in the autumn, July is conventionally the last old-crop contract and December the first new-crop one. A July–December spread therefore prices the coming harvest against existing stocks, not a storage cost. Carry in such markets is measured between contracts a year apart, or as the deviation of the adjacent spread from its seasonal average.

Why it appears here. §5.2 explains the seasonality of the gas curve, and §5.5 the distinction between new crop and old crop. §6.5 notes gas's steep negative carry in the shoulder seasons. §7.3, §11.2 and §14.2 draw the consequences for contract choice, carry measurement and Amaranth's spreads.

Deeper. Geman (2005); Pindyck (2001).

A.31 The gold lease rate

The idea. Gold pays no dividend, but a holder can lend it and earn a fee. The borrowers are dealers, jewellers or miners who need metal now. That fee, the lease rate, plays the part of an income yield in gold's cost of carry. Gold is held overwhelmingly

as an investment, and the stock in vaults is enormous relative to industrial need. The lease rate is therefore usually small, convenience is negligible, and the futures curve sits close to full carry.

Formally. With lease rate ℓ (continuously compounded), the cost-of-carry relation for gold is

$$F_t(T) = S_t\, e^{(r+u-\ell)\tau},$$

so ℓ enters as q does for an index, and u is small. The futures curve therefore implies a lease rate, $\ell = r + u - \ln(F/S)/\tau$. The bullion market used to quote the gold forward offered rate (GOFO). GOFO is a dollar interest rate minus the lease rate, which is the same relation read the other way. A rise in the lease rate signals that metal is scarce for immediate use, and it flattens the curve.

Why it appears here. §5.2 lists the lease rate as gold's income yield and uses it to explain gold's near-full-carry curve. §5.3 cites gold as a curve in contango that could still pay a premium in Keynes's sense.

Deeper. Hull (2021), the chapter on the determination of forward and futures prices; Geman (2005).

A.32 Real options and backwardation in oil

The idea. A real option is an option embedded in a physical decision rather than in a financial contract. An oil producer with oil in the ground owns one: it can pump now or later. Waiting keeps the choice open. An open choice is worth more when the price is more uncertain, just as an option's value rises with volatility. A producer therefore pumps now only if selling today beats the discounted futures price by enough to give up that option. For oil to keep flowing, spot must therefore sit above the discounted futures price. This gives a structural reason for backwardation, independent of hedging pressure.

Formally. Ignore extraction costs. A barrel produced now is worth S_t. A barrel kept for production at T and sold forward is worth $e^{-r\tau}F_t(T)$ today. Keeping the barrel also keeps the option to choose later, worth V_t, which increases with the volatility of the price. Producing now must beat both:

$$S_t - e^{-r\tau}F_t(T) \;\geq\; V_t \;>\; 0.$$

So in an equilibrium with positive production, the curve is in at least *weak* backwardation, with discounted futures below spot. The backwardation is steeper when volatility is high.

Why it appears here. §4.1 and §5.2 cite Litzenberger and Rabinowitz's explanation of why crude oil has spent long periods in backwardation.

Deeper. Litzenberger & Rabinowitz (1995); A. K. Dixit & R. S. Pindyck, *Investment under Uncertainty* (Princeton University Press, 1994).

A.33 Normal backwardation, hedging pressure and the Commitments of Traders

The idea. Keynes argued as follows. Producers of a commodity want to lock in a price for what they will sell, so they sell futures. Consumers hedge less. Speculators must take up the difference, and they will do so only if they expect to profit. The futures price must then sit below the expected spot price, and longs earn a premium for providing insurance. The imbalance between hedgers' shorts and longs is called hedging pressure, and it should drive the premium. Measuring it requires data on who holds what. The US regulator publishes such data weekly.

Formally. Normal backwardation means $\pi_t(T) > 0$ in $F_t(T) = E_t[S_T] - \pi_t(T)$ (§5.4). It is a statement about expected returns, and it cannot be read off the curve. **Hedging pressure** in a market is commonly measured as hedgers' net short position, as a fraction of their total position:

$$\text{HP} = \frac{\text{hedgers' short positions} - \text{hedgers' long positions}}{\text{hedgers' short positions} + \text{hedgers' long positions}}.$$

Positive values mean hedgers are net short, and by Keynes's argument longs should then be paid. The CFTC's **Commitments of Traders** (COT) reports split open interest each week, as of Tuesday, by type of trader:

- the legacy report: *commercial* traders, presumed to be hedgers, and *non-commercial* ones, presumed to be speculators;
- the disaggregated report: producers, merchants, processors and users; swap dealers; managed money; and other reportables.

Traders classify themselves as commercial, so the split between hedgers and speculators is approximate.

Why it appears here. §3 and §4.1 introduce Keynes's theory, and §5.3 separates his sense of backwardation from the curve's. §4.7 and §5.4 weigh the evidence for hedging pressure from Bessembinder (1992) and de Roon, Nijman and Veld (2000).

Deeper. Bessembinder (1992); de Roon, Nijman & Veld (2000); Rouwenhorst & Tang (2012); the CFTC's explanatory notes to the COT reports.

A.34 Futures risk premia: spot, term, insurance and liquidity

The idea. A risk premium is the return a position is expected to earn for bearing risk, over and above what cash earns. Futures returns are already excess returns, so for futures the premium is simply the expected return. The main text splits it in two ways.

- *By the risk borne.* The *spot* premium pays for exposure to the spot price. The *term* premium pays for exposure to changes in the slope of the curve, the extra risk that a position in a longer contract adds.
- *By who pays whom, and over what horizon.* Over long horizons, hedgers pay speculators an *insurance* premium for taking price risk. Over days and weeks, speculators who trade with momentum pay hedgers a *liquidity* premium for taking the other side of their trades.

Formally. In the notation of §5.4 the premium is $\pi_t(T) = E_t[S_T] - F_t(T)$ in price units, or $\pi_t(T)/F_t(T)$ as an expected return to expiry. An *unbiased* futures price has $\pi = 0$. In log terms, write $s_t = \ln S_t$ and $f_t^{(n)}$ for the log price of a contract with n periods to expiry. Holding the one-period contract to expiry earns the **spot premium** $E_t[s_{t+1}] - f_t^{(1)}$. Holding an n-period contract for one period earns $E_t[f_{t+1}^{(n-1)}] - f_t^{(n)}$. The excess of that over the spot premium is the **term premium** for maturity n, compensation for bearing changes in the basis. Horizon tells the insurance and liquidity premia apart. The insurance premium shows up in returns over months, and the liquidity premium over days and weeks.

Why it appears here. §5.4 defines the premium and reports Kang, Rouwenhorst and Tang's reconciliation of the two theories by horizon. §6.4 and §11.3 report Szymanowska et al.'s spot and term premia from sorting on the basis.

Deeper. Szymanowska, de Roon, Nijman & van den Goorbergh (2014); Kang, Rouwenhorst & Tang (2020); Fama & French (1987).

A.35 The Samuelson effect and factor models of the curve

The idea. Front commodity contracts are more volatile than deferred ones, because of mean reversion. A shock to today's supply or demand moves today's scarcity a lot. But traders expect inventories, production and consumption to adjust. They therefore expect the shock to fade by later delivery dates, and later contracts move less. Models of the term structure make this precise by treating the log spot price, or the convenience yield, as a variable pulled back toward a long-run level. The speed of that pull sets how fast volatility decays along the curve.

Formally. In the one-factor model of Schwartz (1997), the log spot price $x_t = \ln S_t$ follows $dx_t = \theta(\bar{x} - x_t)\,dt + \sigma_S\,dW_t$. This is an Ornstein–Uhlenbeck process with mean-reversion speed θ, long-run level $\bar{x}$ and volatility σ_S, where W is a Brownian motion. The log futures price is then linear in x_t, with coefficient $e^{-\theta\tau}$. Its volatility is therefore the spot volatility damped by that coefficient:

$$\text{sd}\left(d \ln F_t(T)\right) = \sigma_S\, e^{-\theta\tau}\sqrt{dt}.$$

Volatility falls with time to expiry: this is the **Samuelson effect**. Without mean reversion ($\theta = 0$), every contract has the spot's volatility, as in financial futures. Gibson and Schwartz (1990) add a second factor, a mean-reverting convenience yield correlated with the spot price. Schwartz (1997) adds a third that makes the interest rate random. These models fit the whole curve with a few state variables. They are the basis for valuing longer-dated commodity claims.

Why it appears here. §4.1 introduces Samuelson, Gibson and Schwartz, and Schwartz, and §5.5 describes volatility by maturity. §7.3 notes that under volatility sizing, deferred contracts' lower volatility changes the contract count.

Deeper. Samuelson (1965); Schwartz (1997); Gibson & Schwartz (1990).

A.36 Squeezes and corners

The idea. A short in a physically delivered contract must either deliver or buy the contract back before expiry. Suppose one trader, or a few acting together, holds a large long position and controls much of the deliverable supply. The shorts then have nowhere to get the goods. They must buy back at whatever price the longs demand. This is a **corner**. A milder version, in which deliverable supply is merely tight, is a **squeeze**. The expiring contract's price rises relative to later months and to the physical market elsewhere. Calendar spreads can then move many times their usual range.

Formally. In the language of A.28, a squeeze is a convenience yield specific to one delivery period. The front contract's implied y jumps while later months' do not, so $F^{(1)} - F^{(2)}$ moves deep into backwardation for reasons that belong to a single expiry. Exchanges and regulators limit the risk in three ways:

- **position limits** and accountability levels in the expiring month;
- delivery rules that widen the deliverable supply through alternative grades and locations;
- emergency powers to order liquidation or change margins.

Why it appears here. §9.6 and §11.4 say to size calendar spreads by how they behave in a squeeze, not by their margin. §11.5 warns that a front contract squeezed near expiry shows a carry that reflects one contract's delivery problem.

Deeper. Geman (2005); the CFTC's rules on speculative position limits.

A.37 The VIX and the volatility risk premium

The idea. The VIX is a number, not an asset. Cboe computes it continuously through the trading day from the prices of S&P 500 options, as a measure of the volatility the options market implies for the next 30 days. Nobody can buy the VIX, so no cash-and-carry trade ties VIX futures to it. Each future is a bet on where the index will be on its expiry day. On average, implied volatility has exceeded the volatility that followed, because option buyers pay for protection against crashes. That difference is the volatility risk premium. It is why the VIX futures curve usually slopes up. It is also why sellers of VIX futures usually make money, until the day they lose a great deal. Selling volatility is selling insurance.

Formally. The VIX is $100\sqrt{V}$, where V is a weighted sum of out-of-the-money put and call prices across strikes K_i, with weights proportional to $1/K_i^2$. V estimates the risk-neutral expected variance of the S&P 500's log return over the next 30 days, annualised. The **variance risk premium** is defined here as $E_t^Q[\text{realised variance}] - E_t[\text{realised variance}]$, the risk-neutral expectation (A.20) minus the real-world one. It is positive on average. Some authors use the opposite sign. A VIX future's price is $F_t(T) = E_t[\text{VIX}_T] - \pi_t(T)$, with $\pi_t(T) < 0$ in the usual case. The futures price then exceeds the expected VIX, and longs pay the premium. **Constant-maturity** VIX products hold a mix of adjacent futures, rebalanced daily so that their average time to expiry stays fixed. That time is about a month for short-term products, and longer for mid-term ones. They pay the roll-down of whichever part of the curve they hold.

Why it appears here. §1.1 and §5.2 make the VIX the case where the futures price is an expectation plus a premium. §5.4, §6.4 and §6.5 report the premium's size and the short position's appeal. §7.3 and §8.3 discuss constant-maturity products, and §14.2 describes the collapse of the inverse VIX note.

Deeper. Implied Volatility in this collection; Simon & Campasano (2014); Eraker & Wu (2017); P. Carr & L. Wu, "Variance Risk Premiums," *Review of Financial Studies* 22(3) (2009), 1311–1341.

A.38 Bitcoin reference rates, perpetual swaps and funding

The idea. Bitcoin trades on many unregulated spot exchanges at slightly different prices. A cash-settled future therefore needs one reference price that no single exchange can move. CME's contracts settle to a benchmark computed once a day from trades on several large exchanges. Most crypto leverage, though, sits in perpetual swaps. These are futures-like contracts with no expiry, traded on crypto exchanges. With no expiry to force convergence, they are held near spot by periodic payments between longs and shorts.

Formally. The **CME CF Bitcoin Reference Rate** (BRR) is a once-a-day dollar price of bitcoin, calculated by CF Benchmarks from trades on a set of constituent spot exchanges during a fixed one-hour window in the London afternoon. Its method is designed to dampen any one exchange's outliers, and CME bitcoin futures settle to it at expiry. A **perpetual swap** makes a **funding payment** at the end of every funding interval, commonly eight hours. The payment is proportional to the gap between the perpetual's price and a spot index, plus a small interest component. When the perpetual trades above spot, longs pay shorts. The annualised funding rate is the perpetual's analogue of the annualised basis: a long perpetual position earns minus the funding rate as carry.

Why it appears here. §2.3 notes that bitcoin futures settle to a reference rate. §4.3 and §5.2 report the large and variable basis of futures and perpetuals over spot. §11.1 measures bitcoin carry from the basis or from the funding rate.

Deeper. Schmeling, Schrimpf & Todorov (2023); CF Benchmarks' published methodology for the BRR.

A.39 Commodity indices and their rolls

The idea. A commodity index is a rule for holding a basket of futures. It says which commodities to hold, in what weights, which contract of each, and when to roll. Funds that track an index hold its futures and roll them on its schedule. The index's rules therefore become real flows in the market. Each index comes in three versions that answer different questions: how the futures prices moved, how a futures position did, and how a fully collateralised position did.

Formally. The three versions are:

- The **spot index** tracks the price level of the nearby contracts and ignores the roll. It is the unadjusted series of §8.3, aggregated across commodities.
- The **excess-return index** compounds the returns of the futures positions actually held, rolling on the index's schedule. It is a return-stitched series (§8.3).
- The **total-return index** adds interest on Treasury bills equal to the notional: $\mathrm{TR}_t/\mathrm{TR}_{t-1} = \mathrm{ER}_t/\mathrm{ER}_{t-1} + r^{\text{bill}}\Delta$.

The main indices differ in weights and roll rules. The **S&P GSCI** weights commodities by world production, which makes it heavily weighted to energy. It rolls from the fifth to the ninth business day of each month, a fifth each day: the "Goldman roll" of §7.4. The **Bloomberg Commodity Index** weights by liquidity and production, with caps on any one sector. The **DBIQ Optimum Yield** indices choose, at each roll, the contract with the most favourable implied roll yield (A.49). The "roll yield" reported in index fact sheets is usually the excess-return index's return minus the spot index's.

Why it appears here. §3 and §4.4 introduce index funds and the financialisation debate, and §6.1 defines excess-return and total-return indices. §7.3 describes optimised rolling, and §7.4 the Goldman roll and the traders who trade ahead of it.

Deeper. Gorton & Rouwenhorst (2006); Mou (2011); the index providers' methodology documents.

A.40 Financialisation, predictable flows and price impact

The idea. From the early 2000s, investors who wanted commodity exposure bought it through index funds that held rolling long futures positions. These funds were large and predictable, which raised two questions. Did their buying push futures prices away from fundamentals? And did their fixed roll schedules let other traders profit by trading ahead of them? Both questions turn on two properties of a market. Price impact is how far a trade moves the price against the trader. Resilience is how fast the price recovers once the trade is done.

Formally. An order to buy X contracts moves the price by an impact $\mathcal{I}(X)$, which increases with X and is larger the less liquid the market. Impact has a *permanent* part, from the information the trade reveals or a lasting shift in supply and demand. It also has a *transient* part that decays after the trade, and **resilience** is the speed of that decay. A trader who knows a large order is coming can buy first and then sell to it. That trader captures part of the order's transient impact, and the order's cost rises by the same amount. A flow that is large relative to the market and fixed in its timing is the easiest to trade against. Spreading it over more days, or varying its timing, reduces the transient impact it pays. **Financialisation** names the broader hypothesis that index investors changed how commodity prices behave, for instance by raising their correlation with each other and with other assets.

Why it appears here. §3 and §4.4 introduce the debate, and §4.7 summarises the opposing findings of Tang and Xiong, Mou, and Irwin and Sanders. §7.4 applies it to the Goldman roll and the US Oil Fund's rolls, and draws the practical rule: roll outside the large funds' windows.

Deeper. Mou (2011); Bessembinder, Carrion, Tuttle & Venkataraman (2016); Tang & Xiong (2012); Irwin & Sanders (2011).

Part V — Risk, margin and sizing. This part covers how exchanges compute the margin they demand, and how leverage and margin are reported. It also covers the volatility-based sizing that the main text uses in place of sizing by margin.

A.41 Value at risk

The idea. Value at risk (VaR) answers one question: how much could this position lose over a given horizon, in all but the worst small fraction of cases? A one-day 99% VaR of 10,000 dollars says that a loss larger than 10,000 dollars should happen on about one day in a hundred. It says nothing about how large the loss is on that day, which is its best-known weakness. Exchanges use VaR because it gives one number per position and per portfolio. US rules also set futures margin as a VaR.

Formally. For a loss $\mathcal{L}$ over a horizon, $\text{VaR}_\alpha = \inf\{\ell : P(\mathcal{L} \le \ell) \ge \alpha\}$, the α-quantile of the loss distribution. Suppose a position with notional $|N|mF_t$ has a daily return that is normal with mean zero and standard deviation σ_d (A.4). Its one-day 99% VaR is then 2.33 standard deviations of its dollar return:

$$\text{VaR}_{0.99}^{\text{1 day}} = z_{0.99}\,\sigma_d\,|N|mF_t, \qquad z_{0.99} = \Phi^{-1}(0.99) \approx 2.33,$$

so 1% daily volatility gives a one-day 99% VaR of 2.33% of notional. With independent days, an n-day VaR is about $\sqrt{n}$ times the one-day figure. *Historical* VaR drops the normal assumption. It applies each day of a historical window to today's positions, and takes the empirical quantile of the resulting losses. The **liquidation period** is the horizon over which a defaulter's positions are assumed to be closed out. Exchange models add two components to the VaR. A **stress** component covers losses in chosen historical or hypothetical crises. A **liquidity** or concentration component adds margin for positions too large to close within the liquidation period.

Why it appears here. §3 and §9.2 describe SPAN 2 as a historical VaR with stress and liquidity add-ons, and §9.2 compares margin with a normal one-day 99% loss. §9.5 and §13.4 size the cash buffer at three times that loss. §12.2 equates exchange margin with a one-day 99% VaR plus add-ons.

Deeper. Value at Risk in this collection; Hull (2021), the chapter on value at risk and expected shortfall; 17 CFR 39.13.

A.42 SPAN, SPAN 2 and the rules for exchange margin

The idea. A clearing house needs a margin rule that is fast, applies to whole portfolios, and gives credit where positions offset. For decades CME's answer was SPAN. It revalues each position under a fixed set of price and volatility scenarios, charges the worst loss, and gives credits for spreads. Its replacement, SPAN 2, is a historical value-at-risk model. Both respond to recent volatility, which makes margin procyclical. Margin rises after a large move, when traders have already lost money, and it can force selling that makes the move worse.

Formally. SPAN (Standard Portfolio Analysis of Risk, CME, 1988) builds, for each product, a *risk array*: the position's loss under 16 scenarios. The scenarios are price moves of up to a full "price scan range" in each direction, each combined with volatility up and down, plus two extreme moves. The *scanning risk* is the worst of the 16. On top of it SPAN makes three adjustments:

- it adds an intra-commodity spread charge, because a calendar spread is not riskless;
- it subtracts an inter-commodity spread credit for related products, such as 5-year and 10-year Treasury futures;
- it adds a delivery-month charge for positions near expiry.

SPAN 2 replaces the scenario grid with a historical VaR of the portfolio (A.41), plus stress, liquidity and concentration components. Under 17 CFR 39.13, a derivatives clearing organization (A.10) must set initial margin to cover at least 99% of price changes over a liquidation period of at least one day for futures. **Procyclicality** is the tendency of margin to rise with realised volatility. The usual dampers are long look-back windows, stress periods kept in the sample, and margin floors.

Why it appears here. §3 records the move to SPAN 2, and §9.2 describes both models and the regulatory minimum. §9.4 and §14.2 cover the silver and nickel margin spirals. §9.6 covers the spread credits that make spreads cheap to margin.

Deeper. CME, SPAN 2 framework rollout; 17 CFR 39.13.

A.43 Leverage, margin-to-equity and managed-futures funds

The idea. Leverage is how much exposure an account carries per dollar of equity. For a single position the main text measures it by notional. That measure is natural, but it misleads across asset classes. A dollar of 2-year Treasury note futures and a dollar of crude oil futures carry very different risk. Professional futures managers therefore report risk-based measures. As a summary of how stretched an account is, they also report the share of equity committed as margin. Most of these managers are registered as **commodity trading advisors** (CTAs), the US category for advisers who trade futures for clients. The funds they run are called managed-futures funds, and many follow trends.

Formally. For positions N_i in markets i, gross notional leverage is $L = \sum_i |N_i|\, m_i F_i / A$, which generalises the notation table's single-market L. Risk-weighted exposure replaces notional with annual dollar volatility, $\sum_i |N_i|\, m_i F_i \sigma_i / A$. This is the sum of the positions' risk budgets before diversification (A.44). With $M_i^!$ the initial margin per contract in market i (as in A.12), **margin-to-equity** is the total initial margin as a fraction of equity:

$$\frac{\sum_i |N_i|\, M_i^!}{A}.$$

For a single market it equals L times the initial margin rate $M^{\mathrm{I}}/(mF)$. A doubling of margin rates doubles it. An account at 50% margin-to-equity is forced to cut positions when margins double. An account at 25% is not.

Why it appears here. §1.1 and §9.3 define leverage as a choice. §9.3 reports managed-futures funds' typical margin-to-equity, and sets the rule of keeping margin below a quarter to a third of equity. §15.1 restates the liquidation distance in terms of L.

Deeper. Carver (2015) and Carver (2023), on leverage and margin in futures portfolios.

A.44 Volatility targeting, risk budgets and the diversification multiplier

The idea. Volatility targeting decides how much risk to take, rather than how many contracts to hold, and lets the contract count follow. An account picks an annual volatility it is willing to bear and divides that budget among markets. In each market it holds as many contracts as it takes to use that market's share. When a market becomes more volatile, its position shrinks. When it calms, the position grows. Markets do not move in lockstep, so positions sized this way add up to less than the target. A multiplier scales them back up.

Formally. With target $\sigma^\star$, risk share w_i, and one contract's annual dollar risk $m_i F_i \sigma_i$, the ideal position is $N_i^\star = A\sigma^\star w_i/(m_i F_i \sigma_i)$ (§10.2). If every position sat exactly at its budget, the portfolio's annual dollar volatility would be $A\sigma^\star\sqrt{\mathbf{w}^\top \mathbf{P}\,\mathbf{w}}$. Here $\mathbf{w}$ is the vector of risk shares and $\mathbf{P}$ the correlation matrix of the markets' returns. The **diversification multiplier**

$$\mathrm{DM} = \frac{1}{\sqrt{\mathbf{w}^\top \mathbf{P}\,\mathbf{w}}}$$

restores the target. It is the reciprocal of the fraction of risk that survives diversification. With n equal shares and average correlation $\bar\rho$ it is $1/\sqrt{1/n + (1-1/n)\bar\rho}$ (A.5). Practitioners estimate $\mathbf{P}$ conservatively and cap the multiplier, because correlations rise in crises. An overstated multiplier overstates every position. The signal s_i of §12.1 scales each position within its budget.

Why it appears here. The notation table defines $\sigma^\star$, w and $N^\star$. §10.2 derives the sizing formula, §10.3 the multiplier and the minimum account, and §12.1 builds the master formula on both.

Deeper. Carver (2015); Systematic Trading Strategies, which works through the minimum account.

A.45 Turnover and no-trade buffers

The idea. A strategy that trades every time its ideal position changes pays costs on every small change, and most of those changes reverse. Turnover measures how much a strategy trades. When costs are proportional to the amount traded, the best policy does not chase the ideal. It leaves the position alone while the position stays within a band around the ideal. When the position leaves the band, the policy trades only back to the band's edge. The band accepts a small tracking error in exchange for a large saving in costs. It matters most for small positions, where rounding to whole contracts makes the target flicker across a boundary.

Formally. Annual turnover can be measured as the year's sum of $|N_t - N_{t-1}|\,mF_t$, divided by average exposure or by equity. Annual cost is turnover times the cost per unit traded (§2.6). The buffered rule of §10.4 sets the band $\big[\,\mathrm{round}(N^\star - b),\ \mathrm{round}(N^\star + b)\,\big]$, where b is the buffer's half-width. If the position is outside the band, the rule moves it to the nearer edge. For a single asset with proportional costs, a classical result in portfolio choice shows that the optimal policy takes exactly this no-trade-region form (Davis and Norman, 1990). Smoothing a signal before sizing, as §11.1 does for carry, reduces turnover by the same mechanism.

Why it appears here. §10.4 and §13.3 implement the buffer. §11.1 reports that smoothing carry roughly halves turnover. §12.1 writes rounding with a buffer into the master formula.

Deeper. Carver (2015); M. H. A. Davis & A. R. Norman, "Portfolio Selection with Transaction Costs," *Mathematics of Operations Research* 15(4) (1990), 676–713.

A.46 Crowding and carry unwinds

The idea. A strategy that many investors hold with leverage carries a risk that its own history does not show: the risk that they all exit at once. Carry trades are the classic case. They earn small amounts steadily, so they attract capital and leverage. When a shock causes losses, margin calls and risk limits force the most leveraged holders to sell. Their selling moves prices

against everyone else in the same trade, and the losses spread. "Up by the stairs and down by the elevator" describes the resulting pattern of slow gains and sudden losses, a negatively skewed distribution (A.7).

Formally. The mechanism is a liquidity spiral, which runs in four steps:

1. A loss reduces holders' equity.
2. Margin requirements, which often rise with volatility (A.42), require positions to shrink by more than the loss alone would.
3. Many traders hold the same position, so the forced sales have price impact (A.40).
4. The price impact produces further losses, and the cycle repeats.

The spiral makes the strategy's losses correlated across markets and across firms, precisely in crises. A volatility estimate from calm periods does not capture this.

Why it appears here. §11.3 describes currency carry crashes, and §11.5 lists crowding among the ways carry signals fail. §7.4's advice not to roll with the crowd rests on the same price-impact mechanism.

Deeper. Brunnermeier, Nagel & Pedersen (2008); M. K. Brunnermeier & L. H. Pedersen, "Market Liquidity and Funding Liquidity," *Review of Financial Studies* 22(6) (2009), 2201–2238.

Part VI — Funds, data, backtests and signals. This part covers the funds that an individual weighs against futures. It also covers the vocabulary of building series, testing strategies, and reading the signals that the main text mentions.

A.47 Exchange-traded products: ETFs, ETNs, and leveraged and inverse funds

The idea. There are three kinds of product to keep apart.

- An **exchange-traded fund** (ETF) holds assets, or futures, and trades on an exchange like a share. Large dealers can swap shares for the underlying holdings and back, which keeps the price close to the value of what the fund holds. The fund charges an annual fee.
- An **exchange-traded note** (ETN) looks the same on a screen but holds nothing for its investors. It is an unsecured debt of the issuing bank, promising the return of an index minus a fee. Its holders bear the issuer's credit risk and its terms, including the right to redeem the note early.
- **Leveraged and inverse funds** promise a multiple of an index's *daily* return. Over longer periods their return depends on the path the index took, not just on where it ended.

Formally. The **expense ratio** is the annual fee as a fraction of assets, deducted continuously from the fund's value. An ETN's **indicative value** is the issuer's or exchange's running estimate of what the note is worth under its formula. Prospectuses typically give the issuer the right to **accelerate**, or redeem, the note at that value after specified events.

Take a fund that resets daily to β times the index's return. The index has drift a and volatility σ, and rebalancing is continuous. The fund's log growth is $\beta a - \frac{1}{2}\beta^2\sigma^2$. By comparison, β times the index's own log growth is $\beta(a - \frac{1}{2}\sigma^2)$. The gap, $-\frac{1}{2}(\beta^2 - \beta)\sigma^2$ a year, is the **volatility decay**. For an inverse fund ($\beta = -1$) it is $-\sigma^2$, and for a twice-levered fund it is also $-\sigma^2$. A fund that holds commodity futures also earns the futures' carry, so in contango it lags spot by the roll-down.

Why it appears here. §1.7 and §10.5 compare futures with ETFs, and §10.5 mentions the decay of inverse funds. §14.2's inverse VIX note was an ETN that the issuer redeemed after its indicative value collapsed.

Deeper. M. Avellaneda & S. Zhang, "Path-Dependence of Leveraged ETF Returns," *SIAM Journal on Financial Mathematics* 1 (2010), 586–603; the appendix of Systematic Trading Strategies.

A.48 US tax treatment: Section 1256 and partnership funds

The idea. For a US taxpayer, regulated futures contracts are taxed under a special regime that ignores how long a position was held. Gains are taxed as if mostly long-term, which is favourable for anything held less than a year. Open positions are treated as sold at the year-end price, so the tax cannot be deferred. Some commodity funds that hold futures are organised as partnerships. They pass the futures' tax treatment through to their holders, on a partnership form rather than the usual one.

Formally. Section 1256 of the Internal Revenue Code (26 U.S.C. 1256) covers regulated futures contracts. Gains and losses on them are treated as 60% long-term and 40% short-term capital gains, whatever the holding period. Every open position is **marked to market** at its fair value on the last business day of the tax year. The gain or loss is recognised then, and the cost basis is reset. A commodity pool organised as a publicly traded partnership reports each holder's share of its income

on **Schedule K-1** rather than Form 1099. The holder is taxed on that share each year, whether or not anything is distributed. Equity ETFs, by contrast, follow the ordinary holding-period rules.

Why it appears here. The chapter's opening paragraph calls futures often the most tax-efficient choice for a US taxpayer. §10.5 lists the 60/40 treatment and partnership forms in its comparison of micros, standard contracts and funds.

Deeper. 26 U.S.C. 1256; IRS Publication 550; the tax section of Systematic Trading Strategies.

A.49 Roll-down, roll yield and the roll gap

The idea. Three different quantities go by similar names. Confusing them is the root of the folk account of contango.

- **Roll-down** is the return a contract earns as it ages along an unchanged curve, sliding from its current maturity toward a shorter one. It is realised carry.
- The **roll gap** is the price difference between the old and new contract on the day of a roll. It is not a return at all, because the roll swaps one zero-value position for another.
- Index providers' **"roll yield"** is a residual, defined below.

A related distinction sits underneath the continuous series of §8: whether a position keeps its contract count or its notional at each roll.

Formally. With the curve unchanged in time-to-expiry terms, a contract held for Δ years earns roll-down of $\kappa_t \Delta$ (§6.2). The *implied roll yield* of contract j is $(\ln F^{(j)} - \ln F^{(j+1)})/(T_{j+1} - T_j)$, the annualised slope of the curve next to it. The roll gap at a roll from the first contract to the second is $F^{(2)} - F^{(1)}$, a difference between two prices on the same date. **Index roll yield** is the excess-return index's return minus the spot index's (A.39), and it includes curve shocks. Positions differ in what they hold constant at a roll:

- A **constant-count** position keeps N contracts through the roll. Its profit is tracked by the back-adjusted series.
- A **constant-notional** position buys $F^{(1)}/F^{(2)}$ new contracts for each old one, keeping NmF unchanged. It is tracked by the ratio-adjusted and return-stitched series.

Why it appears here. §6.2 and §6.3 separate roll-down from the roll gap. §7.3 uses implied roll yield to select contracts. §8.2 and §8.4 interpret the adjusted series as constant-count and constant-notional positions. §12.3 lists the three meanings of roll yield.

Deeper. Koijen, Moskowitz, Pedersen & Vrugt (2018); Carver (2023), on rolling and carry measurement.

A.50 Non-synchronous closes, forward-filling and old floor sessions

The idea. A daily price series records each market at its own close, and closes are not simultaneous. Tokyo, London and Chicago settle hours apart. Suppose news arrives after one market has closed. It shows up in that market's return the next day, but in the other markets' returns the same day. Same-date returns therefore understate how closely markets move together. Forward-filling carries the last price over a day on which a market was closed. It papers over the gap by inventing a zero return, followed by a catch-up return. Older data adds a third clock problem. Before electronic trading, many contracts traded only in floor sessions whose hours and settlement times differed from today's.

Formally. Suppose each market's daily return is the sum of independent hourly increments, and two markets' increments in the same hour have correlation ρ. Suppose their daily windows overlap for a fraction ϕ of the day. Their same-date correlation is then about $\phi\rho$, biased toward zero. Part of the true correlation appears instead between one market's return and the other's next-day return. Summing same-day and adjacent-day covariances, or using weekly returns, largely removes the bias. A forward-filled series replaces a missing F_t with F_{t-1}. The resulting zero return lowers measured volatility and correlation. Forward-filling across a roll also joins the prices of two different contracts. **Open outcry** was face-to-face trading in an exchange's pit, with its own session hours. CME closed most of its futures pits in 2015.

Why it appears here. §8.6 warns about markets in different time zones, forward-filling across rolls, and floor sessions in old histories. §2.1 and §8.6 warn against mixing settlement and last-trade clocks.

Deeper. M. Scholes & J. Williams, "Estimating Betas from Nonsynchronous Data," *Journal of Financial Economics* 5(3) (1977), 309–327; T. W. Epps, "Comovements in Stock Prices in the Very Short Run," *Journal of the American Statistical Association* 74(366) (1979), 291–298.

A.51 Backtest biases: look-ahead, survivorship, out-of-sample decay and capacity

The idea. A backtest simulates what a rule would have earned. It flatters the rule in five predictable ways.

- **Look-ahead bias** uses information that was not available when the decision was made.
- **Survivorship bias** tests only on markets that still exist, which were selected for having survived.
- **In-sample** results, from the same data used to choose the rule, overstate what it will earn on new data, the **out-of-sample** period, because some of what was fitted was noise.
- **Publication decay** is the tendency of published anomalies to weaken once traders exploit them.
- **Capacity** limits returns: any strategy's returns shrink as its capital grows, because its own trades move prices.

Formally. A decision at t may depend only on the information set $\mathcal{F}_t$ (A.3). Using any value published after t, such as day-t open interest released on day $t+1$ (A.17), is look-ahead. Survivorship bias is the difference between two averages: the performance of a universe selected by its state at the end of the sample, and that of the universe as it existed at each date. An in-sample estimate is computed on the data used to select or fit the rule, and an out-of-sample estimate on data not used. Suppose M variants are tried. The best in-sample result is then biased upward, even if no variant has an edge, by an amount that grows with M. Publication decay is the fall in a strategy's returns after publication. Capacity is the capital a strategy can deploy before its own price impact (A.40) consumes the expected return.

Why it appears here. §4.2 cites Bhardwaj, Gorton and Rouwenhorst's out-of-sample update. §7.4 and §11.3 question how much survives publication, costs and capacity. §7.5 and §14.1 forbid look-ahead in roll rules, and §8.6 warns about survivorship in a universe of markets.

Deeper. R. D. McLean & J. Pontiff, "Does Academic Research Destroy Stock Return Predictability?" *Journal of Finance* 71(1) (2016), 5–32; D. H. Bailey, J. M. Borwein, M. López de Prado & Q. J. Zhu, "Pseudo-Mathematics and Financial Charlatanism," *Notices of the AMS* 61(5) (2014), 458–471; Foundations of Econometrics in this collection.

A.52 Cross-sectional and time-series signals

The idea. A signal can rank markets against each other, or judge each market on its own.

- A **cross-sectional** strategy goes long the markets with the highest signal and short those with the lowest, at each date. Its net exposure to the asset class is roughly zero, and it earns only the difference between winners and losers.
- A **time-series** strategy goes long or short each market according to the sign or size of its own signal, so it can be long everything at once.

The two can use the same carry or momentum measure and still behave quite differently, because one bets on relative performance and the other on absolute performance.

Formally. Given signals $s_{i,t}$ across n markets, a cross-sectional rule sets positions proportional to $s_{i,t} - \bar{s}_t$, where $\bar{s}_t = n^{-1}\sum_i s_{i,t}$, or to ranks, so the positions sum to zero. A time-series rule sets positions proportional to $s_{i,t}$ itself, or to its sign. In the empirical literature, "in the cross-section" means comparing different markets at the same date. "Over time", or "in the time series", means comparing one market with itself at different dates.

Why it appears here. §6.4 reports that the cross-section of commodity returns is explained by the term structure. §11.3 reports that carry predicts returns "in the cross-section and over time". §11.4 gives both versions of the carry strategy.

Deeper. Koijen, Moskowitz, Pedersen & Vrugt (2018); T. J. Moskowitz, Y. H. Ooi & L. H. Pedersen, "Time Series Momentum," *Journal of Financial Economics* 104(2) (2012), 228–250.

A.53 Trend-following, momentum and price-level signals

The idea. Trend-following bets that recent price moves continue: hold markets that have risen, and short those that have fallen. Momentum is the academic name for the same tendency, measured in returns. Many practitioners' rules are expressed in price levels instead. A breakout rule buys when the price exceeds its highest level of the last n days. The 52-week high is the highest price of the past year. Chartists add support and resistance: price levels at which they believe buying or selling has clustered before. Level-based rules are sensitive to how a continuous futures series is built, because adjusted series change past levels.

Formally. Write P_t for the level of whichever series a rule reads.

- A *time-series momentum* signal is the sign or size of the past return over a look-back window, for example $\ln P_t - \ln P_{t-252}$ for a year.
- A *moving-average crossover* compares exponentially weighted averages of price over a fast window and a slow window.
- A *breakout* signal fires when $P_t > \max_{t-n\le k<t} P_k$.
- A distance such as "5% below the 52-week high" is $P_t / \max_{t-252<k\le t} P_k - 1$. Because it is a ratio, ratio adjustment preserves it and back adjustment does not (§8.3).

- Support and resistance are statements about traded prices, so they need each contract's actual prices.

Why it appears here. §5.4 mentions speculators who trade with momentum. §8.3 and §8.5 choose the series for momentum, breakout, 52-week-high and support-and-resistance rules. §11.4 combines trend with carry.

Deeper. Trend-Following in Financial Markets and Momentum in Financial Markets in this collection; Moskowitz, Ooi & Pedersen (2012).

A.54 Scheduled data releases

The idea. Some of the information that moves futures arrives on a published timetable. For energy and grains the most important releases are government inventory and crop reports. They update the market's estimate of supply and, through the theory of storage (A.28), the slope of the curve. Price moves concentrate around the releases, and liquidity thins just before them. A roll or large order placed at the minute of a release pays for that timing.

Formally. The US Energy Information Administration (EIA) publishes the **Weekly Petroleum Status Report**, with US crude and product inventories, usually on Wednesday mornings. It also publishes a weekly natural gas storage report on Thursdays. The US Department of Agriculture publishes the **World Agricultural Supply and Demand Estimates** (WASDE) monthly, with projections of supply, use and ending stocks by crop. What moves the price is the *surprise*: the release minus the market's expectation. The expected part is already in the price (A.3).

Why it appears here. The daily routine of §13.4 checks the calendar for these releases, alongside central-bank meetings and inflation data.

Deeper. The EIA and USDA release calendars, published on their websites.

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